

Fast and Stable Virtual Human Video Generation: A Survey

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Abstract Virtual-human video generation produces controllable face and body videos from speech, text, motion, or visual input. Practical systems must balance computational efficiency with the preservation of identity, appearance, and motion over extended generation. Although recent methods address both requirements, their different architectures and evaluation protocols make the reported results difficult to compare. This survey reviews the literature through two complementary perspectives: *Fast*, concerning generation cost and latency, and *Stable*, concerning long-horizon consistency. We organize methods into four routes according to the component responsible for generated appearance: generative adversarial network (GAN)-based, diffusion-based, autoregressive, and render-based methods. Within each route, we examine the principles, advantages, and limitations of its acceleration mechanisms. Across routes, we identify five recurring approaches to drift control: persistent identity anchors, bounded multiscale memory, position re-indexing, self-history training, and motion inheritance. The reviewed evidence shows why reducing one component’s cost can expose another bottleneck, while bounded memory alone does not establish bounded visual drift. We further review datasets and evaluation protocols, distinguish reported performance from architectural expectations, and identify open problems in joint efficiency and stability evaluation. The resulting organization provides a basis for comparing design choices under explicit task, hardware, and measurement conditions.

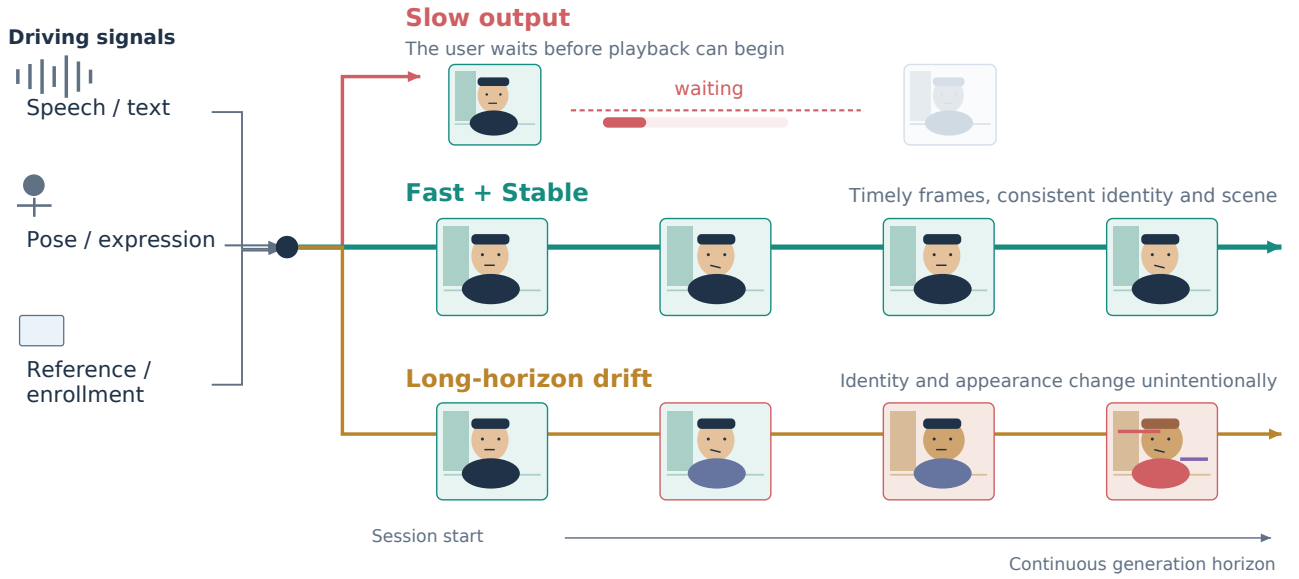
1 Introduction

Virtual-human video generation aims to synthesize a human subject whose appearance and behavior follow specified driving signals, such as speech, text, pose, expression, or video. The output may show a face, an upper body, or a full body, while the subject may be specified by one reference image, an enrollment recording, or a learned identity representation. Across these settings, the central requirements are to preserve the intended appearance and reproduce the motion conveyed by the input [123, 167, 195, 217].

Practical deployment introduces two further requirements. The first is computational efficiency: generation must fit the available compute and latency budget, particularly when serving many users or following a live signal [58, 76, 193]. The second is long-horizon stability: identity, appearance, synchronization, and motion should remain consistent as generation continues [102, 152]. We refer to these perspectives as *Fast* and *Stable*. Short-clip visual quality alone establishes neither property. We therefore examine how each family of methods reduces computation, maintains temporal consistency, and trades these objectives against representation quality and deployment cost.

Figure 1 illustrates the distinction: delayed output limits responsiveness, whereas unintended changes in identity or scene limit continuity. A useful evaluation must measure both, while allowing changes explicitly requested by the driving signal.

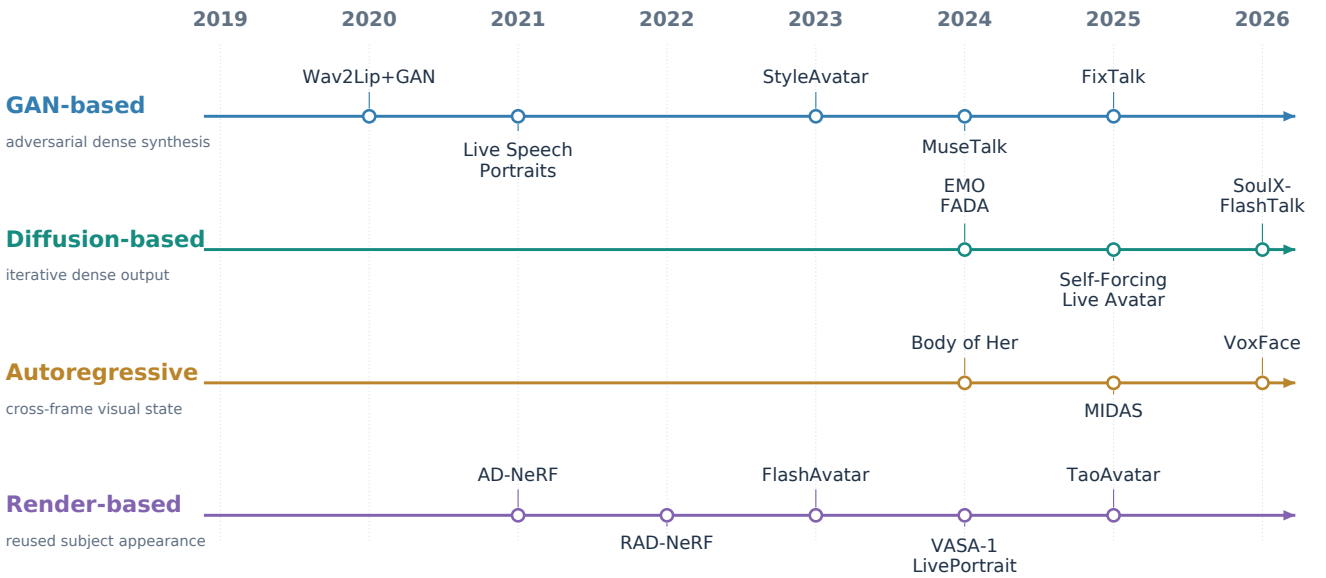
Figure 2 places representative methods on the four ownership routes over time. Recent work reduces different costs. Distillation can reduce the number of diffusion evaluations [116, 160, 207, 208]. Caching can reuse computation across denoising steps or adjacent frames [107, 118, 121]. Autoregressive models can generate cross-frame visual state sequentially and reuse



FAST Cost, first-output latency, steady-state throughput

STABLE Identity and appearance over an evaluated horizon

Fig. 1 Fast and Stable are independent requirements for virtual-human generation. The central path emits timely frames while preserving identity and scene appearance. The upper branch delays playback; the lower branch illustrates unintended identity and appearance drift during continuous generation. Frame glyphs are schematic.



Years follow archival publication or arXiv v1, whichever the supplied chronology records first. Selected examples only; no speed ranking.

Fig. 2 Chronology of representative methods per route. Years denote the earliest public appearance of the cited work (archival publication or arXiv v1, whichever occurred first). Selected methods mark Fast or Stable milestones within each route; the chronology is illustrative, is not exhaustive, and does not establish priority or compare speed across incompatible settings. Methods shown: Wav2Lip+GAN [138], Live Speech Portraits [113], StyleAvatar [178], MuseTalk [220], FixTalk [162], EMO [167], FADA [231], Self-Forcing [75], Live Avatar [76], SoulX-FlashTalk [152], Body of Her [4], MIDAS [26], VoxFace [191], AD-NeRF [59], RAD-NeRF [166], FlashAvatar [188], VASA-1 [193], LivePortrait [58], and TaoAvatar [24].

cached history [26, 93, 192]. Subject-specific renderers can reuse an enrolled scene or reference representation across frames [58, 59, 188, 193]. These mechanisms change different cost terms and should not be treated as equivalent: a measured frame rate does not by itself establish first-output latency or long-session stability, and a local speedup can move a bottleneck to the decoder, the condition encoder, or the serving system rather than remove it.

These developments motivate an organization based on the component responsible for generated appearance, which we call the *visual owner*. Systems with similar outputs can have different visual owners and computational costs, whereas systems covering different body regions can share the same acceleration mechanism. Identifying this component provides a common starting point for discussing efficiency and stability within each method family.

1.1 Scope: Fast and Stable

The main subjects of this survey are systems that produce human-centered RGB video and introduce mechanisms for reducing runtime cost or supporting extended generation. We cover face, upper-body, and full-body synthesis, including both generalizable and subject-specific methods. Render-based systems are included when they contribute to efficient appearance construction or frame production, for example through neural radiance fields (NeRFs) or 3D Gaussian splatting (3DGS) [83, 126]. Methods whose final output consists only of a mesh, motion coefficients, or vertex trajectories are discussed as supporting components when they clarify the design of an RGB-generation pipeline.

We distinguish the systems under comparison from foundational and contextual references. Foundational methods explain the origin of a representation or mechanism; contextual references clarify an interface, such as motion prediction followed by rendering. Neither role implies inclusion in a route’s deployment comparison. Inclusion in a primary route requires sufficient architectural evidence to identify the visual owner. Works with undisclosed ownership, or with a visual synthesis mechanism outside the four families considered here, are omitted from the route comparisons. Missing timing or stability measurements do not prevent an otherwise supported architectural assignment; such measurements are marked not reported (NR).

The following two cross-cutting lenses organize the analysis.

Fast concerns the computation and time required to produce visible output. We examine representation

size, model-call count, cost per call, computation reuse, and decoding or rendering overhead. Throughput and first-output latency are recorded separately under the source’s hardware and timing conditions.

Stable concerns consistency over an evaluated continuous duration. We examine identity and appearance drift together with the memory, training, and recovery mechanisms intended to control it. A long demonstration establishes an observed duration; quantitative stability additionally requires measurements over that duration.

Interactivity is a related but distinct property: a system can respond to new conditions during generation, such as a dialogue turn, interruption, or change of pose [76, 102, 199, 213, 215]. This capability does not establish stability over a long session; an interactive system may operate briefly, reset frequently, or accumulate drift. We include interactive systems when their generation mechanisms fall within the stated scope and examine their efficiency and stability evidence separately. Interaction behavior has its own surveys and evaluation protocols [7, 14, 46, 211] and is not ranked here.

Fast and Stable are independent evidence axes. A fast method may drift within a minute; a stable long-horizon method may run far below playback speed. We accept a joint claim only when each stated property has independent evidence. Section 2 defines the terms and evidence boundaries used throughout the survey.

1.2 Relation to Existing Surveys

Prior surveys provide complementary views of the area. Work on efficient generative modeling reviews distillation, caching, attention, and inference optimization [151]. Long-video surveys organize temporal extension around generation paradigms and resource or quality challenges [94]. Human-centered surveys organize talking heads, portrait animation, and human video by task or representation [123, 144, 148, 195]. Graphics-oriented reviews cover avatar construction and learned appearance priors [239]. Recent surveys consider interactive generative video, organized around the interaction loop [211], and real-time interactive avatars, organized by method family and by production axes such as rendering latency, anatomical coverage, and personalization [46].

Efficiency-oriented surveys do consider speed and long-horizon stability together, but for general video generation rather than virtual humans [94, 151]. Within the virtual-human setting, the closest treatment reviews real-time interactive avatars and comes near our concerns by making rendering latency one of its organizing axes, alongside anatomical coverage, personalization, and input modality [46]. Its grouping is by representation family, however, and it treats long-horizon identity

Identify the visual owner; stop at the first satisfied rule.

- | | | |
|--|----------------------------|------------------------|
| 1. Does a subject-specific scene or identity-conditioned warp carry appearance before rendering? | $\xrightarrow{\text{Yes}}$ | Render-based |
| ↓ No: test the next rule. | | |
| 2. Does native autoregression predict evolving dense or discrete visual state from prior visual state? | $\xrightarrow{\text{Yes}}$ | Autoregressive |
| ↓ No: test the next rule. | | |
| 3. Does diffusion or flow matching generate dense pixels, visual latents, or visual chunks? | $\xrightarrow{\text{Yes}}$ | Diffusion-based |
| ↓ No: test the next rule. | | |
| 4. Does an adversarially trained feed-forward generator produce the final dense or regional imagery? | $\xrightarrow{\text{Yes}}$ | GAN-based |
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Scope and evidence: A method enters the comparison only when its documented architecture supports one of these assignments. Otherwise it is omitted from the route comparisons. Auxiliary controllers and generic VAE decoders do not determine the route.

Fig. 3 Assignment to the four visual-generation routes. For methods meeting the scope and evidence requirements, the first satisfied rule determines the primary route. The order resolves hybrid cases without ranking quality or efficiency. Methods lacking a supported assignment are excluded from route comparisons. AR denotes autoregressive; VAE denotes variational autoencoder.

drift as an open challenge rather than a systematized design axis. The remaining reference points organize differently again: a task taxonomy groups systems with different inference loops and temporal state, an acceleration taxonomy groups caches that reuse different objects [123, 151, 195], and an interaction taxonomy foregrounds the dialogue loop rather than the generation cost [211]. We therefore ask three questions. Which component owns generated appearance? How is that route made fast? How is it kept stable over a long horizon? Answering these per route, through an ownership-based classification and route-specific efficiency and drift-control guidance, is what distinguishes this survey.

1.3 Ownership Taxonomy

Figure 3 assigns each included system to one of four routes by identifying its visual owner. The rules are applied in order: render-based, autoregressive, diffusion-based, then GAN-based. This convention resolves hybrids, such as an autoregressive motion controller driving a subject-specific renderer. It is not intended to classify every possible video generator. Section 2.3 defines the rules and the evidence required for an assignment.

- **GAN-based methods** (Section 3) use an adversarially trained feed-forward generator for dense or regional imagery [54, 81, 138, 163, 220]. Regional prediction and compact control reduce per-frame work, while temporal consistency depends on the generator’s conditioning and any temporal components.
- **Diffusion-based methods** (Section 4) use diffusion or flow matching to generate dense pixels, visual latents, or visual chunks [65, 68, 106, 167]. Acceleration reduces iterative calls, per-call cost, or repeated

work; stability mechanisms include reference anchors, bounded memory, and training with generated history.

- **Autoregressive methods** (Section 5) predict evolving visual tokens or latents from previous visual state [4, 26, 187, 192]. Token volume and serial prediction determine much of the cost, while retained history can support continuity and propagate errors.
- **Render-based methods** (Section 6) reuse a subject-specific mesh, neural field, Gaussian set, or identity-conditioned warp [58, 83, 99, 126, 193]. Reused appearance can reduce repeated synthesis and anchor identity, with enrollment, tracking, deformation, and compositing contributing additional costs and failure modes.

Driving modality, anatomical coverage, personalization, representation, and deployment mode are comparison axes rather than additional routes. Secondary diffusion, autoregressive, GAN, or rendering modules are recorded as boundary components and are not double-counted in route totals.

1.4 Contributions

This survey makes four contributions:

- **A classification by visual ownership.** We define four method families and state how hybrid systems are assigned, separating visual generation from auxiliary motion and control modules.
- **An analysis of computational efficiency.** For each family, we relate representative acceleration mechanisms to the costs they address and discuss their advantages, limitations, and remaining bottlenecks.

- **A comparison of stability mechanisms.** We organize drift control into five recurring patterns and examine their use and evaluation across the four families.
- **An evaluation framework and research agenda.** We review datasets and metrics, specify the conditions needed to interpret efficiency and stability results, and identify open problems in state management, personalization, and deployment.

Throughout, we preserve each source’s measurement boundary, mark undisclosed fields as not reported (NR), and compare source-reported performance only within compatible protocols. The manuscript does not claim a common-hardware re-benchmark.

1.5 Organization

Section 2 defines the task, route ownership, and the cost and measurement language. Sections 3–6 review the four routes under common Fast and Stable lenses, each ending with route-specific design guidance. Section 7 examines datasets, metrics, and reporting protocols. Section 8 develops cross-route design guidance and future directions.

2 Preliminaries and Common Evaluation Language

This section introduces a common formulation for the systems reviewed below. We first distinguish the generated representation from its decoder or renderer, then define route assignment and the measurements used to compare efficiency and stability.

2.1 Problem Formulation and System Boundary

Let $V = (I_1, \dots, I_T)$ denote the generated human video. A system may use enrollment evidence e (e.g., subject video, calibrated views, or a pre-built avatar), runtime conditions $c_{1:T}$ (e.g., audio, text, pose, expression, or a driving video), and persistent deployment state h_t . We write

$$(I_{t:t+k-1}, h_{t+k}) = \mathcal{G}(e, c_{\mathcal{V}_t}, h_t; \theta), \quad (1)$$

where k is the emitted frame or chunk length and $\mathcal{V}_t$ is the portion of the conditioning stream consumed at emission time. This formulation covers one-shot, subject-specific, and generalizable systems without assuming that a reference image is supplied at every runtime step.

Three output requirements recur across tasks. *Identity fidelity* asks whether the intended subject remains

recognizable; *driving fidelity* asks whether speech, expression, pose, gaze, and gesture follow their controls; and *temporal fidelity* asks whether motion and appearance remain coherent across emitted units. Their measurement depends on anatomy, modality, personalization, and the availability of a paired reference, so no single metric is valid for every method.

For analysis, we expose an intermediate representation r and a decoding or rendering operation D :

$$\begin{aligned} (r_{t:t+k-1}, h_{t+k}) &= G(e, c_{\mathcal{V}_t}, h_t), \\ I_{t:t+k-1} &= D(r_{t:t+k-1}). \end{aligned} \quad (2)$$

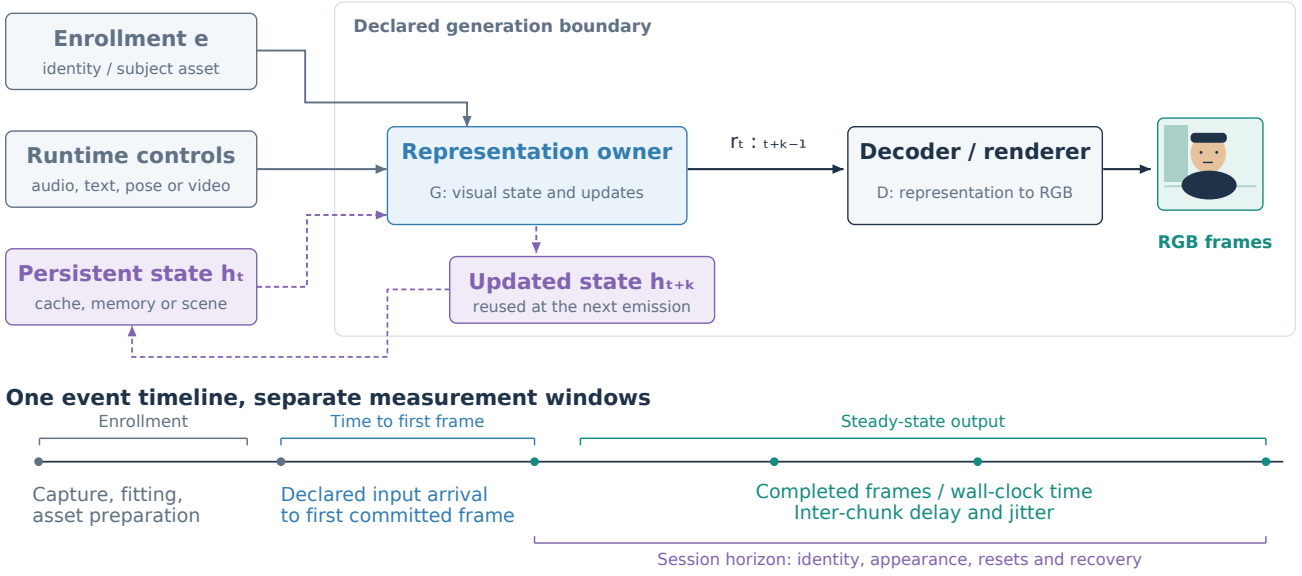
Equation (2) is a bookkeeping view, not a mandatory two-network architecture. In some systems, G and D are fused; in others, e is compiled into a subject-specific scene or the parameters of D before runtime. Any subject evidence consumed at decoding is therefore included in r or in those precompiled parameters under this notation. The distinction remains useful because it identifies the representation and state-update owner, the decoder or renderer boundary, and the cost paid during enrollment versus repeated inference.

Figure 4 makes this accounting explicit: G owns the generated representation and persistent-state update, while D maps that representation to observable frames. The clocks are separate reporting boundaries, so a steady-state throughput result cannot substitute for enrollment, first-output, or session-horizon evidence.

2.2 Efficiency and Stability Evidence

The Fast and Stable perspectives require different evidence. Efficiency is assessed from the computation and measured timing of a declared pipeline, including conditioning, synthesis, decoding, and system overhead. Stability is assessed from changes in identity, appearance, synchronization, and motion over a declared generation horizon, together with the state and recovery policy used during that run.

For long-horizon results, we distinguish an author’s duration claim, a qualitative demonstration, and quantitative measurements over time. These support different conclusions. A bounded cache establishes a bound on stored context, while drift must be measured from the generated output. Similarly, sequential emission and responsiveness to new input describe operating capabilities; neither establishes long-horizon stability. Enrollment duration is recorded separately from generation duration.



Every clock reports its included components. G and D may be fused; throughput cannot substitute for TTFF or stability.

Fig. 4 Analytical generation boundary and independent reporting clocks. The representation owner G updates visual state; decoder or renderer D produces observable RGB frames. Enrollment, time to first frame, steady-state output, and the evaluated session horizon retain their own measurement boundaries. G and D need not be separate networks.

2.3 Ordered Route Ownership

For a system within the scope of Section 1.1, we inspect the documented representation, its temporal update, and its path to RGB. We then apply the following rules in order and assign the first matching route (Figure 3). This precedence resolves hybrid architectures; it does not rank quality or efficiency.

1. **Render-based.** A subject-specific explicit or neural scene, including a mesh, neural field, Gaussian set, or identity-conditioned warp, owns appearance and is queried or updated before rendering. A learned model that predicts compact motion, geometry, deformation, or Gaussian attributes remains a controller.
2. **Autoregressive.** Otherwise, a native autoregressive model owns cross-frame dense or discrete visual state, such as visual tokens or visual latents. A downstream diffusion process or renderer may be a local emission head, but it does not own the evolving visual sequence.
3. **Diffusion-based.** Otherwise, diffusion or flow matching owns dense pixels, dense visual latents, or sequential dense visual chunks. A generic variational autoencoder (VAE) decoder does not displace this ownership.
4. **GAN-based.** Otherwise, an adversarially trained feed-forward generator owns final dense or regional visual synthesis. Adversarial loss attached only as a regularizer is insufficient.

An assignment requires architectural evidence from the primary paper or its accompanying technical description. If that evidence is insufficient, or none of the four rules applies, the work is excluded from route comparisons rather than assigned an additional category. The taxonomy is therefore an organization of the reviewed method families, not an exhaustive partition of all video synthesis algorithms.

Supporting modules are identified by their role in the complete pipeline. A diffusion motion controller driving a subject-specific renderer remains part of a render-based system; an autoregressive dialogue model emitting pose controls does not become a visual AR method. Missing performance fields are marked NR independently of this architectural decision. Historical and contextual examples are identified explicitly and do not supply deployment evidence for another route.

2.4 Route-Specific Cost Structures

The same visible frame can arise from different cost structures, so each route chapter opens with a route-specific cost decomposition. The following summaries fix the shared vocabulary.

GAN-based synthesis. After training, a GAN generator can synthesize dense appearance without an iterative sampling loop [54, 81, 82]. One feed-forward evaluation describes call count, not absolute latency: preprocessing, network size, output resolution, compositing, and

hardware still determine runtime. Route-specific speed claims therefore require a measured boundary rather than the presence of a discriminator or a single-pass architecture.

Diffusion and flow matching. Diffusion [65] and flow-matching [50, 106] models traverse a noise-to-data path through repeated function evaluations. Their cost depends on the generated unit (frame, clip, or chunk), strict number of function evaluations (NFE), cost per evaluation, conditioning passes, latent codec [88, 137, 146], and system scheduling. Step reduction [116, 147, 160], caching, and parallelism reduce different terms and should not be collapsed into a single speed label.

Autoregressive visual state. Autoregressive models factor a sequence and predict future units from prior state [13, 143, 192], over discrete tokens [171] or continuous latents [23]. Runtime depends on token or block granularity, serial depth, attention, cache updates, local decoding, and output rate. Key-value (KV) caching can avoid recomputing past projections while still growing memory; bounded windows, recurrent summaries, compression, eviction, or resets are separate Stable mechanisms. Serial depth itself can be attacked with parallel or speculative emission [93].

Render-based synthesis. Render-based systems amortize a subject-specific appearance representation and repeatedly update or query it using compact controls. Their costs include enrollment or scene construction, upstream tracking, controller evaluation, deformation or field queries, rasterization or volume integration, neural decoding, compositing, and input/output (I/O). A renderer call is not universally cheap or one-step: neural radiance field (NeRF) ray sampling and Gaussian sorting expose different cost terms [83, 126, 127]. Under our accounting rule, a fixed scene can anchor identity without proving long-session stability of the complete pipeline.

2.5 Conditions and Intermediate Representations

Runtime conditions specify what the person should do. Common signals include speech audio, text, pose, expression, gaze, and a driving video; pose and expression may themselves be estimated by upstream trackers whose cost belongs to the deployed pipeline [18, 204]. Enrollment evidence specifies who or what is rendered and may be a single reference, a short monocular recording, calibrated multiview capture, or a pre-built asset.

Intermediate representations determine the interface between control and appearance. Landmarks and keypoints are interpretable, but they omit appearance. Facial action units [49], three-dimensional morphable

models (3DMMs), Faces Learned with an Articulated Model and Expressions (FLAME), the Skinned Multi-Person Linear Model (SMPL) and its expressive extension SMPL-X, and blendshape parameters expose compact geometric controls [99, 112, 133]. Flow fields and identity-conditioned warps transport source features. Learned motion latents trade semantics for capacity, whereas visual tokens or latents carry dense appearance state. Neural fields and Gaussian primitives encode subject-specific scenes [83, 126]. Dimensionality alone does not determine speed. Compact controls may feed an expensive renderer, while a dense latent may use a fast decoder.

For comparison, we record anatomical coverage (face, upper body, full body, hands), personalization regime, representation owner, decoder/renderer, and evaluation boundary. These axes remain orthogonal to the four primary routes.

2.6 Common Acceleration and Measurement Terms

Generic acceleration acts on distinct Fast terms [151]. The following categories are the shared skeleton of every route’s Fast analysis; each category still requires a measured system boundary:

- **Fewer evaluations:** distillation, consistency objectives, solver reduction, block prediction, or speculative emission reduce strict NFE or serial calls [93, 116, 160, 207].
- **Cheaper evaluations:** representation compression, sparse computation, lightweight backbones, quantization, and optimized kernels reduce the cost of each call. For attention, I/O-aware FlashAttention reorganizes exact-attention computation in 16-bit floating point (FP16) or bfloat16 (BF16) to reduce high-bandwidth-memory traffic [41, 42], whereas FlashAttention-3 also provides a Hopper 8-bit floating-point (FP8) path whose speed lever is lower-precision Tensor Core execution and whose block quantization and incoherent processing control quantization error [149]. These standalone kernel results do not establish an end-to-end virtual-human speedup.
- **Reuse:** feature, condition, reference, or KV caches reuse a precisely named object across denoising steps, frames, chunks, or autoregressive history [107, 118, 121].
- **Decoder/renderer reduction:** efficient codecs, bounded ray samples, primitive pruning, rasterizer design, and regional synthesis reduce visible-output cost.

- **System optimization:** compilation, mixed precision, memory layout, I/O, pipelining, and parallel execution change the measured pipeline boundary.

The following terms are not interchangeable. *Throughput* is completed output per wall-clock time after the disclosed operating condition is reached. *Frames per second (FPS)* is frame throughput at a stated resolution and boundary; it is not content frame rate or latency. *Real-time factor (RTF)* is wall-clock time divided by generated media duration under a stated convention, so values below one are faster than playback. *First-output latency* is the umbrella term for the interval from a declared input-arrival event to the first committed output of any type. *Time to first frame (TTFF)* is reserved for the case in which that first committed output is an observable frame. *Inter-chunk latency* measures successive committed outputs and includes the declared buffering policy. *Number of function evaluations (NFE)* counts evaluations of the iterative vector field or denoiser for the stated generated unit.

Every reported number should identify hardware, device count, resolution, precision, batch or concurrency, warm-up, measurement window, and included components. When these fields or the evaluated duration are absent from the primary source, we report them as not reported (NR) rather than infer deployment capability from architecture or title.

3 GAN-Based Methods

Adversarially trained feed-forward generators produce a frame or image region in a single generator evaluation [54, 81, 82]. This structure avoids an iterative sampling loop, but runtime still depends on the network, conditioning, and output-processing stages. We review three ways of reducing this work: regional synthesis, compact control, and reusable appearance. We then examine temporal consistency, which requires particular attention when the visual generator operates independently on each frame.

3.1 Scope, Ownership, and Cost Boundary

A method is assigned to this route when an adversarially trained feed-forward generator owns final dense or regional visual synthesis after the render-, autoregressive-, and diffusion-ownership tests in Section 2.3. A discriminator is normally used only during training, so removing it from the deployed graph is not a route test. Conversely, an adversarial loss may regularize a warper, neural field, or diffusion decoder without making that system GAN-primary. The relevant questions are what representation

is generated, which module owns appearance, and how RGB is reconstructed.

The training objective and inference path must both be checked when assigning a method. Wav2Lip reports a base generator without the visual-quality adversarial term and a Wav2Lip+GAN variant with a separate quality discriminator [138]; only the latter is included in the GAN comparison. MuseTalk reuses a pretrained variational autoencoder (VAE) and a U-Net architecture associated with latent diffusion, but performs one-call regional prediction with reconstruction, synchronization, and face/lip adversarial losses [220]. It has no diffusion objective or iterative sampler. Thus, architectural ancestry alone does not determine the route.

For a feed-forward visual path, a useful first-order deployment decomposition is

$$C_{\text{frame}} = C_{\text{condition}} + C_{\text{generator}} + C_{\text{decode/compose}} + C_{\text{system}}. \quad (3)$$

The first term covers audio, identity, pose, expression, and reference encoding. The second covers the network that predicts pixels or dense visual features. The third covers decoding, parsing, blending, refinement, and super-resolution. The final term covers data movement, precision conversion, scheduling, and I/O. The methods below each attack a different term; none establishes time to first frame (TTFF) or steady-state throughput unless the source defines the timer and system boundary.

3.2 Regional Synthesis: Shrinking $C_{\text{generator}}$

The most direct saving is to restrict synthesis to the region that actually changes. Wav2Lip conditions a feed-forward face decoder on identity, a lower-half-masked pose prior, and speech, and generates a dense RGB face crop frame by frame; a frozen five-frame synchronization expert supplies a training loss rather than recurrent inference state [138]. Under the paper’s LRS2-derived protocol, the base and GAN variants expose a small synchronization–fidelity trade-off measured by lip-sync error distance (LSE-D), lip-sync error confidence (LSE-C), and Fréchet Inception Distance (FID): 6.386/7.789/4.887 versus 6.469/7.781/4.446. The source reports no runtime measurements, but the design pattern of synthesizing the mouth and keeping the canvas became the template for a large lip-editing family, including VideoReTalking’s editing pipeline [27], DInet’s deformation-inpainting for high-resolution dubbing [221], landmark- and appearance-prior variants [232], memory-augmented mouth retrieval [132, 165], and style-space lip synchronization [57, 86], with ReSyncer unifying style-based synchronization across tasks [56].

MuseTalk moves regional synthesis into a pretrained VAE latent space, which shrinks the generated tensor a second time: its multimodal U-Net predicts a lower-face latent in one call, and a precomputed mask is reused for compositing [220]. The authors report 30 FPS for 256×256 output on one V100 with preloaded data (component inclusion NR). The supplement identifies occasional jitter from single-frame generation, the canonical Stable weakness of stateless regional synthesis, taken up in Section 3.6.

Training-only critics can improve a regional generator without adding a deployed inference module. TalkLip uses a frozen AV-HuBERT-derived lip-reading expert and reports improved word intelligibility over Wav2Lip on its LRW/LRS2 protocols [175]. Synchronization experts, lip-reading experts, and quality discriminators therefore offer additional supervision without a corresponding critic evaluation at runtime. Their training cost and the generator’s own inference cost remain separate considerations.

3.3 Structured Control and Codebooks: Shrinking $C_{\text{condition}}$ and Bounding the Output Space

A second family separates identity from compact motion controls. MakeItTalk splits content and speaker-aware landmark motion [235], while PC-AVS separates pose, content, and identity in latent space [233]. EDTalk factorizes mouth, pose, and expression into orthogonal basis banks; its shared adversarial image generator produces RGB, and a normalizing flow samples compact pose-bank weights [163]. Emotional adaptation provides another use of disentangled control [53].

StyleGAN-based reenactment instead controls a pretrained generator by editing latent directions, masking style channels, or refining inversions [10–12, 20]. Variants address continuous emotion [6], semantic expression [131], cross-domain heads [80], and text-guided editing [16]. These methods make the control interface compact, but do not necessarily make control extraction inexpensive: inversion and feature estimation can add work, and the full image generator remains on the inference path.

Codebooks represent generated appearance through a learned set of reusable features. FlowVQTalker combines autoregressive normalizing-flow controllers for compact 3DMM expression and pose coefficients with an adversarially trained vector-quantized image generator, which decodes a spatial code sequence into a 512×512 face [164]. A grid-based codebook provides a related efficiency-oriented representation for person-specific reenactment [177].

Retrieval reuses detail compatible with the source identity. FixTalk compresses fourth-layer detail features from $512 \times 32 \times 32$ to a 1×512 token and retrieves aligned detail before generation; its Enhanced Motion Indicator suppresses identity leakage from driving features [162]. The reported throughput is 27.601 FPS on an A100 (80 GB) at 512×512 , with complete timer composition NR. This supports the reported operating point, but does not independently quantify the speed contribution of retrieval or compression.

3.4 Personalized Compositional Generators: Amortizing Appearance

Appearance reuse can be made stronger through subject-specific training, trading enrollment cost for per-frame savings. This is the same trade that defines the render route, executed here inside an adversarial generator. Live Speech Portraits is the route’s early real-time landmark: a person-specific pipeline from streaming audio through predicted motion to adversarial image synthesis, reported at 30 FPS with over 27 FPS attributed to generation on a consumer GPU [113]. StyleAvatar scales the idea to a compositional generator: two StyleGAN modules for static facial-UV and background features, a StyleUNet for non-facial foreground, Neural Textures with a learned mask for regional composition, and a second StyleUNet for final RGB [178]. With two 16-bit TensorRT StyleUNets, the authors report a source-labeled “live system” rate of 35 FPS (28 ms/frame) on one RTX 3090, with 20 ms for the model-only GPU boundary. The cost structure is instructive: static regional features are computed once (enrollment, reported at roughly two to three hours of person-specific training plus about four more for teeth convergence), so the per-frame path touches only the dynamic StyleUNets, a direct reduction of $C_{\text{generator}}$ purchased with C_{enroll} . RAM-Avatar extends person-specific adversarial synthesis to full-body control with real-time intent [45].

3.5 Fast Design Guidance

Table 1 consolidates the evidence. Four levers recur, each acting on a different term of Equation (3):

1. **Synthesize only what changes** ($C_{\text{generator}} \downarrow$). Face-crop or lower-face prediction reduces the generated area; latent-space prediction, as in MuseTalk, further compresses the representation. The saving depends on crop size and codec compression. Compositing and boundary blending remain necessary, with seams and mouth interiors requiring separate quality evaluation.

2. **Drive with compact controls** ($C_{\text{condition}} \downarrow$). Basis banks, 3DMM coefficients, latent directions, and codebook indices replace dense driving features. Cost: expressiveness is bounded by the control space; motion outside it cannot be generated.
3. **Reuse appearance features** ($C_{\text{generator}} \downarrow$). Person-specific static feature banks, as in StyleAvatar, trade enrollment for repeated savings. Reference-conditioned retrieval, as in FixTalk, can instead support one-shot operation. Their preparation costs and generalization requirements should therefore be reported separately.
4. **Optimize the fixed path** ($C_{\text{system}} \downarrow$). Compilation, reduced precision, and kernel fusion can reduce deployment overhead. StyleAvatar provides an FP16 TensorRT example. The gains depend on the hardware and implementation; its model-only 20 ms remains distinct from system latency.

These mechanisms address different components and can be combined, although their total benefit depends on the remaining pipeline. Face detection, alignment, audio encoding, and I/O may become substantial once generation is inexpensive. The FPS values in Table 1 use different resolutions, devices, personalization regimes, and timing boundaries, so they describe individual systems rather than a controlled ranking.

3.6 Stable Design Guidance

Framewise generation avoids feedback of a previously synthesized frame when that frame is not reused as an input. This removes one source of recursive error, but does not enforce temporal consistency. MuseTalk reports single-frame jitter; variation in pose estimates or reference features can also produce inconsistent output. Recurrent controllers and retrieval memories introduce additional state whose errors must be evaluated independently of the image generator.

The drift-control patterns in Section 4.7 help distinguish these cases. A fixed reference or person-specific feature bank serves as an identity anchor (L1). FixTalk’s retrieval memory and FlowVQTalker’s coefficient history supply persistent information, although they do not by themselves establish bounded multiscale temporal memory (L2). Position re-indexing (L3) is relevant only where a sequential component uses positions. Evidence for self-history training (L4) and explicit motion inheritance (L5) is limited in the systems reviewed here. None of the papers in Table 1 reports a joint minute-scale, time-binned evaluation of identity, synchronization, motion, and visual degradation. The long-session benefit of

avoiding visual feedback therefore remains a hypothesis to test.

3.7 Limitations and Outlook

Regional methods require evaluation that separates synthesized content from preserved source pixels, including artifacts at their boundary. Compact controls should be assessed for expressive coverage and extraction cost, while personalized generators require enrollment reporting alongside runtime. For stability, a duration-binned comparison could test whether avoiding visual feedback reduces drift and how much framewise jitter remains. The reviewed sources do not provide that joint comparison. Reporting the generated region, resolution, personalization, preparation cost, and timing boundary would make these trade-offs easier to assess.

3.8 Summary

GAN-based methods reduce repeated synthesis through feed-forward prediction, regional output, compact control, and reusable appearance. Their temporal behavior depends on both the image generator and its conditioning components. The reviewed sources demonstrate several efficiency mechanisms, but their heterogeneous timing boundaries and limited longitudinal evaluation leave the joint efficiency–stability trade-off unresolved.

4 Diffusion-Based Methods

Diffusion and flow models support generation conditioned on identity, speech, text, pose, and previous video [65, 68, 105, 106, 167]. Their iterative inference motivates methods that reduce sampling calls, compress the generated representation, or reuse computation. Extending generation across chunks introduces a second challenge: the model must maintain consistency while consuming its own outputs. We first review the main acceleration mechanisms, then discuss streaming generation and five recurring approaches to drift control. Architecture alone does not determine their speed relative to the other routes.

4.1 Scope and Cost Model

The reverse chain of a conventional denoising diffusion probabilistic model (DDPM) serially invokes the denoising network many times [65]. Flow matching avoids simulating an ordinary differential equation (ODE) while

Table 1 Representative GAN-based visual generators. Timing values retain the source paper’s hardware, resolution, and component boundary and are not directly comparable. NR denotes not reported in the primary paper.

Method	Visual owner / scope	Primary Fast mechanism	Source timing boundary	Stable boundary
Wav2Lip+GAN [138]	Feed-forward RGB face-crop generator; identity-general	Face-crop rather than full-canvas synthesis; training-only sync and quality critics	FPS, hardware, resolution, TTFF, and timer NR	Stateless framewise generation; evaluated duration NR
MuseTalk [220]	U-Net-generated lower-face VAE latent; no subject adaptation	One-call regional latent prediction; reusable blend mask	30 FPS, 256 ² , one V100, preloaded data; included stages NR	Single-frame jitter reported; quantitative long-horizon evidence NR
TalkLip [175]	Feed-forward 96 ² RGB face-crop generator; identity-general	Training-only lip-reading expert improves intelligibility	FPS, latency, hardware, parameters, and timer NR	Stateless framewise generation; long-horizon evidence NR
Live Speech Portraits [113]	Person-specific adversarial portrait generator	Compact predicted motion; person-specific appearance	Source-reported 30 FPS system rate on a consumer GPU; timer boundary NR	Recurrent motion prediction; long-horizon evidence NR
EDTalk [163]	Adversarial RGB generator controlled by mouth/pose/expression banks; one-shot	Compact reusable bases and progressive freezing	Inference timing NR; about 7 h disentanglement training on two 3090 GPUs	Compact controller state only; long-horizon evidence NR
FlowVQ Talker [164]	Adversarial VQ image generator; full 512 ² portrait	Compact coefficient control and learned spatial codebook	Inference FPS, latency, hardware, memory, and timer NR	Coefficient history in the controller; long-horizon evidence NR
FixTalk [162]	Adversarial dense portrait generator; one-shot	Identity-suppressing EMI and compressed memory-retrieved EDI detail	27.601 FPS on A100 80 GB; 512 ² data, full timer boundary NR	Retrieval memory (weak L2); TTFF and long-horizon evidence NR
StyleAvatar [178]	Person-specific compositional StyleUNet stack; full portrait	Static regional features, wavelet-domain processing, FP16 TensorRT	Source-labeled “live system”: 35 FPS / 28 ms on one RTX 3090; 20 ms model-only component	Subject-specific enrollment as L1 anchor; Stable drift evaluation NR

EDI: Enhanced Detail Indicator; EMI: Enhanced Motion Indicator; FP16: 16-bit floating point; FPS: frames per second; GAN: generative adversarial network; GB: gigabyte; GPU: graphics processing unit; L1/L2: drift-control patterns defined in Section 4.7; RGB: red-green-blue; TTFF: time to first frame; VAE: variational autoencoder; VQ: vector quantization.

Table 2 Design guidance for the GAN route. Mechanisms are supported by the sources in Table 1; costs are the recurring trade-offs those sources disclose or exhibit.

Target	Recommended levers	Known costs
Low per-frame latency, general identity	Regional (crop or latent) synthesis; one-call generator	Seams, mouth-interior quality; preprocessing untimed
High throughput at high resolution	Compact control + codebook or retrieval; compiled FP16/INT8 path	Control space bounds expressiveness; hardware-specific
Maximum speed for one subject	Person-specific static feature banks; amortized appearance	Hours-scale enrollment; no zero-shot use
Minute-scale stability	Fixed identity reference; temporal training or compact temporal control	Frame-wise jitter and controller drift; joint long-horizon evaluation NR in the reviewed systems

constructing its training targets and can learn straighter probability paths, but generation still requires integrating the learned vector field [50, 106]. With a diffusion Transformer (DiT) in video-latent space, each invocation is itself governed by the number of tokens, model width and depth, and the temporal and spatial extent of attention [134, 203]. A useful first-order decomposition of deployed generation cost is

$$C_{\text{gen}} = N_{\text{FE}} C_{\text{denoise}} + C_{\text{condition}} + C_{\text{decode/render}} + C_{\text{system}}. \quad (4)$$

Here, the first term counts the number of function evaluations (NFE) and depends on sampling steps, classifier-free guidance (CFG) branches, solver stages, and any extra network calls used to refresh a cache. The second covers speech, text, reference-image, and motion encoders; the third includes VAE decoding, super-resolution, and any separate neural renderer; and the last includes kernel scheduling, device communication, data transfer, playback buffering, and I/O. Equation (4) explains why few-step sampling does not by itself establish end-to-end speed. Once N_{FE} is small, the bottleneck can move to a single large DiT call, the codec, condition encoding, or communication. Component-level gains from caching, quantization, and parallelism therefore cannot be multiplied mechanically into a system-level speedup.

For deployed systems, two service clocks matter beyond raw throughput. Time to first frame (TTFF) controls when playback can begin; it includes input acquisition, condition encoding, first-chunk denoising, first decode, and buffering, so audio buffering imposes a TTFF floor that no denoiser acceleration can remove. Steady-state inter-chunk latency and jitter control whether playback continues without stalls. A nominal playback rate and a real-time factor are neither of these. We project *Fast* onto NFE, per-call computation, decoding, and system overhead, with TTFF and steady-state throughput as its service observables; we project *Stable* onto bounded state and identity or appearance drift

under self-generated history. Reported speeds retain the source timing boundary.

We define the route by what the diffusion or flow component actually generates. Dense visual owners generate image or video latents; a VAE or super-resolution module then reconstructs RGB. FADA, RAP, StreamAvatar, SoulX-FlashTalk, and Avatar V are representative [48, 103, 152, 161, 231]. Compact-motion boundary cases instead predict expression, head pose, FLAME coefficients, or a skeleton and pass these controls to an independent warper, Gaussian renderer, or parametric avatar: IF-MDM, INFP, AudioRTA, Avatar Forcing, EmbodiedHead, and FloodDiffusion belong to this boundary group [15, 85, 91, 201, 219, 238], and we do not rank motion-space throughput against dense-video diffusion. Routed hybrids follow the ownership test: OmniTalker’s flow-matching backbone predicts spectrogram and facial-motion codes decoded by Vocos and a warp-based GAN renderer, so it is not direct RGB video diffusion [186]; ARIG, Teller, and REA-Listener attach compact motion generation to warp- or Gaussian-based renderers and are treated in Section 6 [60, 226, 228]. Conversely, X-Streamer’s Actor applies DDIM sampling to every sequential dense LTX-VAE video chunk, so it remains on this route despite its autoregressive system organization [190], whereas MIDAS and Body of Her give a native autoregressive backbone ownership of cross-frame state with only a local diffusion head and belong to Section 5 [4, 26].

The terminology around “flow” needs one clarification. Normalizing flows use invertible transformations with tractable Jacobians; their connection to diffusion lies at the level of probability-density evolution [130]. In current virtual-human systems, flow matching or rectified flow means regressing a velocity field from noise toward data and sampling it with an ODE solver or a distilled approximation; “simulation-free” characterizes the training objective, not one-call inference [106]. Sampling steps convert to strict NFE only when solver stages, CFG branches, and auxiliary cache calls are counted.

Across these architectures, a stable task prior organizes many of the mechanisms below: *appearance changes slowly while the driving signal changes quickly*. Identity, clothing, background, and reference style are low-frequency state that can be compressed, cached, or kept as persistent anchors. Phonemes, mouth shape, transient expressions, and gesture must track the current input. READ aligns a temporally compressed visual latent with a SpeechAE [173]; RAP introduces hybrid audio attention to compensate for strongly compressed video tokens [48]; SyncCache reuses heavy DiT residuals within one reverse trajectory while recomputing light audio blocks at every step [117]. The same asymmetry

motivates long-video systems to preserve a reference sink and recent motion while compressing or evicting older history.

The main acceleration mechanisms can now be located in Equation (4). Few-step distillation, consistency training, and distribution matching reduce N_{FE} . Compressed latents, smaller backbones, sparse attention, and low-precision execution reduce $C_{denoise}$. Pre-encoding and cross-step, cross-block, or cross-modal caches remove repeated conditioning and attention work. Lightweight or causal codecs reduce $C_{decode/render}$. Fixed windows, fused kernels, and asynchronous pipelines reorganize C_{system} . Deployed systems usually combine several levers, and meaningful evaluation must identify the active levers and track where the bottleneck moves, rather than assigning the joint system result to a single module [76, 117, 231].

Figure 5 makes the two projections operational. Panel A separates component costs from candidate levers; Panel B illustrates bottleneck migration as successive levers are applied; and Panel C separates identity anchors, compressed memory, rolling motion, and current conditions, the state slots that Section 4.7 organizes into patterns L1–L5.

4.2 Fast I: Compact Representations and Efficient Backbones

Representation-level acceleration follows two fundamentally different paths. One diffuses in a low-dimensional motion, expression, or parametric space and delegates appearance to a separate decoder or renderer. Its per-step state is small and easy to retain, but its expressiveness is bounded by the intermediate representation and often by an identity-specific renderer. The other continues to generate dense visual content, but applies a highly compressed spatiotemporal VAE so that the DiT processes far fewer video tokens. It preserves a more open appearance model, while paying for audio–visual alignment, recovery of local high frequencies, and VAE decoding. FPS measured for these two paths does not denote the same computation.

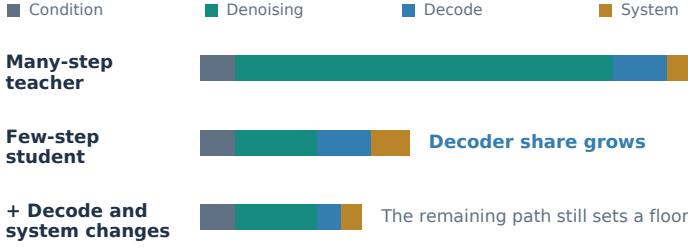
Compact state can move the expensive stochastic process out of pixel space. IF-MDM diffuses in a 20-dimensional implicit facial-motion space before an RGB generator synthesizes the portrait. INFP uses a conditional DiT with only four blocks to generate speaking and listening motion latents (no inference timing reported). AudioRTA distills a 40-step DDIM teacher into one-step expression-latent prediction and renders through the mesh/3D-Gaussian decoder of URAvatar. Avatar Forcing applies block-causal diffusion forcing to

A Cost terms and their levers



B Bottleneck migration

Schematic costs, not measurements



Component gains do not multiply into an end-to-end speedup.

C State for continuous generation

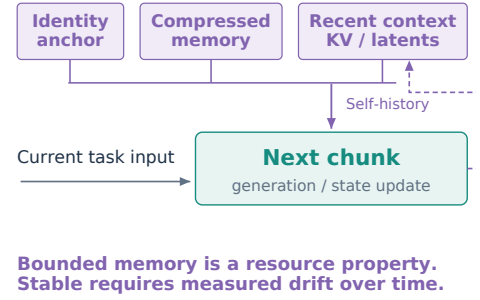


Fig. 5 Fast and Stable projections for diffusion-based visual generation. (A) Acceleration targets distinct terms: condition encoding, denoiser evaluations, cost per call, decoding, and system overhead. (B) Schematic cost bars illustrate bottleneck migration after step reduction and subsequent decoder/system optimization; they are not measurements. (C) Persistent anchors, compressed memory, recent context, and current controls support continued generation. Bounded state does not by itself establish bounded drift.

a 512-dimensional motion latent per frame and predicts reactive dyadic behavior from both parties' signals. EmbodiedHead conditions on the per-frame listening/speaking state, predicts a 115-dimensional FLAME motion vector with a rectified-flow DiT, and passes it to GAGAvatar [85, 91, 201, 219, 238]. FloodDiffusion studies streaming, time-varying text conditions in skeleton-motion space [15]. These methods show why compact state is efficient: the diffusion process runs over tens of dimensions rather than dense latents. Their motion-module or renderer-inclusive speeds should nevertheless not be compared directly with open-domain dense-video pipelines, and their appearance ceiling is set by the downstream renderer (Section 6).

The dense path reduces cost by compressing space and time. Stable Video Diffusion shows that latent-video quality depends not only on the denoiser, but also on image pretraining, video-data curation, and temporal-layer training [9]. HunyuanVideo, Wan, and CogVideoX organize large video generation around a spatiotemporal VAE, DiT, and text encoder [90, 172, 203]; they expose the shared computational substrate: VAE compression determines token volume, full spatiotemporal attention determines much of the per-call cost, and chunked decoding determines whether frames can be emitted before all latents are available. LTX-Video co-designs a high-compression VAE with a token-matched DiT, illustrating that reducing token count can change the computation more substantially than shrinking the

network [61]. For virtual humans, however, aggressive compression is especially risky around teeth, lip boundaries, eyelids, and rapid expressions.

If temporal and spatial compression factors are r_t and r_s , the video-token count scales approximately as $THW/(r_t r_s^2)$ [61, 134]. Global self-attention is quadratic in this count, and audio cross-attention depends on the number of audio tokens. Greater compression reduces attention, MLP computation, KV storage, and communication. It also increases the phonetic changes represented by each temporal token and allocates fewer spatial tokens to small facial structures. Compression must therefore be co-designed with audio downsampling, detail recovery, and decoder throughput. READ and RAP illustrate two solutions: READ uses a Temporal VAE to extend each visual token over a spatiotemporal block and trains a SpeechAE at the corresponding time scale (eight solver steps; the reported 4.421 s is backbone time for one 4.84 s, 121-frame sample) [173]; RAP combines an LTX-VAE latent, hybrid audio attention, and latent inheritance, reaching 42.41 and 39.87 FPS on one A800 at 512×512 under the authors' protocol [48]. Both show that once the visual timeline is compressed, frame-level speech features cannot simply be injected without accounting for how several phonetic events map to one visual token.

Dense generation can also reduce the cost of a long reference. Avatar V lets generated tokens attend to a long reference video through asymmetric sparse ref-

erence attention; reference K/V projections and their rotary position embedding (RoPE) results are computed in the first denoising step and reused for the remaining 23, followed by a one-step adversarial super-resolution stage. The paper reports an approximately threefold latency reduction relative to its unoptimized implementation, with no absolute latency or FPS [103]. The case shows that long-reference tokens can become a second attention bottleneck and that cross-step conditioning caches matter even after denoising is shortened.

After DiT distillation, the VAE can become dominant, the canonical bottleneck migration of Figure 5B. PersonaLive offers TinyVAE for higher decoder throughput, but records degradation in identity, motion, and temporal metrics; MotionStream’s highest reported 29.5 FPS configuration on one H100 also uses TinyVAE [102, 153]. Wan’s causal VAE can encode and decode arbitrarily long video in chunks while caching adjacent codec features [172]. Compact representations should be evaluated with reconstruction quality, decoder latency, mouth detail, identity, synchronization, and the actual end-to-end timing boundary, not only compression ratio or denoiser floating-point operations (FLOPs).

4.3 Human-Centric Conditioning and Expression Control

Audio conditioning is not a single objective. EmotiveTalk separates audio information relevant to lip motion from information associated with emotion and expression, then uses emotional video diffusion to expand the temporal and expressive signal. The design makes explicit a distinction that matters for both Fast and Stable evaluation: phonetic accuracy, motion naturalness, and identity fidelity are separate targets, and an audio encoder or diffusion sampler may improve one while leaving the others unchanged. Conditioning-side disentanglement therefore belongs in the diffusion cost ledger, including its extra encoders, attention paths, and temporal expansion, rather than being counted as a free quality module [174]. The same principle motivates emotion-specialized or mixture-of-expert controllers, but routing complexity and expert capacity must be reported separately from the visual generator.

4.4 Fast II: Few-Step and One-Step Distillation

Few-step methods shorten the reverse trajectory, but at least three operations are often conflated. Trajectory rectification or consistency training makes distant noise

states easier to connect; distribution matching distillation (DMD) trains a short-path generator by contrasting real- and generated-distribution scores; and CFG distillation folds conditional and unconditional branches within a step into one student call. These mechanisms alter the path, student distribution, and calls per step, respectively, and should not all be called one-step generation. In a video DiT, a sampling step may invoke several CFG branches, while a streaming method may add a clean-latent call solely to update KV. Strict NFE must count actual inference-time network evaluations.

The generic lineage requires precise objectives. Consistency Models map points on the same probability-flow ODE trajectory to a common output; consistency distillation depends on a pretrained diffusion trajectory, whereas standalone consistency training learns directly from data [160]. Latent consistency models and their LoRA form made few-step sampling broadly practical for latent image diffusion [116]. DMD estimates a Kullback–Leibler gradient direction from the difference between real-data and generated-distribution scores and adds a perceptual regression term anchoring the student to its teacher’s conditional sample [207]; DMD2 addresses stale generated-score estimates and unstable optimization through two-timescale updates, adversarial supervision on real data, and backward simulation of inference-relevant states [206]. These generic image results motivate virtual-human students, but do not settle identity, synchronization, or self-history errors.

A reproducible call ledger should separate the main denoiser, auxiliary score networks used only during training, and the decoder. For each inference step, the main-network count is the product or sum induced by CFG branches, solver substeps, and extra cache-refresh calls. Teacher and critic evaluations during training do not belong in deployed NFE, whereas an inference-time clean-latent forward does. If a paper reports steps but not solver or guidance calls, the survey retains “steps” rather than inventing NFE; if it reports an NFE ratio without wall-clock measurements on matched hardware, the ratio is not rewritten as a system speedup.

FADA provides a relatively explicit ledger. Its PeR-Flow student reduces a 25-step DDIM teacher to six denoising steps. Because the teacher executes three multi-CFG branches per step, the authors count 75 denoising-network evaluations (NFE-D). The Balanced student retains the three branches and uses 18 NFE-D, whereas the Fast student replaces them with a learnable CFG token and uses 6 NFE-D. The corresponding $4.17\times$ and $12.5\times$ reductions refer to denoiser calls; the paper reports neither inference hardware nor wall-clock time [231]. Trajectory shortening and branch folding are

thus distinct levers: a six-step model can still repeat a large network three times at every step.

Live Avatar illustrates why cache calls also belong in this accounting. Its undistilled baseline uses 80 NFE. DMD removes CFG and yields four denoising steps, but a conventional Self-Forcing implementation adds one clean-cache refresh and therefore totals five NFE. History Corrupt instead keeps historical KV at the same noise level as the current block, eliminating that clean refresh and reducing the deployed ledger to exactly four NFE [76]. The mechanism simultaneously avoids one call and produces local, step-matched noisy KV for a pipeline that assigns one timestep to each device.

LLIA and TurboTalk represent direct few-step training and progressive one-step distillation. LLIA trains two- and four-step students with consistency and adversarial objectives, then combines them with TensorRT, 8-bit integer (INT8) execution, and VAE-U-Net overlap; at 384×384 on an RTX 4090D, 78 FPS is measured for a 12-frame steady-state chunk, with a source-defined approximately 140-ms initial/state-switch response using a four-frame startup segment [210]. TurboTalk first obtains a four-step DMD student and progressively distills three-, two-, and one-step students; its claimed $120\times$ ratio compares approximately 120 NFE per InfiniteTalk chunk with the final author-counted 1 NFE per chunk, with no wall-clock result [109]. NFE ratios are valuable algorithmic evidence, but they do not substitute for deployment timing.

Block-autoregressive distillation must additionally close the gap between ground-truth histories used in training and self-generated histories seen at deployment. This is a Stable concern (pattern L4 in Section 4.7) that surfaces inside the distillation objective. MotionStream’s Self Forcing, StreamAvatar’s and LiveTalk’s on-policy rollouts, SoulX-FlashTalk’s multi-block retrospective self-correction, and AsymTalker’s asymmetric distillation all expose the student to some of its own errors [28, 114, 152, 153, 161]. SoulX-FlashTalk trains a four-NFE student over rollouts of up to five blocks and reports 32 FPS with a 0.87 s startup time at 720×416 on eight H800 GPUs [152]. AsymTalker gives the teacher a ground-truth continuity reference and the student a self-generated one, then reports FID and ArcFace curves through 600 s [114]. A streaming student must match not only an isolated teacher clip, but also the state distribution induced by its own rollout.

Virtual-human-specific rewards or timestep schedules can protect signals that generic DMD discards. Lip Forcing observes that teacher CFG affects fidelity and lip synchronization differently along the trajectory; it activates CFG only in an intermediate teacher-score window and weights DMD updates with Sync-

Net confidence. The deployed student performs two denoising NFE without CFG plus one clean-latent cache forward; its 1.3B and 14B variants report 31.58 and 15.11 FPS on one H100 at 512×512 , excluding speech preprocessing, detection, and paste-back [30]. Hallo-Live uses VideoAlign, AudioBox, and a SyncNet proxy in preference-based DMD and reports joint audio–video throughput of 20.38 FPS with 0.94 s reported latency on two H200 GPUs [95]. Such rewards may preserve synchronization or identity while inheriting evaluator bias. A convincing comparison should report sampling steps, CFG branches, actual NFE, matched-hardware wall-clock time, teacher–student quality gaps, and long-horizon degradation under self-generated history.

4.5 Fast III: Computation Reuse and System Parallelism

When several denoising calls remain, adjacent steps contain exploitable redundancy. LightningCP runs the full U-Net at selected key timesteps, reuses high-level decoder features at the others, and assigns finer attention to the foreground; on its Hallo setup, fixed-clip U-Net latency falls from 23.692 to 7.528 s ($3.15\times$), though the baseline uses one A800 whereas the proposed system uses an unspecified number of A800 GPUs [111]. Follow-Your-Emoji-Faster uses a timestep-aware information cache under the same 30-step DDIM schedule, reducing U-Net denoising for 64 frames at 256×256 from 6.28 to 2.36 s ($2.66\times$; hardware NR) [120]. Both results are fixed-clip denoiser acceleration rather than FPS. The same lever generalizes: flow caching skips recomputable segments of the velocity field in autoregressive video generation [121], and speculative decoding, which drafts cheap candidate continuations verified by the full model, has been transplanted from language models [93] to autoregressive video diffusion [71].

The slow-appearance/fast-speech asymmetry permits spatially and modally selective reuse. SyncCache’s spatial-aware probe applies a soft portrait mask to the difference between outputs of the first Transformer block at adjacent denoising steps and accumulates that error; only a threshold crossing triggers a full forward and refresh. Its modality-dynamic cache operates within one reverse trajectory: after a full-computation step, selected heavy DiT inter-block residuals are reused at later steps, while lightweight audio blocks are recomputed at every step. On eight A800 GPUs, the authors report a HunyuanVideo-Avatar path decreasing from 524 to 127 s ($4.12\times$) and a Wan path from 113 to 30.15 s ($3.75\times$) [117].

Two cache axes are orthogonal, and matching each cache to the update frequency of the state it stores is the

route’s general reuse rule. *Denoising-time reuse* shares features between sampling steps of the same output chunk; static references, background, and slowly varying high-level features are plausible candidates, but feature distributions change from the high-noise to detail-refinement regime. *Sequence-time reuse* carries KV or clean latents from one output chunk to the next; it enables continuation, but can propagate a mouth, pose, or texture error. Spatially, the background tolerates less frequent refresh than the mouth; across modalities, identity references tolerate longer reuse than current speech. Avatar V caches reference K/V that is invariant across denoising steps; SoulX-LiveAct’s Neighbor Forcing aligns historical and current chunks at the same diffusion step so that existing KV can be reused across chunks, while ConvKV compresses older history [103, 229]. Hybrid KV compression schemes extend the same idea for autoregressive video diffusion [79]. Cache evaluations should expose key-step selection, refresh thresholds, hit rates, warm-up cost, peak memory, and quality as cache depth increases [76, 117, 229].

System design determines whether algorithmic savings become a stable output pipeline. LLIA overlaps VAE and U-Net execution under TensorRT and INT8. StreamAvatar overlaps DiT and VAE on two H800 GPUs; its approximately 1.20 s time to first output decomposes into 0.48 s input buffering, 0.33 s for the first DiT output, and 0.39 s for the first VAE output, and its 25 FPS is the target playback rate rather than measured full-pipeline throughput [161]. FlowAct-R1 combines a three-NFE base streaming cycle without CFG, selective FP8, frame-level hybrid parallelism, kernel fusion, and asynchronous DiT/VAE execution, reporting 25 FPS and about 1.5 s TTFF at 480p on an “NVIDIA A100 platform” (GPU count NR) [179]. SoulX-FlashTalk assigns pipeline stages to speech processing, four DiT steps, VAE, and history encoding [152].

Live Avatar’s Timestep-forcing Pipeline Parallelism (TPP) partitions denoising time rather than network layers. Four H800 GPUs each execute one fixed DiT timestep and retain rolling KV at that timestep’s noise level; only the latent is sent to the next GPU, and a fifth H800 decodes with a causal VAE. The paper’s Table 3 progresses from 3.66 FPS for single-GPU DMD to 10.16 FPS with four-GPU timestep pipelining, 20.88 FPS after parallel VAE execution, and 45.2 FPS with FP8, FlashAttention-3, fused kernels, and compilation. The final configuration reports 1.21 s TTFF excluding network transfer, with approximately 3 s estimated when networking is included [76]. The paper also contains a conflicting 20 FPS overview figure and a 0.94 s TTFF for a default three-latent block setting; we retain the full Table 3 configuration and this within-paper

ambiguity. Pipeline fill raises steady-state throughput, but coupling four denoising steps to four DiT devices does not reduce total compute per generated video. It also does not reduce TTFF, which is set by the first traversal.

Table 3 makes the comparison problem explicit. Motion-parameter prediction is lighter than dense-video synthesis, TinyVAE and device count can alter the throughput of the same student, and NFE ratios exclude decoders and I/O. The useful object of comparison is therefore a quality–speed Pareto curve under a fixed task, frame size, hardware budget, and timing boundary, with algorithmic call reduction, component reuse, and system parallelism reported separately.

4.6 Streaming Generation: Where Fast Meets Stable

Extending a clip generator to continuous output is the point where the route’s Fast and Stable problems become one design problem. Chunkwise emission bounds per-chunk latency (Fast), but it forces the model to condition on its own committed history (Stable), and the chunk size trades the two directly: smaller chunks reduce buffering delay and TTFF while increasing scheduling, cache-update, and decoder-startup overhead; larger chunks improve parallel efficiency while raising TTFF. Equal FPS can therefore coexist with very different service behavior, which is why TTFF and inter-chunk latency are separate columns in our tables.

Diffusion Forcing provides the mechanism-level starting point. It independently samples a noise level for every token, unifying prediction, filtering, and imputation under a causal sequence model; at inference, a configurable two-dimensional noise schedule controls the denoising progress at each position [22]. Self-Forcing changes the training distribution: the student consumes its own generated history during an on-policy rollout and matches the distribution of the full autoregressive trajectory while maintaining a rolling KV cache [75]. This addresses the discrepancy between ground-truth history in training and self-history at deployment (pattern L4 in the next subsection) and is not itself an NFE-reduction method. CausVid distills a bidirectional video-diffusion teacher into a student that generates chunks in order, using ODE initialization, asymmetric DMD, and rolling KV (9.4 FPS with four sampling steps on one H100, generic video) [208]. Causal Forcing first constructs an autoregressive teacher aligned with the deployed architecture and then applies asymmetric DMD with self-rollout (17.0 FPS on one H100, generic video) [237]. Rolling Forcing pushes the same lineage to real-time long-video generation with rolling-window joint denoising [108]. Foundation-model stream-

Table 3 Fast lens (cost projection): representative diffusion-based virtual-human systems. Values preserve the source protocol; differences in task, resolution, hardware, device count, model size, and timing boundary make the rows unsuitable for direct ranking. NR denotes not reported in the primary paper.

Method	Efficiency lever	Steps / NFE	Reported speed or latency	Measurement boundary and limitation
FADA [231]	PeRFlow and multi-CFG folding	6 steps; 75 $\rightarrow$ 18/6 NFE-D	$4.17 \times / 12.5 \times$ NFE-D ratio	Hardware, FPS, and wall-clock time NR
LLIA [210]	2-step consistency, INT8, pipeline	2/4 steps; strict NFE NR	78 FPS steady state; source-defined approximately 140 ms initial/state-switch four-frame response	RTX 4090D, 384 ² ; 12-frame steady-state and 4-frame startup settings
RAP [48]	Compressed LTX latent, hybrid attention	Strict NFE NR	42.41 / 39.87 FPS	One A800, 512 ² ; complete timing boundary NR
LightningCP [111]	Feature cache, parallel denoising	Steps / NFE NR	Hallo U-Net 23.692 $\rightarrow$ 7.528 s/clip	One-GPU baseline versus unspecified multi-A800 method; not end-to-end
Follow-Your-Emoji-Faster [120]	Timestep-aware cache	30 DDIM steps; NFE NR	U-Net 6.28 $\rightarrow$ 2.36 s	64 frames, 256 ² ; inference hardware NR
PersonaLive [102]	4-step micro-chunks, optional TinyVAE	4 denoising steps; amortized NFE NR	15.82 FPS; 0.253 s mean inter-chunk latency	One H100, 512 ² , default SD-VAE; not first-frame latency
Motion Stream [153]	Self Forcing, 3-NFE student, TinyVAE	3 NFE/chunk	29.5 FPS; 0.39 s reported latency	One H100, 480p; complete timing boundary NR; general motion task
Stream Avatar [161]	3-step student, two-GPU DiT/VAE pipeline	3 steps; strict NFE NR	About 1.20 s to first output	Two H800, 928 $\times$ 704; 25 FPS is playback setting
Hallo-Live [95]	Preference DMD, asynchronous audio/video	Steps / NFE NR	20.38 FPS; 0.94 s reported latency	Two H200; resolution and complete boundary NR
Lip Forcing [30]	2-step SW-DMD, rolling KV, TAE	2 denoising NFE + 1 cache call	31.58 / 15.11 FPS (1.3B / 14B)	One H100, 512 ² ; excludes speech preprocessing, detection, and paste-back
FlowAct-R1 [179]	Base stream, FP8, parallel and asynchronous decode	3 base-cycle NFE, no CFG; refinement NFE NR	25 FPS; about 1.5 s TTFF	480p, A100 platform; GPU count, repair/MLLM timing inclusion, and full boundary NR
Live Avatar [76]	4-step DMD, noisy KV, TPP, FP8/VAE pipeline	4 NFE	45.2 FPS; 1.21 s TTFF	Five H800, 720 $\times$ 400; TTFF excludes network; about 3 s network-inclusive end-to-end latency
SoulX-FlashTalk [152]	4-NFE DMD, multi-block correction, pipeline	4 NFE	32 FPS; 0.87 s startup	Eight H800, 720 $\times$ 416
SoulX-FlashHead [212]	1.3B DiT, compressed latent, DMD	Exact steps / NFE NR	Lite 96 FPS; Pro 10.81 FPS	Paper conflicts between one H20 and RTX 4090 for the Lite result
SoulX-LiveAct [229]	DMD, Neighbor Forcing, ConvKV, FP8	3 sampling steps; strict NFE NR	20 FPS; 0.94 s latency (definition NR)	Two H100 or H200; codec and conditioning boundary NR
Avatar V [103]	CFG/DMD distillation, reference KV, USP	24 base-DiT steps + 1-step SR	About 3 $\times$ relative latency reduction	Eight GPUs of unspecified model; no absolute latency, FPS, or RTF

Abbreviations: CFG, classifier-free guidance; DDIM, denoising diffusion implicit model; DiT, diffusion Transformer; DMD, distribution matching distillation; FP8, 8-bit floating-point format; FPS, frames per second; GPU, graphics processing unit; INT8, 8-bit integer format; KV, key-value; MLLM, multimodal large language model; NFE, number of function evaluations; NFE-D, number of function evaluations of the denoising network; RTF, real-time factor; SD-VAE, Stable Diffusion variational autoencoder; SR, super-resolution; SW-DMD, Sync-Window DMD; TAE, Tiny AutoEncoder; TPP, Timestep-forcing Pipeline Parallelism; TTFF, time to first frame; U-Net, generic U-shaped convolutional architecture; USP, Ulysses Sequence Parallelism; VAE, variational autoencoder.

ing variants now package these mechanisms at scale: Wan-Streamer targets end-to-end real-time interactive generation, with its second version raising resolution at unchanged latency [72, 73], and Vidu S1 makes the same real-time streaming claim for a production system [215].

Virtual-human systems add task state to this lineage. PersonaLive advances micro-chunks through four noise levels [102]. StreamAvatar and LiveTalk distill a bidirectional teacher into a chunkwise student and add reference or identity sinks plus self-generated history [28, 161]. EchoTorrent generates chunks in order with a reference sink and tail-only forcing [124]. SoulX-FlashTalk uses multi-block self-correction; SoulX-LiveAct maintains fixed multiscale KV; Lip Forcing introduces synchronization-aware distillation with an explicit clean-cache call; and FlowAct-R1 extends chunkwise generation to full-body behavior with MLLM action planning [30, 152, 179, 229]. Sliding-window systems orga-

nize the same state for offline long inputs: StableAvatar’s Dynamic Weighted Sliding Window (2.32 min per 81 frames at 480 $\times$ 832) and InfiniteTalk’s 81-frame chunks with nine context frames extend duration without targeting service latency [169, 200].

One Fast implication deserves explicit statement: the conditioning pipeline sets a latency floor that no visual-path acceleration can cross. Speech encoders with wide receptive fields, block-level audio buffering (Teller collects a 200 ms speech block before computing [228]), and window-based speech pre-encoding all contribute delay ahead of the first DiT call; StreamAvatar’s 1.20 s first output includes 0.48 s of pure input buffering [161]. Reports that time only the denoiser therefore understate TTFF by construction, and reproducible streaming protocols should jointly report input-chunk duration, TTFF, steady-state inter-chunk latency and jitter, play-

back buffer, device count, and module-specific timing endpoints.

4.7 Stable: Five Drift-Control Patterns

Bounded windows prevent memory from growing with video duration, but they remove global context, and chunkwise generation feeds the model its own errors. The route’s countermeasures recur so consistently that we name them as five patterns, referenced throughout this survey as L1–L5:

L1: Persistent identity anchor. Retain a reference image, an early high-quality latent, or a dedicated attention sink for the subject. Examples include StreamAvatar’s Reference Sink, LiveTalk’s three identity sinks, EchoTorrent’s reference sink, and LPM 1.0’s sink/reference token [28, 124, 161, 213]. The anchor constrains recursive appearance drift; its known cost is a static-scene prior that can block legitimate changes.

L2: Bounded multiscale memory. Combine recent motion context with compressed or selected older state under a fixed budget. SoulX-LiveAct holds two reference, two long-term (ConvKV-compressed), and two short-term latent chunks; FlowAct-R1 maintains a fixed structured memory bank; PersonaLive appends motion-distance-selected keyframes (capacity and eviction NR, so selection alone does not establish bounded state) [102, 179, 229].

L3: Position re-indexing. Prevent cached positions from leaving the training range during unbounded rollout. JoyAvatar-Flash, Lip Forcing, and LPM 1.0 use variants of pre-RoPE storage or dynamic positions; Live Avatar’s Rolling RoPE continually corrects the relative positions of sink and current chunk [30, 76, 96, 213].

L4: Self-history training. Expose the student to its own generated history during training so that deployment-time state is in-distribution. MotionStream, StreamAvatar, SoulX-FlashTalk, SoulX-FlashHead, Hallo-Live, and AsymTalker use on-policy rollouts, retrospective correction, or asymmetric teacher–student context [95, 114, 152, 153, 161, 212]; Hallo2 corrupts conditioning history with patch-drop and noise for the same purpose in a segmented setting [39].

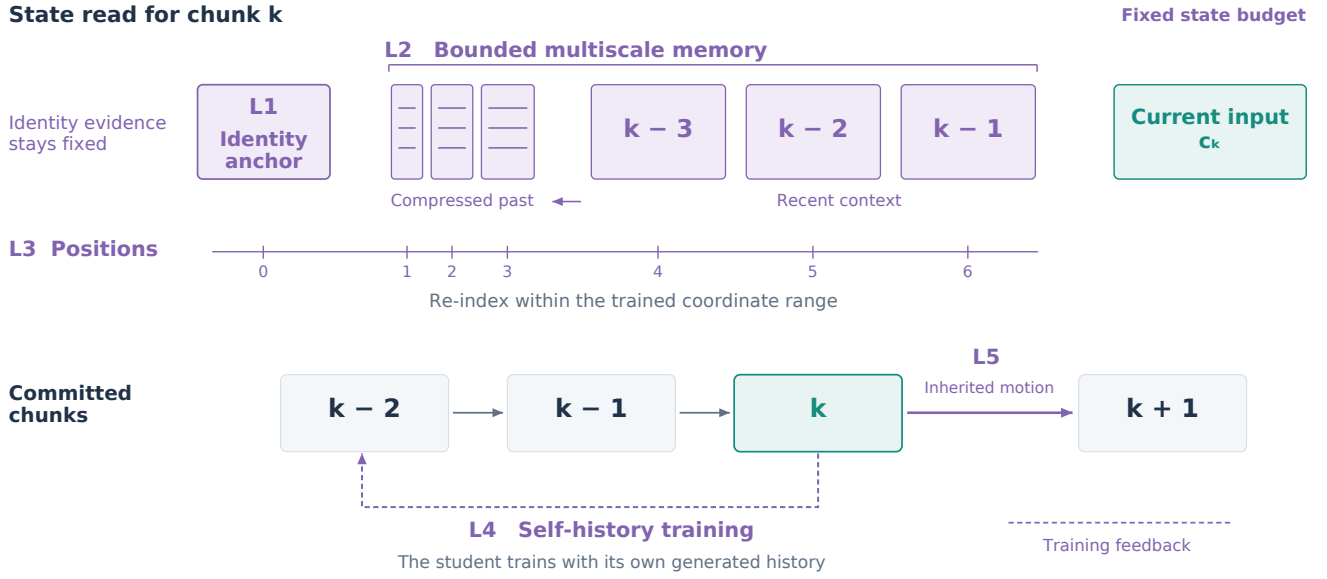
L5: Motion inheritance. Propagate continuity across chunks through an explicit motion prior or inherited latent. READ, RAP, InfiniteTalk, JoyAvatar-Flash, and ViSA are representative [48, 96, 173, 197, 200]. Inheritance delays drift but can reduce motion diversity or preserve an early error.

Figure 6 locates the five patterns on a shared rolling timeline, distinguishing runtime state reuse from training-time feedback. Generic long-video work clarifies how these patterns interact. Self-Forcing++ samples

five-second segments from long self-generated rollouts for distributional supervision (L4 at scale), exposing training to states far from the sequence origin; its standardized quantitative horizon is 100 s [40]. LongLive combines short-window attention, a persistent frame sink (L1), KV recaching, and streaming long tuning; its recache is event-triggered at a prompt-switch boundary, rebuilding KV from recent generated video and the new prompt to remove stale semantics [199]. The reported 20.7 FPS uses a 1.3B model at 832×480 on one H100, with quantitative tests at 30 s (single prompt) and 60 s (scheduled prompt switches) and a 240 s demonstration. Mapped to virtual-human state: a short window preserves immediate mouth and motion continuity, an identity sink stabilizes appearance, compressed long-term KV recovers information lost by local attention, event-triggered recaching removes stale semantics, and RoPE re-indexing controls positional extrapolation. Together, these mechanisms show that computation can continue with finite memory; they do not by themselves demonstrate recovery of identity, synchronization, and motion over a long session [161, 199, 229].

From a state-estimation perspective, the patterns have complementary roles: the anchor (L1) supplies identity and scene priors, the recent window preserves velocity and mouth continuity, and long-term memory (L2) retrieves poses or events forgotten locally. An ever-growing cache increases computation and error contamination, and a constant cache can forget remote events. A confidence-based design that updates, compresses, or resets memory adaptively remains an open problem. Stable evaluation should measure identity, appearance, synchronization, and motion in predeclared duration bins, and report error trends, legitimate changes blocked by the anchor, generation faults, and recovery length.

Live Avatar illustrates the joint design of sampling, deployment, and bounded state. It combines an Adaptive Attention Sink (L1), Rolling RoPE (L3), and History Corrupt, which uses step-matched noisy KV and removes a clean-cache call. Its window retains four recent blocks of three latents each and one persistent sink; after the first chunk, the first generated latent replaces the original reference sink. The long-horizon test loops roughly seven-minute speech recordings and measures windows at 0–10, 100–110, 1000–1010, and 10000–10010 s. The reported Q-Align aesthetic score, image quality, Sync-C, and DINOv2 cosine similarity do not show marked degradation in those windows [76]. This supports stability at sampled points under repeated audio. The static-scene prior, unsampled intervals, and absence of an external baseline under the same protocol limit broader conclusions.



These patterns manage different failure channels. Finite memory alone does not establish bounded drift.

Fig. 6 Five complementary drift-control patterns on a rolling timeline: L1 persistent identity anchors, L2 bounded multiscale memory, L3 position re-indexing, L4 training with self-generated history, and L5 motion inheritance. Dashed training feedback is distinct from runtime motion transfer. Slots and local indices are schematic.

Interactive systems provide further examples of state management, although interaction capability alone does not establish stability. LPM 1.0 supports speaking, listening, idling, and reflecting and combines a 2-NFE generator with a 1-NFE refiner over pre-RoPE KV (L1+L3). Its benchmark accepts inputs up to one hour and its demonstration timeline reaches 60,000 s, but duration-binned quantitative metrics are NR [213]. FlowAct-R1 uses fixed structured memory and generated-history replacement during training (L2+L4) for full-body behavior with MLLM action planning [179]. As a contextual example from the render route, ARIG maintains 512 interaction summaries and reports improved source metrics relative to a capacity of four; this cache capacity is not translated into a measured generation duration [60].

4.8 Design Guidance

Table 5 summarizes three practical considerations. First, the benefit of further step reduction depends on how much runtime remains in the denoiser; once decoding or conditioning dominates, optimizing those stages may be more effective. There is no universal NFE threshold for this transition. Second, pipeline overlap primarily improves steady-state throughput, whereas smaller initial chunks and shorter input buffers can reduce TTFF. Third, rolling caches, chunk overlap, and memory compression affect both per-chunk cost and the history avail-

able for maintaining consistency. Their efficiency and stability effects should therefore be evaluated together.

4.9 Limitations and Outlook

Online diffusion-based virtual humans are produced by a stack of mechanisms rather than a universally faster sampler, and a gain at one layer can be cancelled by the next bottleneck. Future work should report the quality–speed–horizon Pareto frontier rather than one favorable scalar, with the model-call ledger (sampler, steps, CFG branches, strict NFE), component timings, TTFF, steady-state chunk time, jitter, and peak memory as distinct fields; cache methods should disclose warm-up, refresh policy, and memory growth with duration.

For stability, the reviewed evidence lacks a matched-compute comparison of all five patterns and their combinations. Such a comparison should examine both drift reduction and restrictions on motion diversity or legitimate scene changes. Memory updates based on estimated reliability are a further research direction. Evaluation should track identity, synchronization, visual quality, background drift, motion, and failures in predeclared time bins; training studies should compare ground-truth, generated, and mixed histories under matched conditions. Section 8 also considers authorization and provenance for long-running systems with cached identity state.

Table 4 Stable lens: state, stability mechanism (classified by patterns L1–L5), and evidence horizon for representative diffusion-based virtual-human systems. The final column gives the source timing context; protocols are not directly comparable. NR denotes not reported in the primary paper.

Method	Generation / context	Stability mechanism (pattern)	Quantitative / demonstration-only / claimed horizon	Timing context
Hallo2 [39]	Segmented long video with patch-drop/noise augmentation	Robustness to corrupted history (L4); separate 4K enhancer	Quantitative: 4 min at 512 ² , enhancer off; claimed: hour-long	NFE, FPS, latency, and inference hardware NR
StableAvatar [169]	Dynamic weighted sliding window; window length NR	First-frame/history conditioning (L1/L5) and full-sequence training	Quantitative: Long100, 100 clips of 2–5 min	2.32 min per 81-frame sample at 480 × 832; offline
PersonaLive [102]	Four-step micro-chunk window	Motion-distance keyframe selection (L2); bank cap/eviction NR	Quantitative: LV100 (100 clips, at least 1 min); demonstration-only: 2700 generated frames	One H100; 0.253 s is mean inter-chunk latency; duration scaling NR
Motion Stream [153]	Three-frame chunks; one fixed sink and one rolling recent window	Self Forcing (L4), fixed-state rolling KV (L1), cache-relative RoPE (L3)	Quantitative: 241 frames (about 15 s); claimed: indefinite	One H100, 480p; general-motion task; fixed sink restricts scene changes
JoyAvatar-Flash [96]	4-step steady state; 8/7/6/5 startup steps	PSB, MCI (L5), URCR (L3)	Quantitative: NR; demonstration-only: 120 s; claimed: unbounded (theoretical)	Single-GPU DiT 16.2 FPS; end-to-end latency NR
Stream Avatar [161]	3-step local window, two-device pipeline	Reference Sink (L1), RAPR (L3), self-history (L4)	Quantitative: 20 s protocol; demonstration-only: about 150 s	About 1.20 s first output; two H800
LiveTalk [28]	4-step rolling window	Three identity sinks (L1), two rolling slots, on-policy distillation (L4)	Claimed: several minutes; demonstration-only: 100 s	Unspecified single GPU, 24.82 FPS / 0.33 s
EchoTorrent [124]	4 NFE chunkwise generation	Reference sink (L1) and tail-only forcing (L4)	Demonstration-only: 1000 s	Eight A100, 10.50 FPS; timing boundary NR
Live Avatar [76]	Three-latent chunks; four-block rolling, per-step local KV	History Corrupt (L4), AAS (L1), Rolling RoPE (L3)	Quantitative: over 5 min and sampled to 10000–10010 s with looped speech	Five H800, 45.2 FPS / 1.21 s TTFF; static-scene prior
LPM 1.0 [213]	2-NFE generator + 1-NFE refiner	Sink/reference token (L1), pre-RoPE KV (L3), duplex conditions	Duration-binned metrics: NR; benchmark inputs up to 1 h; demonstration-only: 60,000 s timeline; claimed: unbounded	0.35 s is one DiT component
AsymTalker [114]	4-step block generation	TRE and asymmetric GT/self-history distillation (L4)	Quantitative: FID/ArcFace at 5/50/150/300/600 s	One A100, 66.02 FPS for 30 s at 512 ² ; full boundary NR
Lip Forcing [30]	2 denoising NFE + clean-KV call; 7-latent cache	Sink (L1), pre-RoPE KV and dynamic RoPE (L3), SW-DMD (L4)	Quantitative: up to 6 min	One H100, 31.58/15.11 FPS; speech pre-encoded offline
SoulX-FlashTalk [152]	Five history + 28 new frames per cycle	Multi-block retrospective self-correction (L4)	Quantitative: over 5 min; demonstration-only: 1000 s	Eight H800, 32 FPS, 0.87 s startup
SoulX-FlashHead [212]	Blockwise generation with trailing 8 s speech context	Oracle-guided self-history (L4) and reference latent (L1)	Demonstration-only: 60 s	Hardware reports conflict (H20 vs. RTX 4090)
SoulX-LiveAct [229]	Same-step Neighbor Forcing; fixed 2/2/2 context	ConvKV-compressed history (L2) and reference anchor (L1)	Claimed: hour-scale/unbounded mechanism; demonstration-only duration: NR	Two H100/H200, 20 FPS; 0.94 s definition NR
Avatar V [103]	About 6.4 s recurrent chunks; two-frame overlap	Appearance anchor (L1), motion propagation (L5), reference-KV reuse	Claimed: unbounded length; quantitative duration: NR	Eight unspecified GPUs; no absolute latency or FPS
FlowAct-R1 [179]	Chunkwise generation with fixed structured memory	Reference/long/short memory (L2), Memory Refinement, generated-GT replacement (L4)	Quantitative: longest rollout NR; claimed: arbitrary duration	25 FPS / about 1.5 s TTFF; repair/MLLM timing inclusion NR

Abbreviations: 4K, source-described 4K-resolution enhancement; AAS, Adaptive Attention Sink; DiT, diffusion Transformer; DMD, distribution matching distillation; FID, Frechet Inception Distance; FPS, frames per second; GT, ground truth; KV, key-value; L1–L5, drift-control patterns defined in Section 4.7; MCI, Motion Condition Injection; MLLM, multimodal large language model; NFE, number of function evaluations; PSB, Progressive Step Bootstrapping; RAPR, Reference-Anchored Positional Re-encoding; RoPE, rotary position embedding; SW-DMD, Sync-Window DMD; TRE, Temporal Reference Encoding; TTFF, time to first frame; URCR, Unbounded RoPE via Cache-Resetting; VAE, variational autoencoder.

Table 5 Design guidance for the diffusion route. Levers are supported by the sources in Tables 3 and 4; costs are the recurring trade-offs those sources disclose.

Target	Recommended levers	Known costs
Faster fixed clips	Step-cache reuse; CFG folding; compressed latents	Cache staleness near detail steps; compression harms mouths and teeth
Playback-rate streaming	Few-step DMD student + chunkwise rolling KV; DiT/VAE overlap; FP8 and fused kernels	Multi-GPU budgets; TinyVAE quality loss; L4 training complexity
Low TTFF	Small first chunk; staged startup steps; time the conditioning path	Input buffering floor; startup-quality gap
Minute-scale stability	Reference anchors and training with generated history; position handling for sliding windows	Static-scene bias; motion-diversity loss; memory policy requires validation
Hour-scale operation	Bounded memory, anchors, and position management; evaluate refresh and recovery	Evidence depends on sampled windows and input protocol; no duration guarantee

4.10 Summary and Transition

Diffusion-based virtual humans combine representation compression, fewer sampling evaluations, computation reuse, and deployment optimization. Streaming methods additionally manage generated history through anchors, memory, position handling, training, and motion transfer. These mechanisms offer complementary benefits, but their joint effect depends on the task and measurement protocol. When a native autoregressive backbone instead predicts the evolving visual state and diffusion supplies only a local emission head, the system belongs to the autoregressive route discussed next.

5 Autoregressive Methods

The term *autoregressive* (AR) appears at several levels of a virtual-human system. A dialogue model may predict speech, a controller may predict motion, and a visual model may predict frame tokens. This section reserves the AR route for the last case. A native AR model must own cross-frame dense or discrete visual state, as in MIDAS and Body of Her [4, 26]. An AR speech, language, pose, landmark, or motion branch does not establish visual ownership when another component owns appearance, and a sequential scheduler does not make a dense diffusion process autoregressive. This narrow definition separates system organization from the mechanism that produces visible temporal content.

The literature nevertheless advances along two largely independent axes. The first is *state ownership*: causal state progresses from language and behavior,

through motion and sparse control, to decodable visual representations. The second is *interaction scope*: one-way animation expands toward online dyadic response, full-duplex interaction, and persistent world grounding. A world-model label belongs to the second axis unless the causal model also predicts the evolving visual representation. This distinction permits highly interactive systems to inform AR system design without relabeling their flow-, diffusion-, or renderer-owned visual generation paths.

Autoregressive visual state connects efficiency directly to history management. Reusing context can avoid repeated computation, while a long context increases memory and may retain generation errors. KV caching, quantization, parallel decoding, and speculative execution provide candidate acceleration mechanisms [71, 93]. Their benefits for a virtual-human pipeline depend on the visual representation, local decoder, and quality constraints, which must be evaluated alongside temporal drift.

5.1 Scope, State Ownership, and Cost Boundary

The ordered test in Section 2.3 first excludes methods whose appearance is owned by a subject-specific mesh, neural field, Gaussian set, or identity-conditioned warp. Among the remaining systems, the AR route is selected only when the evolving visual tokens or latents are predicted natively from prior visual state; a downstream diffusion process may refine the current representation without changing that assignment.

For one emitted frame or block, a useful route-specific accounting is

$$C_t = C_{\text{condition}} + C_{\text{AR}}(h_t) + C_{\text{local-emission}} + C_{\text{decode/render}} + C_{\text{system}}, \quad (5)$$

where h_t names the persistent visual state available at step t . The AR term includes token prediction, attention, and state update; a local continuous-output head retains its own denoising or flow evaluations; and the final terms include codec decoding, rendering, transfer, compositing, and input/output (I/O). Consequently, one AR-backbone update per frame does not imply only one network evaluation or one complete system call per visible frame, and a reported frame rate is interpretable only with the hardware, resolution, precision, and included stages defined in Section 2.6.

5.2 Mechanism Map: Compression, State, and Emission

AR visual generation exposes three coupled design choices [4, 26, 192]. First, the representation fixes se-

rial granularity. A frame may contain many discrete tokens [171], a compact set of continuous tokens [23], or one latent block. Spatial compression and within-frame parallel prediction reduce different work, but tokenizer and decoder costs remain. Second, persistent state controls repeated computation and resource growth. KV reuse avoids recomputing past projections; windows, summaries, eviction, and resets can bound the retained state. A codec feature cache is not interchangeable with an AR-backbone context. Third, local emission converts predicted state into an observable frame. A diffusion or flow head may improve continuous output, but its calls and final decoder remain on the critical path [4, 26].

These choices create a characteristic efficiency tension. More aggressive visual compression shortens the AR sequence but may move cost or reconstruction error into the codec. A longer cache supplies more context while increasing memory and preserving erroneous history; truncation bounds memory but can discard identity, pose, or conversational events. Parallel prediction inside one frame reduces serial depth, yet cross-frame self-conditioning still exposes the model to its own errors. Fast and Stable therefore require separate measurements rather than an architectural label.

5.3 Two Axes of Autoregressive State and Interaction Scope

Response and behavior. At the response level, UniTalker jointly predicts conversational speech and facial-landmark tokens before an emotion-guided renderer produces the visible sequence [70]. OmniResponse extends the causal loop to online multimodal dyadic interaction by coordinating verbal responses and non-verbal facial coefficients, but a pretrained renderer still maps those coefficients and a portrait to pixels [115]. These systems make response timing and behavior autoregressive. They do not transfer appearance ownership to the AR state.

Motion and control. Motion-level systems move the causal variable closer to the image while retaining the same boundary. Teller autoregressively predicts residual-vector-quantized motion tokens and delegates portrait synthesis to a renderer [228]. ARIG predicts continuous head-motion trajectories conditioned on speaking and listening states, again followed by a separate portrait renderer [60]. DyStream combines an AR network with a five-step flow-matching head to produce a 512-dimensional identity-agnostic motion code; a reference-conditioned appearance volume, dense warp, and decoder determine visible identity and pixels [21]. Even *Talker-T2AV* [205], whose formulation jointly generates continuous audio patch latents and compact LIA-X

facial-motion latents, uses a frozen identity-conditioned decoder to produce the portrait. Joint audio-motion autoregression is therefore not equivalent to visual-state ownership.

Unified speaking and listening. UniLS extends the motion-level AR boundary from one-way speaking animation to unified speaking and listening. It first learns an autoregressive motion prior and then conditions the sequence on separate speaking and listening audio streams, allowing listener response motion to be generated within the same causal formulation. The emitted state remains compact motion or facial parameters that a downstream renderer maps to RGB. Under the ownership test of Section 2.3, UniLS is therefore an AR motion controller at the render boundary, not a native visual-state owner. Its contribution is to broaden the interaction scope of motion AR while leaving the appearance, decoder, and rendering costs in the downstream system ledger [34].

Visual-state prediction. Visual autoregression differs from motion autoregression in the representation that the causal model predicts. UniMo interleaves 2D video tokens and quantized 3D human-motion tokens in one causal sequence, using modality-specific embeddings and decoders [129]. It satisfies the visual-state criterion, although its evidence concerns general human video and motion rather than a conversational-avatar deployment. MIDAS predicts compressed dense frame representations [26], while EARTalking performs frame-by-frame in-context visual autoregression [187]. These methods make the visible sequence itself part of the predicted causal state.

5.4 Representative Visual-State Autoregression

UniMo, MIDAS, Body of Her, and EARTalking satisfy the visual-state criterion under different task and decoding assumptions. UniMo supplies a general human video-motion example, whereas the other systems target digital humans or talking heads. Their inclusion establishes architectural membership; the completeness of their deployment evidence is assessed separately in Table 7.

MIDAS is the clearest retained example of an AR visual owner followed by a local iterative head. A Qwen2.5-3B backbone predicts the next continuous frame representation, using a Deep Compression Autoencoder (DC-AE) [23] with 64× spatial compression and at most 60 visual tokens per frame; an approximately 0.5-billion-parameter PixArt- α head then applies four denoising steps to the current representation before DC-AE decoding [26]. The six-frame logical block exposes all of

Table 6 Autoregressive state at different pipeline levels. Native visual AR systems are distinguished from contextual examples whose AR modules predict behavior or motion, or condition a diffusion/flow visual generator. The latter illustrate interfaces and are excluded from the native visual AR comparison in Table 7.

Method	Causal history / emitted unit	Interaction / grounding	Role in this section
UniTalker [70]	Speech and facial-landmark tokens	Conversational dyad	Context: control followed by rendering
OmniResponse [115]	Verbal response and facial coefficients	Online multimodal dyad	Context: behavior followed by rendering
Teller [228]	Residual-VQ portrait-motion tokens	Streaming audio driven	Context: render-based system with AR motion
ARIG [60]	Continuous interactive head motion	Speaking–listening conversation	Context: render-based system with AR motion
DyStream [21]	Identity-agnostic motion code, locally emitted by flow matching	Dual-audio dyadic conditioning	Context: render-based system with AR motion
Talker-T2AV [205]	Shared causal history over committed audio–motion patches; current continuous latents locally emitted	Audio-driven talking portrait	Context: audio–motion prediction followed by rendering
UniLS [34]	Autoregressive compact motion conditioned on two audio streams	Unified speaking and listening	Context: motion prediction followed by rendering
UniMo [129]	Interleaved 2D video tokens and quantized 3D motion tokens	General human video and motion	Native visual AR; general-task bridge
Body of Her [4]	Framewise visual state with local diffusion emission	Human–agent audiovisual history	Native visual AR
MIDAS [26]	Compressed dense frame representation	Multimodal interactive digital human	Native visual AR
EARTalking [187]	Framewise visual latent state	Audio-driven talking head with frame control	Native visual AR
X-Streamer [190]	Dialogue and interaction state	Human-world audiovisual interaction	Context: diffusion-based visual generation
Wan-Streamer v0.2 [73]	Language/state tokens and causal interaction history	Scene-grounded full duplex	Context: diffusion route (flow matching)
Wan-Streamer v0.3 [74]	Language-form speech and behavior events	Explicit world context and full-duplex event stream	Context: diffusion route (flow matching)

its condition tokens to subsequent frame tokens. The implementation separately retains a five-frame DC-AE feature/KV history and truncates AR frame context to the most recent eight blocks. These are two state objects with different owners, and the paper does not disclose the AR cache implementation or reset behavior. On timing, the reported 480 ms block length is a media span and the 25 FPS identity-specific output is a media rate; inference hardware, measured FPS, time to first frame (TTF), and a component ledger are not reported. The paper states that identity-specific post-training can support up to one hour and shows a qualitative four-minute singing case, but neither is accompanied by time-binned stability measurements. MIDAS therefore supports the mechanism claim that compressed AR visual state can drive a four-step local emission head. It does not establish measured end-to-end Fast or Stable performance [26].

Body of Her provides a second, framewise instance. Its Transformer decoder conditions on prior human and agent audio–video histories, current text or trajectory controls, and identity, thereby owning cross-frame behavior and visual evolution; a compact diffusion head samples all continuous tokens of the current frame in parallel, after which an unspecified video decoder produces pixels [4]. The paper describes a roughly 8-billion-parameter model emitting 640×360 synchronized audio–video at a 24 FPS media rate and states that the next frame is generated within 42 ms, but reports neither inference

hardware nor the diffusion step count, strict NFE, or included components, so the 42 ms statement cannot enter a common speed comparison. Context capacity, cache growth or eviction, reset behavior, and any quantitative Stable duration are also not reported, and the demonstrated customized agents use 9 and 5.5 hours of subject-specific data. The method supports a route and feasibility claim, not a quantified speed, generalization, or long-session claim [4].

EARTalking provides a stronger framewise case. Its GPT-style causal model carries prior visual context and generates the next 3D-VAE frame representation with frame-level control; a sink-frame anchor, sliding KV window, and cyclic position policy make its persistence contract more explicit than those of the earlier retained systems [187]. The frame-local generation and final decoder still belong in the cost ledger, but the evolving decodable visual state is owned by the causal backbone. EARTalking consequently passes the route test without implying direct pixel-space next-token prediction.

The route boundary remains consequential for evidence interpretation. DyStream reports 29.24 FPS and 34.2 ms audio-packet-to-motion delay on one H200, but these are motion-level measurements for a render-owned system [21]. Conversely, native visual owners may disclose a causal state contract without publishing a complete end-to-end timing ledger. Table 7 therefore keeps route membership separate from the completeness of Fast and Stable evidence.

Table 7 Evidence boundaries for representative native visual AR systems. Contextual examples with other visual owners are excluded. Media rate and media span remain distinct from measured throughput and latency, and the rows are not directly rankable. NR denotes not reported in the primary paper.

Method / route	Visual state and emission	Persistence and bounds	Source timing boundary	Claimed / quantitative / demonstration-only horizon
MIDAS [26]; native AR visual owner	Qwen2.5-3B next-frame state; at most 60 tokens/frame; four-step, approximately 0.5B local diffusion head; DC-AE decode	Five-frame codec history and eight-block AR truncation; cache implementation and reset NR	Inference hardware, measured FPS, TTFF, and inter-block latency NR; 480 ms is media span and 25 FPS is media rate	Claimed: one hour, identity-specific author statement; quantitative: NR; demonstration-only: four minutes, qualitative
Body of Her [4]; native AR visual owner	Framework Transformer state; current-frame continuous tokens sampled in parallel by a compact diffusion head; video decode	Context capacity, cache growth/eviction, and reset NR	Author reports a 24 FPS media rate and 42 ms/next frame at 640×360 ; inference hardware and included components NR	Claimed: NR; quantitative: NR; demonstration-only: NR
UniMo [129]; native visual AR, general-task bridge	Interleaved causal sequence of Cosmos 2D visual tokens and quantized 3D motion tokens; modality-specific embeddings and decoders	Prior visual and motion tokens form causal context; deployment cache, window, eviction, and reset contract NR	General human video-motion generation; no matched conversational-avatar TTFF or component ledger	Claimed or quantitative conversational long-horizon evidence NR; not a real-time dyadic deployment
EARTalking [187]; native AR visual owner	Framework 3D-VAE visual state with frame-level control; local frame emission and decode	Sink-frame anchor, sliding KV window, and cyclic positions; reset and recovery behavior NR	Reported media and runtime statements are not used as common end-to-end latency without a matched component ledger	Claimed horizon and qualitative demonstrations are not accompanied by a minute/hour time-binned stability protocol

Abbreviations: AR = autoregressive; B = billion parameters; DC-AE = Deep Compression Autoencoder; FPS = frames per second; KV = key-value; TTFF = time to first frame.

5.5 Toward World-Grounded Full-Duplex Streaming

World-grounded conversational systems illustrate why interaction scope should be separated from visual ownership. X-Streamer’s Thinker maintains dialogue and interaction state, while its Actor applies DDIM sampling with v-prediction to each dense LTX-VAE video block [190]. Sequentially committing those blocks does not transfer visual ownership to the Thinker. We discuss this diffusion-based system here to clarify the interface between causal interaction state and dense visual generation.

Wan-Streamer v0.2 strengthens the system-level case without changing that ownership assignment. It retains the v0.1 native-streaming formulation in which user and agent text, audio, and video share a causal timeline and generated audio-video latents are committed to subsequent history [73]. A token-causal Thinker owns perception, language/state update, and the conditioning KV cache; a multi-GPU Ulysses-style Performer owns flow-matching generation of the 640×368 audio-video latents. The contribution is a resolution- and serving-topology upgrade that preserves an approximately 200 ms model-side signal-to-signal latency, not a new visual-AR mechanism.

Video = World + Event Stream, or Wan-Streamer v0.3 [74], makes the persistent context explicit. A world record represents relatively stable scene, character, acoustic, and voice conditions, while an event stream contains time-localized speech, behavior, camera, and environmental changes. The event factorization is causal: language-form speech and open-vocabulary be-

havior are predicted token by token. Continuous audio and video latents, including motion, appearance, and scene evolution, are still generated jointly by conditional flow matching and committed to the next unit’s history. Its evaluated specialization is full-duplex user-agent interaction rather than symmetric generation of two visible participants, and broader roaming or manipulation adaptation remains future work.

The three systems form a scope progression from audiovisual human-world interaction, through scene-grounded full-duplex serving, to explicit world-event factorization. X-Streamer and the Wan series are not one technical lineage, while Wan-Streamer v0.2 and v0.3 are successive versions rather than independent architectural votes. Collectively they show that closing the causal interaction loop does not necessarily transfer dense visual ownership from diffusion or flow matching to the AR state.

5.6 Fast Design Guidance

The Fast levers below are tied to the cost terms in Equation (5). Evidence from native visual owners supports route-specific claims; motion-, diffusion-, and flow-owned systems are used only to expose serving opportunities or missing component boundaries.

- **Compress the serial visual state** ($C_{AR} \downarrow$). MIDAS limits a frame to at most 60 DC-AE tokens, while UniMo interleaves compressed video and quantized motion tokens. Compression can reduce serial

work but may move cost or error into the codec, particularly for fine facial detail and rapid articulation.

- **Parallelize local emission** (serial depth $\downarrow$). Body of Her samples all continuous tokens of the current frame in parallel through a compact diffusion head, and MIDAS uses a four-step local head after the AR frame state. EARTalking similarly separates cross-frame causal state from frame-local realization. The local head’s strict number of function evaluations (NFE), tensor shape, and decoder cost remain part of end-to-end latency.
- **Reuse and bound persistent state** ($C_{AR}(h_t)$ growth control). Standard KV reuse avoids recomputing past projections. MIDAS truncates AR context to eight blocks while separately retaining five codec frames; EARTalking combines a sink-frame anchor with a sliding KV window and cyclic positions. These mechanisms bound repeated work, but their eviction, correction, and reset costs require direct measurement.
- **Separate control-path and emission-path serving** ($C_{system} \downarrow$). Wan-Streamer v0.2 keeps its token-causal Thinker on one GPU and places high-resolution flow-matching generation in a Ulysses context-parallel Performer [73]. This is not native visual AR evidence, but it places the causal state update and the expensive local emitter on distinct serving paths organized by a pipelined unit schedule, without changing the route assignment.
- **Draft or speculate only against a verified visual contract**. Speculative decoding is established for language models and has been studied for general AR video [71, 93]. A virtual-human deployment must additionally verify identity, lip synchronization, and visual-state acceptance when drafted tokens or frames are committed; none of the retained native visual systems reports that ledger.

The route boundary also exposes bottleneck migration. DyStream reports motion-level packet delay while its warp, decoder, and display boundary remain undisclosed; X-Streamer reports a multimodal system rate without a matched decomposition of its dense diffusion Actor; and Wan-Streamer v0.2 reports both model-side and network-inclusive interaction latency for a flow-owned visual path [21, 73, 190]. These values answer different questions. A reproducible AR report should provide AR calls per visible frame, token or block count, local-head NFE, cache-update cost, decoder time, I/O boundary, and peak state memory under one declared pipeline.

5.7 Stable Design Guidance

The AR route holds persistent state by construction, so every drift-control pattern from Section 4.7 (Figure 6) has a natural causal-state interpretation. Persistent or bounded state is not bounded error, however, and none of the retained native visual systems reports a complete time-binned stability protocol.

- **L1 (identity anchor)**. EARTalking’s sink frame preserves a fixed early visual reference inside an otherwise moving window. MIDAS and Body of Her condition on identity, but their retained-state and reset semantics do not establish an equivalent pinned visual anchor.
- **L2 (bounded multiscale memory)**. MIDAS combines a five-frame codec history with an eight-block AR context, and EARTalking uses a sliding KV window around a persistent sink frame. UniMo retains prior visual and motion tokens as causal context but does not disclose a bounded deployment cache, eviction, or reset contract. These designs expose different state objects; correction and recovery remain incompletely reported.
- **L3 (position re-indexing)**. EARTalking’s cyclic position policy prevents unbounded position growth under a sliding window. The other retained visual owners do not disclose an equivalent rollout position contract, so position handling cannot yet be compared independently of cache size.
- **L4 (self-history training)**. MIDAS corrupts training context with noise; Body of Her trains on dataset histories but infers from predicted ones, with no scheduled sampling or rollout curriculum disclosed. Wan-Streamer commits generated latents into the next causal unit, but this is flow-owned visual history and therefore boundary evidence rather than a native-AR training solution. Exposure to self-generated visual errors remains the route’s largest verified gap.
- **L5 (motion inheritance)**. Prior visual state can carry motion continuity, optionally coupled to explicit motion tokens as in UniMo. This implicit continuity should be distinguished from an explicit motion-transfer mechanism. The reviewed sources do not isolate the relative effects of motion priors, adaptive anchors, and error-triggered refresh under a common protocol.

The reviewed AR systems specify useful state-management mechanisms, but their joint long-horizon evaluation remains incomplete. Some diffusion systems provide time-binned observations (Table 4); equivalent measurements would help assess native visual AR under its own task conditions. Following Section 7.7, an

Table 8 Design guidance for the AR route. The minute- and hour-scale rows are mechanism recommendations; matched quantitative evidence remains absent.

Target	Recommended levers	Known costs
Low serial latency	Token compression; parallel within-frame emission	Codec error on facial detail
High throughput	KV caching; speculative decoding (untested for virtual humans)	Draft-model identity fidelity unknown
Minute-scale stability	Pinned visual anchors; bounded visual history; self-history training; adaptive keyframe refresh	Accumulated error; anchor over-constraint; refresh discontinuity; no matched quantitative evidence
Hour-scale operation	Hierarchical bounded state; explicit eviction/reset/recovery; resource and failure logging	Long-range forgetting; reset discontinuity; unmeasured memory and error trajectories

AR evaluation should report identity, synchronization, motion, appearance, memory, evictions, resets, failures, and recovery over a declared duration.

5.8 Limitations and Outlook

The reviewed visual AR systems span video-motion tokens, compressed frame representations, and framewise controlled generation. Their tasks, personalization requirements, tokenizers, and local emission heads differ. No system in Table 7 provides both complete end-to-end timing and a joint time-binned stability evaluation. Consequently, architectural feasibility is supported more broadly than a common deployment comparison.

The most useful next step is interface-level co-design: visual tokenizers should report how compression changes both AR work and decoder cost; state managers should expose cache growth, eviction, correction, and reset; local heads should disclose strict NFE; and serving evaluations should charge condition encoding, decoding, and I/O to the same event timeline. Sub-linear-state sequence models [55, 202] are a natural research direction for the route’s Stable problem, offering constant-size state with different forgetting behavior than KV eviction. Response-, motion-, and world-level AR can be excellent deployment choices, but their appearance owner and emission costs remain assigned to the diffusion, flow, or rendering routes that produce visible content.

Autoregression can organize behavior, motion, or the visible sequence itself. Only the last role defines the visual AR route. Its main design choices concern representation compression, history reuse, and local frame emission; each affects both runtime and error propagation. Future comparisons should evaluate these choices

through explicit state-update policies and matched efficiency and stability measurements.

6 Render-Based Methods

6.1 Scope, Ownership, and Cost Boundary

Render-based methods separate a reusable appearance representation from the time-varying controls that animate it. We assign a system to this route when a subject-specific mesh, neural field, Gaussian set, or identity-conditioned image-space representation owns appearance and its renderer produces the visible frame [58, 59, 83]. The controller may itself be deterministic, autoregressive, diffusion-based, or flow-based; generating compact motion with one of these models does not change which component owns image synthesis. This ownership test also distinguishes the present route from direct pixel or video-latent generation in Sections 3–5.

Reusing appearance connects the route’s efficiency and stability mechanisms. An enrolled asset reduces repeated appearance construction and supplies a persistent identity reference, corresponding to L1 in Section 4.7. Frame production still requires deformation, rendering, and often neural decoding or inpainting. Enrollment, tracking, and compositing must therefore be included when assessing the complete pipeline. For a session of T observable frames, we write

$$C_{\text{session}} = C_{\text{enroll}} + \sum_{t=1}^T [C_{\text{track/cond}}(t) + C_{\text{motion}}(t) + C_{\text{repr}}(t) + C_{\text{render}}(t) + C_{\text{compose}}(t) + C_{\text{system}}(t)]. \quad (6)$$

Here, C_{enroll} covers capture, calibration, tracking, reconstruction, rigging, per-subject optimization, and asset storage; $C_{\text{track/cond}}$ covers runtime acquisition and estimation of pose, camera, gaze, audio, or other controls; and C_{motion} is the cost of predicting the compact time-varying state. The remaining terms cover appearance deformation or attribute updates, rendering, masks/inpainting/super-resolution, and system overhead such as transfer, scheduling, encoding, and I/O. Thus, renderer-only throughput cannot be reported as end-to-end latency, and a high per-frame rate cannot hide a costly enrollment stage.

Figure 7 projects Equation (6) onto four recurring interface choices. It is an ownership and cost audit rather than a measured universal pipeline: enrollment and reused appearance are session-level, whereas the control-to-output path repeats per frame, and individual

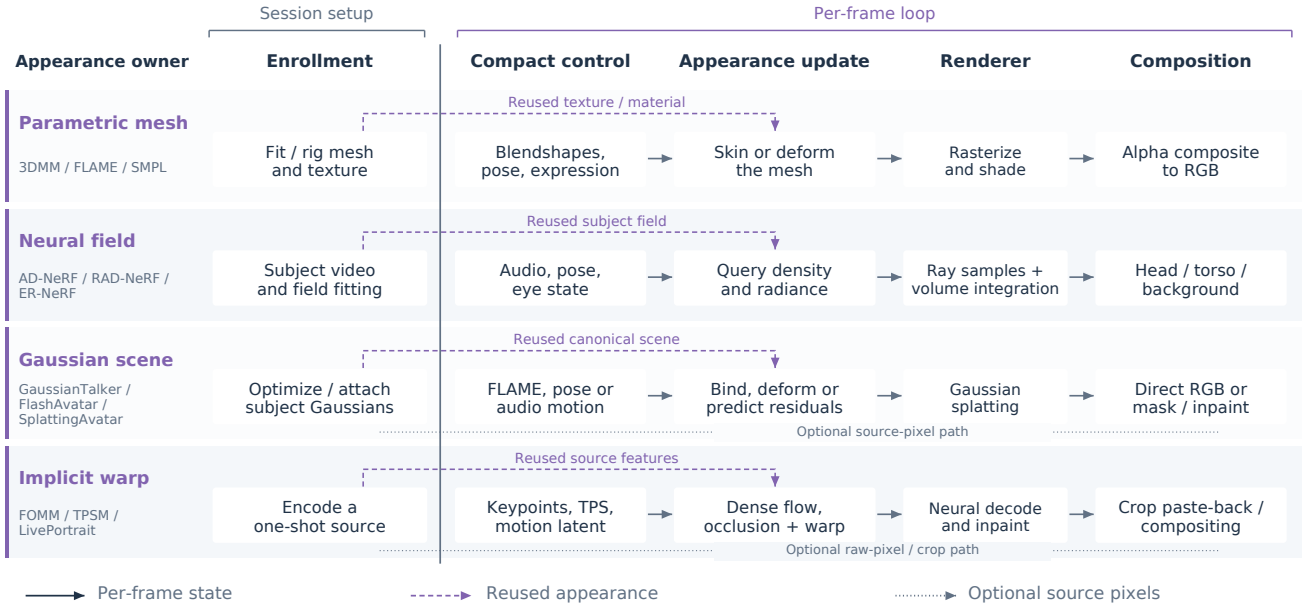


Fig. 7 Interface decomposition of representative render-based appearance owners. Columns separate enrollment from per-frame control, appearance update, rendering, and composition. Solid arrows denote per-frame state; dashed arrows denote reused appearance; dotted arrows denote optional source-pixel paths. The rows summarize parametric, neural-field, Gaussian, and implicit-warp alternatives supported by representative primary papers [58, 59, 98, 99, 155, 188, 209, 223]. They are not a speed, quality, or Stable ranking. 3DMM, FLAME, NeRF, RGB, SMPL, and TPS are defined in the surrounding text.

methods may omit or relocate the optional source-pixel and compositor paths.

Codec avatars provide an early example of person-specific learned appearance for real-time telepresence under studio capture conditions, including deep appearance models [110], expressive extensions [31], and audio- and gaze-driven animation [145]. Subsequent work explores different enrollment budgets, from multi-view capture to monocular video and single-image reconstruction. These settings offer different trade-offs between acquisition effort, subject fidelity, and rendering cost.

6.2 Explicit Controllable Appearance Representations

Explicit backends expose a geometric or scene representation that is updated by pose, expression, viewpoint, or audio-derived controls. Parametric models such as 3DMM, FLAME, SMPL, and the expressive SMPL-X provide interpretable control interfaces [8, 99, 112, 133]; they do not, on their own, establish photorealism or real-time operation. RC-SMPL sits at the lightweight end of this parametric interface: it animates an SMPL-X body mesh whose texture and normal maps are accumulated in real time from a single RGBD camera, trading multi-camera capture for progressive per-subject enrollment; the source claims real-time operation but reports neither frame rate nor inference hardware [157]. Likewise, NeRF and 3D Gaussian Splatting (3DGS) define rendering

families [83, 126], while capture coverage, representation size, deformation, compositing, and hardware determine the operating point of a particular avatar.

6.2.1 Neural-field backends

Early talking-head neural fields made the appearance asset target-specific. AD-NeRF learns separate audio-conditioned head and torso fields from a 3–5-minute target video sampled at 25 FPS, under a static-camera/background setting [59]. In its reported five-minute setting, training takes about 36 hours and rendering one 450×450 frame takes about 12 seconds on an RTX 3090. This is a frame-rendering measurement, not end-to-end throughput. HeadNeRF instead renders a 32×32 neural feature map and uses a 2D neural renderer to produce a 256×256 image [66]; it reports about 25 ms per frame (over 40 FPS), but the inference GPU is unspecified and the timing begins from supplied latent codes and camera parameters, excluding fitting, tracking, and I/O. FLAME-in-NeRF attaches parametric expression control to the field [5].

Hash-grid and occupancy structures then removed most field-query work without removing subject enrollment. This is the route’s adoption of the Instant-NGP lever, in which a multiresolution hash encoding replaces deep MLP queries [127]. RAD-NeRF combines a 3D spatial hash, an audio feature grid, occupancy pruning, and

a separate 2D torso stage; it reports 40 FPS and about 7 hours of target training on one V100, with output resolution and timer coverage not reported in the source runtime statement [166]. ER-NeRF uses tri-plane hash encodings, regional attention, and a separate 2D torso model [98]; its 34 FPS result is reported for a 450×450 Obama comparison on one RTX 3080 Ti, with included runtime modules unspecified. These reports illustrate efficient personalized field designs, but their different devices, timing boundaries, and enrollment procedures do not isolate a common speedup over AD-NeRF.

SyncTalk retains the personalized neural field but makes audio-visual synchronization the central design target. A tri-plane hash head field is driven by a Face-Sync controller aligning lip motion to audio, a 3D facial-blendshape expression model, and a Head-Sync stabilizer for pose, and a Portrait-Sync generator restores hair detail and blends the head with the torso [136]. It reports 52 FPS without and 50 FPS with the Portrait-Sync stage at 450×450 on one RTX 3090 with preloaded data, after about two hours of per-subject training. The synchronization machinery is a quality lever on the conditioning path rather than a change to C_{render} ; its Gaussian successor SyncTalk++ appears in Section 6.2 below.

Beyond talking heads, AvatarReX extends the personalized neural field to expressive full-body avatars with separately controllable body, hands, and face, learned from multi-view video [230]. Its efficiency lever is a deferred rendering pipeline built on surface rather than dense volume rendering; the source reports 25 FPS at 1024×1024 on one RTX 3090 for animation and rendering, characterized as roughly two orders of magnitude faster than prior NeRF avatars. Enrollment is studio-scale (22 synchronized cameras, about 2000 frames per sequence), the opposite end of the C_{enroll} spectrum from the one-shot methods below.

6.2.2 Gaussian scene representations

Gaussian avatars replace repeated field sampling with splattable primitives, but they differ substantially in acquisition and animation. GaussianTalker binds a target-specific Gaussian face to FLAME and predicts speaker-specific blendshapes; Gaussian rasterization and an inpainting generator synthesize the face, while pixels outside the facial mask are retained from the source video [209]. Its two training stages take roughly 6 and 4 hours on one RTX 4090. The paper reports 130 FPS on that GPU with the sequence preloaded and 36 FPS for an MNN/OpenGL implementation on an Apple M1, an early sign that splatting fits mobile compute budgets. GaussianAvatars optimizes a subject from multi-

view observations and assigns Gaussians to FLAME triangles so that they inherit mesh motion [140]. SplattingAvatar uses a registered mesh for low-frequency deformation and walkable barycentric Gaussians for higher-frequency appearance; its Unity implementation reports over 300 FPS on an RTX 3090 and a steady 30 FPS on an iPhone 13 (resolution and timer window unreported) [150]. FlashAvatar narrows the personalized head to 13,453 surface-embedded Gaussians deformed by tracked FLAME motion and a residual MLP, reporting 304 FPS at 512×512 on one RTX 3090 for animation and rendering only [188]. A broader family varies the binding and deformation design without changing the cost structure: HeadGaS’s blendshape-latent formulation [47], PSAvatar’s point-based shape model [227], deformable body avatars [141], emotional control [198], progressive construction [158], and real-time full-pipeline avatar systems [168], text-driven toonification combining a conditional tri-plane with a Gaussian deformation field (TextToon: 48 FPS on one RTX 4090 and 15–18 FPS on an Apple M1) [159], and real-time hand avatars fit from as few as 5–20 views (GaussianHand: up to 125 FPS, inference hardware NR) [224]. In every case, sparse, rasterizable appearance shrinks C_{render} , while enrollment, tracking, and topology handling remain.

Audio-driven per-subject splatting has become its own competitive subfamily. GaussianTalker (Cho et al.) encodes canonical Gaussian attributes into a shared spatial-aware feature and predicts per-frame Gaussian offsets through a spatial-audio attention module driven directly by speech, reporting 98 FPS (and up to 120 FPS for a lightweight variant) at 512×512 after about one to 1.5 hours of per-subject training, with the inference GPU NR [29]. This is a distinct method from the FLAME-bound GaussianTalker of Yu et al. [209] despite the shared name: the former drives raw Gaussians from audio, whereas the latter animates FLAME-bound blendshapes. SyncTalk++ ports the synchronization-centric SyncTalk design onto a 3D-Gaussian dynamic portrait renderer and adds an out-of-distribution audio expression generator and a torso restorer, reporting 101 FPS at 512×512 on one RTX 4090 after about 1.5 hours of per-subject training [135]. The pair illustrates a transition from neural fields to Gaussian rendering within a related synchronization-oriented line of work. The changed hardware, resolution, and auxiliary modules prevent attributing the difference in reported frame rate solely to the renderer.

Recent audio-driven Gaussian work makes the controller-appearance boundary more expressive. GaussianSpeech couples an audio-conditioned Transformer to personalized 3D Gaussians, including expression-dependent attributes and multi-view supervision;

DGTalker instead navigates a disentangled Gaussian latent space that separates lip motion from broader talking expression. Both retain a Gaussian appearance owner, so their learned audio or latent modules are controllers rather than diffusion-route owners. GGTalker addresses the enrollment-generalization trade with a generalizable Gaussian prior followed by identity-specific adaptation, making explicit the cost of replacing per-subject optimization with a reusable prior. AudioAvatar extends the same direct audio-to-deformation idea to a personalized whole-body avatar: audio-conditioned particle/deformation fields drive facial, hand, and body dynamics before differentiable Gaussian rendering. Its feature-level distillation from a video diffusion model is a training or controller mechanism; the Gaussian representation remains the primary appearance owner. These systems therefore move beyond the conventional audio-pose-renderer interface, but they also move stability risk into OOD audio, large motion, multi-view consistency, and deformation quality [3, 69, 92, 104].

GaussianHeadTalk studies a complementary temporal failure mode: inconsistent Gaussian mapping and noisy facial tracking can produce wobble despite a fixed identity asset [1]. This motivates separate measurements of tracking and deformation error, failed detections, and temporal artifacts, alongside identity and synchronization.

Two recent developments extend the Gaussian recipe along the axes this survey tracks. First, *mobile and wearable deployment*: TaoAvatar delivers full-body talking avatars on augmented-reality glasses by baking a StyleUNet-learned deformation into a lighter MLP and reports 90 FPS on the Apple Vision Pro [24], and SqueezeMe distills multiple Gaussian full-body avatars to run simultaneously on a Quest 3 headset by moving corrective computation out of per-frame networks [77]. These examples support operation on wearable devices under the reported conditions; sustained power and thermal behavior require separate measurement. Second, *one-shot Gaussian construction*: LAM builds an animatable Gaussian head from a single image in one feed-forward pass, explicitly targeting cross-platform real-time animation and rendering [63]; GUAVA extends fast feed-forward construction to upper-body avatars [214]; and VASA-3D lifts VASA-1’s motion space to a Gaussian head built from one portrait [194]. One-shot construction attacks the route’s historical weakness, hours of C_{enroll} , by amortizing enrollment into a generalizable network. The reduced input coverage leaves unseen appearance and subject-specific detail to the learned prior, creating a fidelity trade-off that depends on the task.

The generalizable head-avatar line makes the enrollment-versus-representation trade especially leg-

ible through two works by the same authors. GPAvatar reconstructs a controllable head from one or several images in a single forward pass using a tri-plane feature field with NeRF-style volume rendering, driven by a FLAME point-based expression field and a multi-image attention module; it renders a 128×128 feature map super-resolved to 512×512 and reports about 15 FPS on an A100, which the authors explicitly label pre-real-time [33]. GAGAvatar keeps the one-shot, generalizable, FLAME-conditioned formulation but replaces the volumetric field with 3D Gaussians reconstructed by dual-lifting, reporting 67 FPS at 512×512 on an A100 using 2.5 GB of memory (driving-render only, excluding pre-computable driving parameters) [32]. Both use an A100 and report 512×512 output, but they change the reconstruction and rendering pipeline together. Their timing difference is therefore not a controlled ablation of representation alone. GAGAvatar also serves as a reusable backend for controllers such as REA-Listener (Section 6.4).

6.3 Implicit Image-Space Appearance Reuse

Image-space methods store identity primarily in a source image or its encoded features and predict correspondences to each target pose. FOMM estimates ten keypoints and local Jacobians, constructs dense backward flow and an occlusion map, and uses an inpainting decoder for content that cannot be copied directly [155], building on earlier keypoint-based transfer [154] and refined by articulated region modeling [156]. TPS Motion Model replaces the local affine model with ten thin-plate-spline transformations plus an affine background transformation, combined by multi-resolution masks before inpainting [223]. Both were evaluated as source-to-driving-video animation without runtime disclosure. The lineage acquired deployment weight with face-vid2vid, which learns a 3D-aware keypoint representation for one-shot talking-head synthesis explicitly motivated by video-conferencing bandwidth: transmitting keypoints instead of pixels makes the warp both the codec and the renderer [182]. Latent-space navigation offers an alternative implicit interface [183], and audio-driven variants animate a still image directly from speech [180].

LivePortrait provides an efficiency-oriented instance of this family. It learns implicit keypoints and motion, warps a source feature volume, and decodes with a SPADE-based generator, supporting cross-identity animation and paste-back compositing [58]. The paper reports 12.8 ms for model generation/inference on one RTX 4090 in naive PyTorch, without binding the timer to a stated resolution or decomposing preprocessing,

paste-back, and I/O. Its one-shot appearance interface is reused by several systems discussed in Section 6.4. Large pose changes, disocclusion, mouth interiors, hair, and crop-boundary compositing remain relevant failure cases when evaluating these composed systems.

6.4 Motion Controllers over Reused Appearance

Reusable appearance permits motion control to be studied separately from image synthesis. A compact motion model can drive a one-shot warp or Gaussian asset, changing temporal behavior without changing the visual owner. The controller and renderer have separate costs, and either can determine system latency.

VASA-1 illustrates this separation. A diffusion Transformer generates holistic facial dynamics and head motion in a learned face latent space, conditioned on audio and optional control signals, and drives a one-shot appearance decoder; the paper reports online generation of 512×512 video at up to 40 FPS with 170 ms/95 ms latency on one/two RTX 4090 GPUs under its stated protocol [193]. Sampling compact motion can reduce the size of the iterative state, while the appearance decoder reuses identity information. The cost relative to dense-video generation still depends on the networks, sampling schedule, and decoder. FLOAT replaces the sampler with motion-latent flow matching [87]; Ditto structures the motion space for controllable real-time synthesis [100]; ChatAnyone stacks hierarchical motion diffusion for stylized upper-body video with real-time intent [139]; MirrorMe reports real-time audio-driven half-body animation [122]; and FlowTalk pushes motion-space flow matching toward the same operating point [44]. MoDA stays in this compact-controller family while keeping a diffusion motion generator: it reuses LivePortrait’s disentangled facial representation as the appearance renderer and generates motion in a joint parameter space with a rectified-flow model, reducing denoising from 50 to 10 steps and reporting a real-time factor of 0.846 on in-the-wild inputs (below one indicates faster than playback; inference GPU NR) [101]. It is one-shot from a single reference image, and under the ownership test its LivePortrait renderer, not the motion diffuser, owns appearance. GAIA scales a zero-shot two-stage design (motion generation, then appearance) [62], JoyVASA extends the recipe to animal faces [17], and style-controllable variants [119, 181] enrich the control space. The compact interface also connects to speech-to-3D animation research that predicts vertices or coefficients without owning pixels [51, 225]; such motion-only models are boundary components under the ownership rule.

Interactive-agent systems assemble the same parts with session state. Teller processes 200-ms audio blocks

with an autoregressive residual-vector-quantized motion model; an implicit-keypoint warper and a single-pass enhancer produce RGB, with reported stage times of 7, 106, and 71 ms on a machine containing four H800 GPUs [228]. ARIG predicts continuous head-motion coefficients with a 15-step DiffusionMLP over a framewise autoregressive controller and renders through LivePortrait-style warping; it reports a 31 FPS generation-process rate (hardware and included components NR) and maintains a 512-slot interaction summary cache, an L2 memory instance whose Stable effect is unmeasured [60]. DyStream couples an AR/flow motion module (60 ms speaker-audio buffer) to a reference-conditioned appearance volume and reports 29.24 FPS with 34.2 ms audio-packet-to-motion delay on one H200; the delay ends at facial-motion generation rather than final rendering [21]. REA-Listener predicts listener motion with a lightweight Transformer and renders with GAGA-vatar [32]/3DGS plus inpainting; on one RTX 3090, its six listed stages sum to 33.14 ms/frame, of which motion generation accounts for 0.06 ms and 3D-aware rendering for 21.45 ms [226]. This is the route’s cleanest demonstration that once the controller is compact, *the renderer owns the critical path*. PortraitDirector adds hierarchical disentanglement for controllable real-time reenactment [78], and VASA-Rig connects audio-driven rigging to virtual-reality use [128].

6.5 Fast Design Guidance

The route’s levers map onto Equation (6) as follows:

1. **Amortize appearance at enrollment** ($C_{\text{repr}}/C_{\text{render}} \downarrow$ via $C_{\text{enroll}} \uparrow$). Subject-specific fields and Gaussian sets reuse fitted appearance across frames. One-shot warps and feed-forward Gaussian builders, such as LivePortrait and LAM, reduce preparation cost but rely more on learned priors for unseen content.
2. **Structure the queries** ($C_{\text{render}} \downarrow$). Hash grids and occupancy pruning reduce field-query work, while Gaussian representations permit rasterization. Mobile and wearable examples include SplattingAvatar, TaoAvatar, and SqueezeMe. Their reported frame rates establish particular device configurations, not a measured speed-per-watt ranking.
3. **Keep the controller compact** ($C_{\text{motion}} \downarrow$). Motion-space diffusion or flow reduces the representation processed by an iterative sampler. In REA-Listener, the reported 0.06 ms motion stage is small relative to 21.45 ms rendering; other systems require their own component measurements.

Table 9 Evidence boundaries for representative render-based methods. NR denotes not reported in the primary paper. Source timing is transcribed with its device, resolution, and component boundary; rows are not a speed ranking.

Method	Appearance, anatomy, enrollment	Motion and visible-frame boundary	Source timing boundary	Stable evidence
AD-NeRF [59]	Head/torso fields; target-specific, 3–5-min video	Audio-conditioned fields; separate head/torso render	~12 s/frame, 450 ² , RTX 3090; frame rendering; ~36 h training	NR
HeadNeRF [66]	Parametric head field; latent fitting required	32 ² feature render → 2D decoder to 256 ²	~25 ms/frame (> 40 FPS); GPU NR; excludes fitting/tracking/I/O	NR
RAD-NeRF [166]	Head field plus 2D torso; target-specific	Audio grid, spatial hash, occupancy-pruned render	40 FPS, one V100; resolution/timer coverage NR; ~7 h training	NR
ER-NeRF [98]	Tri-plane head field plus 2D torso; target-specific	Audio control, regional attention, field render	34 FPS, 450 ² , RTX 3080 Ti; component/I/O boundary NR	NR
Gaussian Talker [209]	FLAME-bound face Gaussians; target-specific	Blendshapes → splat/inpaint; outside-face source pixels retained	130 FPS, RTX 4090, preloaded; 36 FPS Apple M1; resolution/full timer NR; ~10 h training	NR
Splatting Avatar [150]	Mesh-embedded head/body Gaussians; tracked mesh required	Mesh deformation → direct splatting	> 300 FPS RTX 3090; 30 FPS iPhone 13; resolution/full timer NR	NR
FlashAvatar [188]	FLAME-surface head Gaussians; 1–3-min target video	FLAME/residual deformation → direct splatting	304 FPS, 512 ² , RTX 3090; animation/render only	NR
TaoAvatar [24]	Full-body Gaussians with baked deformation; multi-view capture	Tracked control → lightweight MLP → splatting	90 FPS on Apple Vision Pro; full pipeline coverage NR	NR
LAM [63]	One-shot feed-forward Gaussian head	Blendshape/joint control → direct splatting	Source reports cross-platform real-time rendering; full timer NR	NR
LivePortrait [58]	Source feature volume; one-shot	Implicit motion, feature warp/decoder; optional paste-back	12.8 ms model generation, RTX 4090; resolution/E2E boundary NR	NR
VASA-1 [193]	One-shot face latent appearance decoder	DiT motion diffusion in face latent space → decoder	Up to 40 FPS at 512 ² , 170/95 ms latency, one/two RTX 4090; protocol-stated boundary	NR
Teller [228]	Source-conditioned image features; personalization NR	AR motion → warp/enhance RGB	Components 7/106/71 ms; total < 200 ms; machine with four H800; resolution/TTFF NR	NR
ARIG [60]	Source-conditioned image features; one-shot	AR coefficients + 15-step DiffusionMLP → LivePortrait-style warp	31 FPS “generation process”; device/resolution/component boundary NR	NR
DyStream [21]	Reference appearance volume; one-shot	AR/flow motion code → dense flow warp/decode	29.24 FPS, 34.2 ms packet-to-motion delay, one H200; render/display boundary NR	180-s pose-drift ablation on ten 3-min clips NR
REA-Listener [226]	GAGAvatar 3DGS head; one-shot	Lightweight Transformer motion → splat + inpaint	Six stages sum 33.14 ms/frame, RTX 3090; render is 21.45 ms	NR
AvatarReX [230]	NeRF body/hands/face; full-body; multi-view (22 cams, ~2000 frames)	Parametric-template control → deferred surface render	25 FPS, 1024 ² , RTX 3090; animation+render; ~2 orders faster than prior NeRF avatars	NR
SyncTalk [136]	Tri-plane hash head field; head+torso; per-subject video (~min)	Face/Head-Sync control → field render + Portrait-Sync blend	52/50 FPS (w/o, w/ Portrait-Sync), 450 ² , RTX 3090, preloaded; ~2 h training	NR
SyncTalk++ [135]	3DGS dynamic portrait; head+torso; per-subject video (~min)	OOD-audio expression + Head-Sync → splat + torso restore	101 FPS, 512 ² , RTX 4090; ~1.5 h training	NR
GaussianTalker (Cho) [29]	Audio-driven 3DGS head; per-subject video (~4 min)	Spatial-audio attention → per-frame Gaussian offsets → splat	98 FPS (120 FPS lite), 512 ² ; GPU NR	NR
GPAvatar [33]	Tri-plane+NeRF head; one-/few-shot, generalizable	FLAME point expression field + multi-image attention → 128 ² feature → SR 512 ²	~15 FPS, A100; author-labeled pre-real-time	NR
GAGAvatar [32]	3DGS head; one-shot single image, generalizable	Dual-lifting + FLAME expression Gaussians → splat + neural refine	67 FPS, 512 ² , A100, 2.5 GB; driving-render only, excludes driving-param acquisition	NR
MoDA [101]	LivePortrait reused appearance; one-shot single image	Rectified-flow motion diffusion (50→10 steps) → LivePortrait warp/decode	RTF 0.846 on wild data; inference GPU NR; trained 8×H20	NR

AR: autoregressive; DiT: diffusion Transformer; E2E: end-to-end; FLAME: Faces Learned with an Articulated Model and Expressions; FPS: frames per second; GPU: graphics processing unit; I/O: input/output; MLP: multilayer perceptron; NeRF: neural radiance field; NR: not reported in the primary paper; RGB: red–green–blue; 3DGS: 3D Gaussian Splatting; TTFF: time to first frame.

Method / family	Enrollment evidence / fitting	Source timing	Hardware and timing boundary
AD-NeRF · field	3–5 min video; ~36 h fit	~12 s / frame	RTX 3090; 450 ² ; frame rendering
RAD-NeRF · field	Target video; ~7 h fit	40 FPS	V100; resolution / full timer NR
ER-NeRF · field	Target-specific video	34 FPS	RTX 3080 Ti; 450 ² ; components NR
SyncTalk · field	Target video; ~2 h fit	50 FPS	RTX 3090; 450 ² ; with Portrait-Sync
GaussianTalker (Yu) · GS	Target video; ~10 h fit	130 FPS / 36 FPS	RTX 4090 preloaded / Apple M1
FlashAvatar · GS	1–3 min target video	304 FPS	RTX 3090; 512 ² ; animation + render
SyncTalk++ · GS	Target video; ~1.5 h fit	101 FPS	RTX 4090; 512 ² ; full timer NR
SplattingAvatar · GS · B	Tracked mesh required	>300 FPS / 30 FPS	RTX 3090 / iPhone 13; full timer NR
TaoAvatar · GS · B	Multi-view capture	90 FPS	Apple Vision Pro; full timer NR
AvatarReX · field · B	22-camera multi-view capture	25 FPS	RTX 3090; 1024 ² ; animation + render
GPAvatar · field	One / a few images	~15 FPS	A100; 512 ² ; source: pre-real-time
GAGAvatar · GS	One image	67 FPS	A100; 512 ² ; driving + render only
LAM · GS	One image; feed-forward asset	Numerical timing NR	Cross-platform real-time claim
LivePortrait · warp	One image	12.8 ms	RTX 4090; model generation only
VASA-1 · latent appearance	One image	Up to 40 FPS	RTX 4090; 512 ² ; online protocol
REA-Listener · GS	One image	33.14 ms / frame	RTX 3090; sum of six listed stages

GS: Gaussian scene. FPS and latency retain source units and boundaries. VASA-1 separately reports 170 / 95 ms latency on one / two RTX 4090s. B: body-inclusive. NR: not reported. These observations do not establish a matched speed ranking or a Pareto frontier.

Fig. 8 Enrollment and runtime evidence for representative render-based methods. Acquisition, per-subject fitting, hardware, and timer coverage remain separate. Timing values preserve source units without converting throughput into latency. B marks body-inclusive systems; GS denotes Gaussian scenes; NR denotes not reported. Incompatible protocols do not support a common speed ranking or Pareto frontier. GaussianTalker denotes the FLAME-bound method of Yu et al. Sources: neural fields [33, 59, 98, 136, 166, 230], Gaussian scenes [24, 32, 63, 135, 150, 188, 209, 226], and reused portrait appearance [58, 193].

4. **Reuse source pixels** (C_{compose} trade). Paste-back and outside-mask retention (GaussianTalker, LivePortrait) avoid re-synthesizing static content, at the price of seams and crop-boundary failures.

Table 9 shows why these levers must be evaluated separately: reported quantities span renderer-only throughput, animation-plus-render time, model-generation time, motion-level packet delay, and block latency, and no cross-paper rank is supported when anatomy, personalization, device, resolution, and compositing differ. The route-specific accounting rule is to include $C_{\text{track/cond}}$ wherever runtime face tracking, FLAME fitting, key-point estimation, or audio conditioning is required. Pre-computed controls and runtime-estimated controls define different boundaries.

Figure 8 complements Table 9 by separating acquisition evidence, per-subject fitting, runtime values, and timing boundaries for 16 representative systems. These heterogeneous observations support an evidence map rather than a common speed ranking or Pareto frontier.

6.6 Stable Design Guidance

An enrolled appearance asset provides a persistent identity anchor (L1 in Section 4.7), but does not guarantee stable rendered output. Tracking, controller state, deformation, disocclusion, and compositing can introduce

temporal errors even when the canonical asset is unchanged. Controller memories, such as ARIG’s 512-slot cache, provide another form of persistent state. Position handling and training with generated histories are relevant when these controllers are sequential; their longitudinal effects are not established by renderer throughput or asset reuse alone.

The evidence base matches this asymmetry. The only duration-structured result in Table 9 is DyStream’s 180-s head-pose drift ablation, which shows its appearance anchor reducing pose drift over ten three-minute clips [21]. This is an L1 validation, not a full stability protocol. No retained paper reports a duration-binned protocol jointly measuring identity, appearance, synchronization, geometry, flicker, memory, and latency. The additional systems in that table, including SyncTalk, SyncTalk++, GaussianTalker, GPAvatar, GAGAvatar, AvatarReX, and MoDA, likewise lack this joint protocol in the reviewed evidence [29, 32, 33, 101, 135, 136, 230].

A useful comparison would evaluate a tracked-and-rendered avatar and an anchored dense generator under the same long-session conditions. Identity retention, motion and deformation error, temporal wobble, and compositing failures should be reported separately. GaussianHeadTalk motivates the distinction between identity similarity and temporal wobble [1]; the reviewed evidence does not establish which complete pipeline is more stable over an hour.

Table 10 Design guidance for the render route. Levers are supported by the sources in Table 9.

Target	Recommended levers	Known costs
Maximum per-subject speed	Multi-view or monocular Gaussian asset; baked deformation	Hours of enrollment; capture requirements
One-shot generality	Implicit warp (LivePortrait) or feed-forward Gaussian (LAM); motion-latent controller (VASA-1)	Disocclusion, large poses, mouth interiors; limited input-view coverage
Mobile/wearable deployment	Splatting + distilled/baked correctives (TaoAvatar, SqueezeMe)	Per-subject assets; thermal/battery limits untimed
Minute-scale stability	Enrolled asset as L1 anchor; validate tracking and controller state	Temporal errors in deformation and composition; joint longitudinal evidence limited
Expressive control	Motion-space diffusion/flow controller	Controller expressiveness bounded by the asset’s deformation space

6.7 Limitations and Outlook

The central trade-off is not simply fidelity versus throughput. Subject-specific neural fields and Gaussians amortize a substantial enrollment cost over many frames; one-shot feature warpers reduce enrollment but depend heavily on source-view coverage and learned disocclusion; parametric controls provide stable, editable interfaces whose base space cannot express motion it does not encode. Explicit scenes must handle topology, primitive count, memory, and view coverage. Image-space reuse must synthesize newly visible content and preserve crop seams.

Three directions follow from these trade-offs. First, joint design of feed-forward appearance construction and compact motion control could reduce both preparation and runtime costs; one-shot construction still has a nonzero setup cost. Second, controller training should be evaluated under generated histories to test whether it reduces drift in recurrent systems such as ARIG and DyStream. Third, matched evaluations should report enrollment data and time, asset size, hardware, precision, all runtime components, and quality over duration. These measurements would show whether a change reduces complete-system cost or shifts work to another stage.

6.8 Summary

Render-based methods reuse an explicit scene or source-conditioned appearance representation under time-varying control. Efficient queries, compact controllers, and appearance reuse support reported desktop, mobile,

and wearable deployments. The reused asset anchors identity, while tracking, deformation, and compositing remain independent sources of temporal error. The comparison in Table 9 therefore records enrollment, runtime, and stability evidence separately.

7 Datasets and Evaluation

7.1 Dataset Evidence for Generation and Long-Horizon Evaluation

A dataset determines which conditional relationships a model can learn and which claims an evaluation can support. For efficient virtual-human generation, the useful axes are therefore not venue and year, but the generated anatomy, conditioning modalities, identity and interaction structure, media properties, split policy, and continuous observation horizon. These axes expose task mismatch: a high-quality face-video corpus may support appearance learning without providing audio, a multi-view capture may support reconstruction without cross-identity generation, and a dyadic collection may represent turn taking without providing a single continuous session long enough to test drift.

Single-person resources span distinct purposes. HDTF provides in-the-wild high-resolution audiovisual portraits, whereas VFHQ is a video-face super-resolution corpus whose source paper does not establish retained or synchronized audio [189, 222]. CelebV-HQ instead targets unconditional face-video generation and attribute editing, with manually annotated appearance, action, and emotion attributes but no source-defined audio-driving protocol [236]. MEAD adds controlled speech, synchronized views, discrete emotions, and intensity levels [176]. TalkVerse scales synchronized single-person audio–video curation and pose annotation, but its source clips remain short [185]. OpenHumanVid supplies large-scale human-centric video with caption, skeleton, and audio conditions for pretraining human video generators [97], and EchoMimicV2’s half-body corpus extends conditioning to hand and gesture coverage [125]. These resources are complementary rather than interchangeable benchmarks for lip synchronization, identity, expression, or temporal behavior.

Interaction and reconstruction data introduce different supervision. ViCo pairs a visible and audible speaker with a listener’s non-verbal facial response and supplies both seen-identity and identity-disjoint test regimes [234]. SpeakerVid-5M organizes single-speaker clips into dialogue, listening, and multi-turn structures, but its pretraining/supervised fine-tuning (SFT) tiers are quality strata rather than a documented leakage-free train/test partition [218]. Seamless Interaction records

Table 11 Evidence-bearing properties of representative datasets. Media frame rate is a capture or preprocessing rate, not model throughput. Aggregate corpus duration is not a continuous generation horizon. NR denotes not reported in the primary paper.

Dataset	Task and modalities	Scale and media	Split evidence	Continuous-horizon boundary
HDTF [222]	In-the-wild audiovisual talking face	362 videos; 15.8 h; 300+ subjects; 10k+ sentences; 720p/1080p	Fixed split and identity separation NR	Longest clip NR; 15.8 h is aggregate corpus time
VFHQ [189]	Video-face super-resolution; audio retention NR	16,827 clips from 7,228 source videos; roughly 700–1000 px faces; 512 ² , ×4 benchmark	50 test sequences exclude corresponding training clips; identity separation NR	Tracks retain 100–2000 frames, but frame rate and total hours are NR
CelebV-HQ [236]	Face-video generation and 83-attribute editing; audio NR	35,666 clips; 15,653 identities; at least 512 ² ; source reports approximately 65 h in text versus 68 h in a table	Fixed identity-disjoint split NR	Each clip is 3–20 s and is truncated at 20 s
MEAD [176]	Controlled emotional audiovisual talking faces; seven synchronized views	60 actors; 281,400 clips; 1920×1080 at a 30 FPS capture rate; 48-kHz audio; neutral plus seven emotions and three non-neutral intensities	Paper-specific 20% holdout; split unit and identity separation NR	38 min 57 s per speaker aggregates many short utterances, not one session
ViCo [234]	Dyadic speaker audio–video to listener facial response	483 pairs; 92 identities; 95 min 22 s; 384 ² face crops; source video 1080p at a 30 FPS capture rate and 44.1-kHz audio	D_{test} identities occur in training; D_{ood} is identity-disjoint	Clips span 1–71 s; no minute-scale drift protocol
SpeakerVid-5M [218]	Single, dialogue, listening, and multi-turn audiovisual organizations	Single branch: 5.2M clips/8.7k h; dialogue branch: 770k pairs/1.8k h; branches are not proven disjoint	Pretraining/SFT are quality tiers; VidChatBench has 500 unseen-speaker pairs, but global source/identity leakage control is NR	Basic clips span 3–14 s; stored order does not establish a continuous long-horizon sample
TalkVerse [185]	Single-person synchronized audio–video with 2D poses and structured annotations	2.3M clips; 6.3k h; 720p/1080p; unified to a 25 FPS content rate	Fixed identity-disjoint split NR	Source clips span 5–50 s; aggregate hours do not establish long-session samples
OpenHuman Vid [97]	Human-centric video with caption, skeleton, and audio conditions	13.2M clips reported by the source; large-scale pretraining corpus	Source-defined filtering tiers; identity-disjoint split NR	Clip-scale samples; not a continuous-session resource
NeRSemble [89]	Multi-view human-head capture and sequence-specific novel-view synthesis	222 participants; 4,734 sequences; 31.7M camera images; 7.5 h; 16 views at 3208×2200 and a 73 FPS capture rate	Dataset split NR; experiments use 12 input and four held-out views within each sequence	Roughly 3 min per participant is divided among 25 short sequences; longest sequence NR
Seamless Interaction [2]	Dyadic speech, face, upper-body, hand, transcript, and interaction metadata	4,065.04 h; 64,739 interactions; 5,098 sessions; 4,284 participants; dual 48-kHz audio and 2160×3840 video at a 30 FPS capture rate	Public train/dev/test plus private test; participant partition is functionally clique-level	Released active segments span 2–10 min; one-hour sessions include unreleased meta time; generation evaluation uses about 20 s

Abbreviations: SFT, supervised fine-tuning; D_{test} and D_{ood} , test and out-of-distribution subsets; 2D, two-dimensional; p, progressive-scan resolution; px, pixels; k, thousand; M, million; kHz, kilohertz; h, hours; min, minutes; s, seconds; dev, development; FPS, frames per second.

long active segments from two participants with separate audio channels and participant-level partitions, while NeRSemble supplies synchronized high-resolution multi-view head capture for sequence-specific novel-view reconstruction [2, 89]. For this survey’s lenses, the relevant property of dyadic data is its *continuous-horizon* value: long uninterrupted sessions are exactly the material a stability protocol needs, independent of any interaction claim.

Table 11 distinguishes corpus scale from continuous observation length. Capture or normalized frame rates describe the data rather than generation throughput, and aggregate hours can combine many subjects, views, or short clips. The listed resources do not establish a matched hour-long continuous-generation benchmark. Some long-horizon evaluations therefore extend shorter inputs through repetition, as in Live Avatar [76], which limits the diversity of conditions tested. A public benchmark of continuous 10–60-minute single-subject and dyadic sessions would help measure drift under sustained, changing inputs.

Dataset governance is also part of evaluation validity. Web-derived collections report filtering and source-tracking procedures, but public availability of source videos does not by itself establish participant consent or redistribution rights [185, 189, 218, 234]. Controlled captures provide stronger acquisition records: NeRSemble reports General Data Protection Regulation (GDPR)-compliant participant agreements and deletion rights, and Seamless Interaction reports adult informed consent and compensation [2, 89]. Their primary papers nevertheless do not specify a complete dataset license. Reproducible reporting should therefore separate media provenance, consent, access terms, and technical split integrity rather than compressing them into a binary *public* field.

7.2 Evaluation Scope and Protocols

Evaluation must preserve the distinction introduced in Section 2.1: visual quality, identity fidelity, driving fidelity, temporal fidelity, and deployment behavior are

different observables. A low image distance does not establish audiovisual synchronization, and high steady-state throughput does not establish low time to first frame (TTFF) or stable long-horizon operation. We therefore group evidence by what is measured and record the protocol that makes each value interpretable.

Four classes of evidence are useful. *Full-reference measures* compare an output with a paired and aligned target. *Distributional measures* compare sets of generated and real samples. *Task-specific evaluators* estimate an attribute such as face identity or audiovisual synchronization. *System and longitudinal measures* characterize the visible pipeline boundary and change over an evaluated session. These classes complement rather than replace one another.

7.3 Reference-Based and Distributional Visual Quality

Pixel errors, peak signal-to-noise ratio (PSNR), and the structural similarity index (SSIM) require a paired reference. SSIM compares local luminance, contrast, and structure before spatial aggregation [184], and its relationship to PSNR is itself protocol-dependent [67]. Its original validation concerns aligned still images; a video-level report must therefore state the frame pairing and temporal aggregation, and it should not interpret frame-wise SSIM as a temporal-consistency result. Learned perceptual image patch similarity (LPIPS) instead aggregates channel-normalized deep-feature differences across layers [216]; ensemble variants improve its robustness to transformations [84]. A lower value means that the candidate is closer to the paired reference under the selected backbone and calibration variant. Comparisons must retain the same network, layer set, preprocessing, crop, dynamic range, and aggregation.

The Fréchet Inception Distance (FID) fits Gaussian statistics to pretrained Inception features from real and generated image sets and computes their squared Fréchet distance [64]. Because this construction discards sample order, FID is an image-distribution measure rather than evidence of temporal stability. Fréchet Video Distance (FVD) applies the same statistical form to spatiotemporal features; the original study selected Kinetics-400 RGB Inflated 3D ConvNet (I3D) logits and showed that finite-sample values depend on sample count [170]. Consequently, an FID or FVD number is comparable only after the reference set, feature network and checkpoint, activation layer, preprocessing, sample count, and, for FVD, clip construction and content frame rate have been fixed. Neither distance isolates identity, lip synchronization, or deployment speed. The same protocol dependence applies to similarity scores computed with other pretrained encoders, such as CLIP-based text or

image similarity [142]: the encoder checkpoint and preprocessing are part of the metric’s identity.

7.4 Human-Specific Fidelity

Human-specific evaluators require equally explicit boundaries. ArcFace is a face-recognition representation trained with an additive angular margin [43]; it is not itself a standardized virtual-human quality score. When a paper reports ArcFace-based cosine similarity (CSIM), the report should identify the embedding model and checkpoint, face detector and alignment, reference image or set, handling of failed detections, and frame or track aggregation. Recognition accuracy reported by the ArcFace source cannot be transferred into a threshold for generated-video identity fidelity.

For audiovisual synchronization, the underlying evaluator is SyncNet, which learned a joint audio–video embedding for lip-sync assessment in the wild [35]. Wav2Lip introduced lip-sync error distance (LSE-D; lower is better) and lip-sync error confidence (LSE-C; higher is better) on top of a pretrained SyncNet [138]; the evaluator differs from the synchronization expert used to train its generator. Cross-paper comparison therefore requires the same evaluator checkpoint, audio–video pairing, temporal window, face and mouth crop, preprocessing, and averaging rule. These scores test the selected audiovisual correspondence protocol; they do not measure expression quality, identity preservation, or latency.

Landmark Distance (LMD) was defined for paired lip-motion evaluation using Dlib detections on generated and ground-truth video [25]. The original construction uses 20 lip landmarks, aligns the mean landmark positions, computes corresponding Euclidean distances, and averages over points and frames; lower is better. LMD is thus a paired landmark-accuracy measure, not a no-reference audio–lip synchronization score. Detector version, landmark subset, crop and coordinate scale, alignment, frame correspondence, failure policy, and aggregation must remain fixed for numerical comparison.

Identity, synchronization, and landmark measures cover different failure modes. They should be paired with task-appropriate motion measures only when the underlying detector, normalization, reference, and aggregation are documented. A learned evaluator’s name alone is not a reproducible protocol.

Table 12 Evidence scope and minimum protocol fields for representative visual-quality measures. PSNR denotes peak signal-to-noise ratio; SSIM, structural similarity index; LPIPS, learned perceptual image patch similarity; FID, Fréchet Inception Distance; and FVD, Fréchet Video Distance. The arrows $\uparrow$ and $\downarrow$ indicate that higher and lower values, respectively, are preferable only within a matched protocol.

Measure	Comparison unit	Reference	Minimum fields to report	Does not by itself establish
PSNR / SSIM $\uparrow$	Aligned image or frame pair	Paired target	Crop and alignment, color space, dynamic range, resolution, implementation, frame aggregation	Temporal coherence, identity, lip sync, or system timing
LPIPS $\downarrow$	Aligned image or patch pair	Paired target	Backbone, calibrated variant, layer set, normalization, crop, resolution, aggregation	Attribute-specific fidelity or temporal behavior
FID $\downarrow$	Image sets	Real image set	Reference set, feature checkpoint/layer, preprocessing, sample count, random sampling	Sample order, temporal stability, identity, or lip sync
FVD $\downarrow$	Video-clip sets	Real clip set	Video encoder/checkpoint/layer, clip length and sampling, frame rate, preprocessing, sample count	Failure localization, long-horizon stability, or runtime

7.5 Whole-Body, Interaction, and Cross-Scene Evaluation

Recent avatar systems expand the evaluation surface beyond face-only fidelity. Whole-body generation calls for separate face lip-sync, hand-keypoint consistency, gesture-audio alignment, body-motion diversity, face-body temporal coherence, and identity consistency across body regions. Listening-oriented systems additionally require response appropriateness, listener-motion diversity, speaking/listening transition smoothness, turn-taking responsiveness, and user preference; these should be reported separately from the latency of receiving the partner’s signal. Cross-scene identity evaluation should distinguish same-scene retention from changes in background, lighting, framing, or viewpoint, because a persistent reference may preserve identity in one scene while failing under a context shift.

Temporal wobble and tracking stability form another independent axis. A render-based system can preserve a fixed identity asset while introducing jitter through tracking, deformation, Gaussian mapping, or compositing. Reports should therefore specify the temporal artifact detector, reference coordinate system, time bins, failed-detection policy, and aggregation, and should not substitute a single-frame identity score or renderer FPS for a longitudinal wobble measurement. These dimensions are complementary to ArcFace, SyncNet, and LMD and should be stratified by anatomy, interaction mode, scene constancy, and failure type [1, 34, 92].

A complete protocol should disclose which of these measurements are paired, no-reference, or human-rated, together with the continuous duration and reset policy. At present, the literature reports subsets under incompatible settings rather than a unified whole-body, dyadic, cross-scene benchmark.

7.6 System Timing

System evaluation follows the boundary in Section 2.6. At minimum, a speed claim should state hardware and device count, output resolution and content frame rate, precision, batch size or concurrency, warm-up, measurement window, and every included component. Frames per second (FPS) is a frame-throughput measure only when frames are counted over a disclosed wall-clock generation boundary; otherwise an FPS value may denote content or capture frame rate. Real-time factor (RTF) compares wall-clock time with generated-media duration; TTFF starts at a declared input-arrival event; and inter-chunk latency measures the time between successive committed chunks under a declared buffering policy. Number of function evaluations (NFE) records iterative model calls for a stated generated unit. None can be inferred from another, and source-reported timing values remain source-reported evidence when hardware or boundaries differ; they must not be plotted as a controlled cross-route ranking.

7.7 Longitudinal and Human Evaluation

Longitudinal evaluation measures how output quality and system behavior change during continuous generation. A reproducible protocol should specify session content and duration, initialization, retained context, and reset policy. Identity and appearance should be measured in predeclared time bins relative to an initial or short-horizon reference, alongside synchronization, motion, temporal artifacts, memory, and latency. Failures, resets, and recovery intervals require separate reporting because averages can conceal them. Ablating the active L1–L5 mechanisms under matched conditions would further distinguish their individual contributions from the behavior of the complete system.

We recommend human studies when automatic measures do not isolate visible artifacts or preference. Re-

Table 13 Minimum reporting record for Fast and Stable evidence. FPS denotes frames per second; RTF, real-time factor; TTFF, time to first frame; NFE, number of function evaluations; and NR, not reported in the primary paper.

Record	Required fields	Interpretation boundary
Measured frame throughput (FPS)	Frames counted, wall-clock elapsed interval, resolution, batch or concurrency, warm-up, hardware, precision, included pipeline	Establishes FPS throughput only for the disclosed generation boundary; does not establish latency or content/capture frame rate
RTF	Wall-clock numerator, generated-duration denominator, clip or stream length, pipeline boundary	Values below one establish faster-than-playback only for that convention
TTFF	Input-arrival event, buffering, first observable frame	First-frame service metric; does not establish steady-state inter-chunk latency or throughput
Inter-chunk latency	Successive committed-output timestamps, commitment interval, buffering, jitter summary	Steady-state service metric; does not establish TTFF or throughput
NFE / serial calls	Generated unit, solver steps, denoiser or vector-field calls, guidance multiplier, decoder calls	Call count does not establish wall-clock speed
Stability protocol	Evaluated continuous duration, persistent state, context bound, time bins, identity and appearance drift variables, failure counts, reset policy, and resource trajectories	A long demonstration or aggregate dataset duration is not quantitative Stable evidence

ports should state the question, stimuli, duration, and presentation order. They should also report rater recruitment, repeated judgments, aggregation, uncertainty, and the statistical test. A preference percentage without these fields is descriptive rather than a transferable quality scale.

The current survey therefore does not claim a unified rerun benchmark. In the absence of frozen checkpoints, inputs, runnable configurations, and independently reproduced measurements on common hardware, it reports source measurements with their original boundaries and uses Tables 12 and 13 as a comparison checklist. This preserves what each source actually establishes while exposing the missing evidence needed for deployment-oriented comparison.

8 Discussion, Challenges, and Future Directions

The four routes offer different ways to distribute appearance, motion, and temporal state across a generation pipeline. Their practical value depends on the task, available resources, and evaluation conditions. The evidence reviewed in Sections 3–7 motivates four common research problems: composing modules with explicit interfaces, measuring complete-system cost, maintaining consistency with bounded state, and evaluating deployment safeguards.

Figure 9 brings together the route-specific Fast levers and Stable patterns reviewed in Sections 3–6, with their evidence gaps kept explicit.

8.1 Composition Requires Explicit Interfaces

Hybrid systems often divide work among a condition or motion model, a visual representation owner, a decoder or renderer, and a compositor. Under the ownership rule in Section 2.3, the route label follows the module that produces or persistently stores dense visible

content [21, 60]. Secondary modules remain part of the system boundary. Replacing one module can change motion control without changing the visual owner; it can also reduce visual-model calls while leaving condition encoding, decoding, and transport on the critical path.

Composition requires both a representation interface and a state-update policy. Each module should specify its coordinate system, identity reference, subject-specific assets, retained history, reset behavior, and recovery procedure. These fields determine when output becomes available and which errors can propagate. For example, Teller’s audio buffering precedes motion prediction and rendering, while LivePortrait’s reused appearance still requires motion estimation and decoding [58, 228].

Compact motion interfaces can reduce the state presented to a dense visual owner, but they also restrict appearance changes to those expressible by the interface. Dense latent interfaces offer more freedom while making identity ownership and failure localization harder to audit. These are testable architectural trade-offs rather than universal advantages of a route. A falsifiable research question is whether a hybrid can expose typed representation, state, and recovery contracts such that the latency and failure contribution of every module is reproducible in isolation and in composition.

8.2 Complete-System Cost and Timing

The source literature reports FPS, wall-clock latency, real-time factor, function evaluations, and relative speedups under different boundaries. Tables 1, 3, 7, and 9 preserve those route-specific boundaries. A renderer-only measurement is not an end-to-end system result [58, 76], and source values remain interpretable only with their original hardware, precision, concurrency, warm-up, generated unit, and included components, as formalized in Section 7.6.

	GAN-based	Diffusion-based	Autoregressive	Render-based
Visual owner	Adversarial dense or regional synthesis	Iterative dense pixels or latents	Cross-frame visual tokens / latents	Reusable subject appearance
Fast levers	- Regional synthesis - Compact controls - Compile the fixed graph	- Few-step distillation - Compress visual tokens - Cache and pipeline	- Compress serial state - Reuse KV, bound context - Parallel local emission	- Amortize appearance - Hash grids / splatting - Compact motion controls
Cost terms	Generator / condition / system cost	NFE / cost per call / repeated work	Serial depth / cache / local-head cost	Enrollment / deformation / render / composition
Stable patterns	Mostly framewise state Fixed-reference L1 anchor Jitter follows the controls	L1–L5 mechanisms Self-history training Time-binned observations	Persistent visual history Bounded cache / positions Exposure to rollout errors	Asset as identity anchor Risks: tracking, geometry, and composition
Evidence gap	No complete time-binned Stable protocol in the reviewed sources	Matched pattern ablations and recovery protocols remain limited	No complete measured Fast + Stable protocol	Longitudinal tests needed for tracking, wobble and recovery

Shared pattern vocabulary: L1 anchor; L2 bounded memory; L3 position re-indexing; L4 self-history training; L5 motion inheritance.

Fig. 9 Cross-route Fast and Stable mechanism map in manuscript chapter order. Fast mechanisms act on route-specific cost terms; Stable mechanisms share the L1–L5 vocabulary. Coverage and evidence gaps summarize the reviewed sources and do not define a numerical ranking.

A deployment record should account separately for subject enrollment or asset preparation. The runtime ledger should time condition acquisition and encoding, motion or state generation, dense visual synthesis, decoding or rendering, compositing, transfer, and I/O. User-visible measures include time to first frame, inter-commit latency and jitter, steady throughput, memory, and recovery cost. The critical path moves with the operating point. This is the recurring finding of every route’s design guidance in this survey. Fewer iterative calls can expose a decoder bottleneck (Section 4.2); a compact controller can leave the renderer owning 65% of the frame time (Section 6.4); multi-device pipelining can raise throughput without reducing the wait for the first committed output [76, 161]. A falsifiable research question is which component owns the critical path at each resolution, chunk size, concurrency level, and session duration, and whether the same conclusion survives when enrollment, I/O, and recovery are included.

8.3 Bounded-State Long-Horizon Generation

The five patterns in Section 4.7 provide a common vocabulary for mechanisms whose evaluation remains uneven. The reviewed diffusion systems contain examples of all five, including Live Avatar’s sampled observations beyond an hour under looped audio [76]. Render-based systems reuse an identity asset but can drift through their controllers or rendering pipeline; DyStream’s 180-s

pose-drift ablation provides a limited longitudinal example [21]. Native visual AR explicitly retains history, yet the reviewed systems lack a complete joint stability protocol. Framewise GAN generators avoid recursive visual feedback when earlier outputs are not reused, but can still exhibit jitter or inherit controller errors.

This comparison suggests studying which mechanisms transfer across representations. Training with generated histories could be evaluated in both native visual AR and recurrent motion controllers. Reliability-aware memory updates could be compared with fixed windows, compression, and periodic resets. A matched study should isolate each mechanism’s effect on identity, synchronization, motion diversity, and legitimate scene changes over predeclared time bins. The current evidence neither establishes a universal ordering of the routes nor supports unbounded-duration stability beyond the conditions actually evaluated.

Audio-to-deformation and expanded Stable axes. Recent audio-driven Gaussian systems expose a shift from the conventional audio → pose or SMPL → renderer cascade toward direct audio → Gaussian or particle deformation → renderer paths. This can preserve high-frequency facial, hand, and body dynamics that an intermediate pose interface may quantize, but it moves complexity into learned deformation fields and their tracking boundary. The resulting Stable problem is multidimensional: appearance identity, motion and deformation, audio–visual synchronization, interaction response, and

system/resource behavior can fail independently. Generalizable Gaussian priors reduce enrollment cost but introduce adaptation and out-of-distribution trade-offs; fixed Gaussian scenes anchor appearance but do not prevent wobble, tracking failure, or compositing drift. These systems make the missing evidence concrete: future reports should pair enrollment and runtime ledgers with time-binned identity, motion, synchronization, wobble, failure, and recovery measurements [1, 3, 69, 92, 104].

8.4 Route Selection Under Deployment Constraints

Figure 10 summarizes the decision path. The design guidance of Sections 3–6 supports a first set of conditional guidelines. These compare route-level mechanism maturity under stated constraints; they do not rank individual methods, and each guideline inherits the measurement caveats of its underlying sources.

When per-subject enrollment is acceptable, render-based systems offer reusable appearance and reported high-throughput or wearable configurations [24, 188]. Selection should account for capture and fitting cost, followed by tracking, deformation, and compositing over the intended session duration.

When one-shot operation is required, compact motion generation over reused appearance provides a practical design pattern [58, 63, 193]. The motion module and renderer must both fit the latency budget, while deformation and disocclusion determine which movements the appearance model can reproduce.

When changes extend beyond a reusable appearance model, dense diffusion offers mechanisms for synthesizing broader visual content. The reviewed streaming examples combine distillation, memory, and substantial serving resources [76, 152]. Persistent anchors can also restrict scene changes, so the intended variability must be evaluated directly. Component profiling should determine whether further optimization targets the sampler, decoder, or conditioning path.

When the edit is confined to a small region, feed-forward GAN synthesis can reduce the amount of generated content [138, 220]. Evaluation should include compositing seams, synchronization, and framewise jitter as well as runtime.

When explicit visual-history modeling is central, native visual AR offers a direct formulation. Its deployment evidence is less complete in the reviewed sources: Table 7 does not contain a system with both full-pipeline timing and a joint time-binned stability evaluation. The representation, local emission head, and state policy should therefore be validated for the intended application.

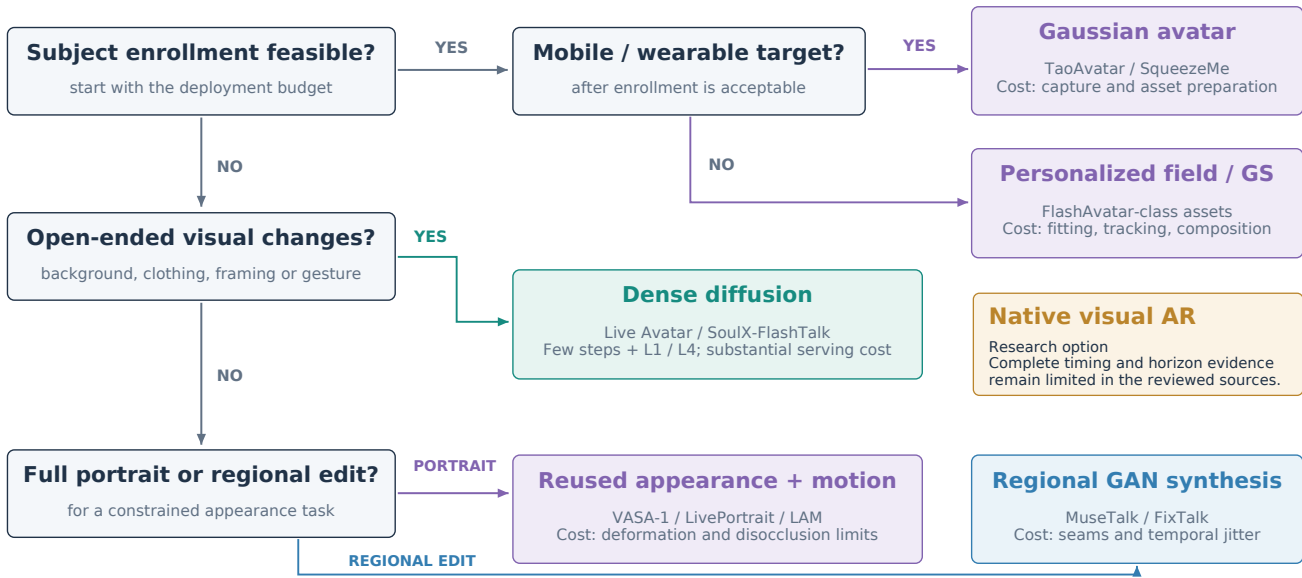
8.5 Trustworthy Deployment Requires Layered Evidence

Fast and Stable systems sharpen deployment risks: low-latency synthesis enables live impersonation, and long-horizon stability with cached identity state extends it. Trustworthiness is not an additional scalar metric. The NIST AI 100-4 report separates provenance-data tracking from synthetic-content detection and discusses watermarks, metadata, filtering, and red-teaming as controls with different lifecycle positions and failure surfaces [19]; it provides no avatar-specific effectiveness or runtime measurement. The C2PA specification defines signed claims, content bindings, and validation states, including segment-level paths for live video [38]. These mechanisms bind a signing credential to covered media bytes and expose specified integrity failures; they do not establish factual truth, verify the depicted identity, or prove authorization to use a likeness, and they carry manifest-stripping, key-management, and privacy risks [36, 37].

Empirical safeguards have equally important boundaries. DeepfakeBench exposes generalization gaps across manipulation types with frame-level evidence [196], and Video Seal demonstrates faster-than-playback video watermarking under a stated offline boundary without adaptive-attack coverage [52]. A watermark hit, a detector score, and a valid provenance record are different observations; none alone proves authorization or harmlessness. A deployment evaluation should declare the attacker model, benign platform transformations, operating point, authorization and revocation procedures, and the latency cost of each control on the serving path. The last point connects trust directly to this survey’s Fast accounting, since provenance signing and watermarking must fit inside the same per-chunk budget the route mechanisms optimize.

8.6 Conclusion

This survey has reviewed efficient virtual-human video generation through visual ownership, computational cost, and long-horizon consistency. The four routes organize different synthesis principles, while the five drift-control patterns reveal shared problems in retaining identity and using generated history. The literature supports a range of acceleration and state-management mechanisms, but differences in task, hardware, and evaluation duration limit direct performance comparisons. Progress requires complete timing records, duration-binned quality measurements, and controlled studies of how efficiency mechanisms interact with temporal stability.



Route-level guidance under source-reported evidence. Validate the selected pipeline under the actual hardware and timing boundary.

Fig. 10 Route-selection guidance under deployment constraints. Leaves summarize mechanism maturity and known costs in the surveyed evidence; example methods illustrate families rather than a controlled ranking. Native visual autoregression remains a research option where complete deployment evidence is limited. Examples follow the route-specific evidence [24, 58, 63, 76, 77, 152, 162, 188, 193, 220].

Data Availability Statement

This survey does not introduce a new dataset. The datasets listed in the paper are available from their respective original sources.

References

- Agarwal, M., Zhang, M., Sevilla-Lara, L., McDonagh, S.: GaussianHeadTalk: Wobble-free 3D talking heads with audio driven gaussian splatting. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pp. 8017–8027 (2026) 29, 32, 36, 39
- Agrawal, V., et al.: Seamless interaction: Dyadic audio-visual motion modeling and large-scale dataset. arXiv preprint arXiv:2506.22554 (2025) 34
- Aneja, S., Sevastopolsky, A., Kirschstein, T., Thies, J., Dai, A., Nießner, M.: GaussianSpeech: Audio-driven personalized 3D gaussian avatars. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 13065–13075 (2025) 29, 39
- Ao, T.: Body of her: A preliminary study on end-to-end humanoid agent. arXiv preprint arXiv:2408.02879 (2024) 2, 4, 12, 21, 22, 23, 24
- Athar, S., Shu, Z., Samaras, D.: FLAME-in-NeRF: Neural control of radiance fields for free view face animation. In: 2023 IEEE 17th International Conference on Automatic Face and Gesture Recognition, pp. 1–8. IEEE (2023) 27
- Azari, B., Lim, A.: Emostyle: One-shot facial expression editing using continuous emotion parameters. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pp. 6385–6394 (2024) 9
- Bai, L., Zhang, T., Shao, S., Tan, D., Zhong, Q., Xie, Z., Li, H., Huang, Q., Shen, D., Ji, T., et al.: MaineCoon: Pursuing a real-time audio-visual social world model. arXiv preprint arXiv:2606.17800 (2026) 3
- Blanz, V., Vetter, T.: A morphable model for the synthesis of 3D faces. In: Seminal Graphics Papers: Pushing the Boundaries, Volume 2, pp. 157–164. Springer (2023) 27
- Blattmann, A., Dockhorn, T., Kulal, S., Mendelevitch, D., Kilian, M., Lorenz, D., Levi, Y., English, Z., Voleti, V., Letts, A., et al.: Stable video diffusion: Scaling latent video diffusion models to large datasets. arXiv preprint arXiv:2311.15127 (2023) 13
- Bounareli, S., Tzelepis, C., Argyriou, V., Patras, I., Tzimiropoulos, G.: Hyperreenact: One-shot reenactment via jointly learning to refine and retarget faces. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 7149–7159 (2023) 9
- Bounareli, S., Tzelepis, C., Argyriou, V., Patras, I., Tzimiropoulos, G.: Stylemask: Disentangling the style space of stylegan2 for neural face reenactment. In: 2023 IEEE 17th International Conference on Automatic Face and Gesture Recognition (FG), pp. 1–8. IEEE (2023) 9
- Bounareli, S., Tzelepis, C., Argyriou, V., Patras, I., Tzimiropoulos, G.: One-shot neural face reenactment via finding directions in gan’s latent space. International Journal of Computer Vision (IJCV) (2024) 9
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al.: Language models are few-shot learners. Advances in neural information processing systems 33, 1877–1901 (2020) 7
- Cai, Y., Chu, X., Gao, X., Gong, S., Huang, Y., Kang, C., Li, K., Liu, H., Liu, R., Liu, Y., et al.: Towards interactive intelligence for digital humans. arXiv preprint arXiv:2512.13674 (2025) 3
- Cai, Y., Wu, Y., Li, K., Zhou, Y., Zheng, B., Liu, H.: FloodDiffusion: Tailored diffusion forcing for streaming

- motion generation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 2295–2304 (2026) **12, 13**
16. Canfes, Z., Atasoy, M.F., Dirik, A., Yanardag, P.: Text and image guided 3d avatar generation and manipulation. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pp. 4421–4431 (2023) **9**
 17. Cao, X., Wang, G., Shi, S., Zhao, J., Yao, Y., Fei, J., Gao, M., Xie, P.: JoyVASA: Portrait and animal image animation with diffusion-based audio-driven facial dynamics and head motion generation. arXiv preprint arXiv:2411.09209 (2024) **30**
 18. Cao, Z., Hidalgo, G., Simon, T., Wei, S.E., Sheikh, Y.: Openpose: Realtime multi-person 2d pose estimation using part affinity fields. *IEEE transactions on pattern analysis and machine intelligence* **43**(1), 172–186 (2019) **7**
 19. Chandra, B., Dunietz, J., Roberts, K., Lee, Y., Fontana, P., Awad, G.: Reducing risks posed by synthetic content: An overview of technical approaches to digital content transparency. NIST Trustworthy and Responsible AI NIST AI 100-4, National Institute of Standards and Technology (2024). DOI 10.6028/NIST.AI.100-4 **39**
 20. Chang, Q., Ding, Y.X., Zhou, K.: Enhancing identity-deformation disentanglement in stylegan for one-shot face video re-enactment. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, pp. 1247–1255 (2025) **9**
 21. Chen, B., Liu, H.: DyStream: Streaming dyadic talking heads generation via flow matching-based autoregressive model. arXiv preprint arXiv:2512.24408 (2025) **22, 23, 25, 30, 31, 32, 37, 38**
 22. Chen, B., Martí Monsó, D., Du, Y., Simchowicz, M., Tedrake, R., Sitzmann, V.: Diffusion forcing: Next-token prediction meets full-sequence diffusion. *Advances in Neural Information Processing Systems* **37**, 24081–24125 (2024) **16**
 23. Chen, J., Cai, H., Chen, J., Xie, E., Yang, S., Tang, H., Li, M., Han, S.: Deep compression autoencoder for efficient high-resolution diffusion models. In: International Conference on Learning Representations, vol. 2025, pp. 96539–96560 (2025) **7, 22**
 24. Chen, J., Hu, J., Wang, G., Jiang, Z., Zhou, T., Chen, Z., Lv, C.: TaoAvatar: Real-time lifelike full-body talking avatars for augmented reality via 3D gaussian splatting. In: Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 10723–10734 (2025) **2, 29, 31, 32, 39, 40**
 25. Chen, L., Li, Z., Maddox, R.K., Duan, Z., Xu, C.: Lip movements generation at a glance. In: Computer Vision – ECCV 2018, pp. 538–553. Springer International Publishing (2018). DOI 10.1007/978-3-030-01234-2_32 **35**
 26. Chen, M., Cui, L., Zhang, W., Zhang, H., Zhou, Y., Li, X., Tang, S., Liu, J., Liao, B., Chen, H., et al.: MIDAS: Multimodal interactive digital-human synthesis via real-time autoregressive video generation. arXiv preprint arXiv:2508.19320 (2025) **2, 3, 4, 12, 21, 22, 23, 24**
 27. Cheng, K., Cun, X., Zhang, Y., Xia, M., Yin, F., Zhu, M., Wang, X., Wang, J., Wang, N.: Videoretalking: Audio-based lip synchronization for talking head video editing in the wild. In: SIGGRAPH Asia 2022 Conference Papers, pp. 1–9 (2022) **8**
 28. Chern, E., Hu, Z., Tang, B., Su, J., Chern, S., Deng, Z., Liu, P.: LiveTalk: Real-time multimodal interactive video diffusion via improved on-policy distillation. arXiv preprint arXiv:2512.23576 (2025) **15, 17, 18, 20**
 29. Cho, K., Lee, J., Yoon, H., Hong, Y., Ko, J., Ahn, S., Kim, S.: Gaussiantalker: Real-time high-fidelity talking head synthesis with audio-driven 3d gaussian splatting. In: Proceedings of the 32nd ACM International Conference on Multimedia, pp. 10985–10994 (2024) **28, 31, 32**
 30. Cho, P.H., Jang, J., Lee, S., Lee, J., Jin, S., Shin, H., Yi, J., Park, Y., Park, C., Kim, S.: Lip Forcing: Few-step autoregressive diffusion for real-time lip synchronization. arXiv preprint arXiv:2606.11180 (2026) **15, 17, 18, 20**
 31. Chu, H., Ma, S., De la Torre, F., Fidler, S., Sheikh, Y.: Expressive telepresence via modular codec avatars. In: European Conference on Computer Vision, pp. 330–345. Springer (2020) **27**
 32. Chu, X., Harada, T.: Generalizable and animatable gaussian head avatar. *Advances in Neural Information Processing Systems* **37**, 57642–57670 (2024) **29, 30, 31, 32**
 33. Chu, X., Li, Y., Zeng, A., Yang, T., Lin, L., Liu, Y., Harada, T.: Gpavatar: Generalizable and precise head avatar from image (s). In: The Twelfth International Conference on Learning Representations (2024) **29, 31, 32**
 34. Chu, X., Liu, R., Huang, Y., Liu, Y., Peng, Y., Zheng, B.: UniLS: End-to-end audio-driven avatars for unified listening and speaking. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 25142–25152 (2026) **22, 23, 36**
 35. Chung, J.S., Zisserman, A.: Out of time: automated lip sync in the wild. In: Asian conference on computer vision, pp. 251–263. Springer (2016) **35**
 36. Coalition for Content Provenance and Authenticity: C2PA harms modelling. Technical Report Version 2.0, Coalition for Content Provenance and Authenticity (2026). Release 2026-03-30; https://spec.c2pa.org/specifications/specifications/2.4/security/Harms_Modelling.html **39**
 37. Coalition for Content Provenance and Authenticity: C2PA security considerations. Technical Report Version 2.4, Coalition for Content Provenance and Authenticity (2026). Release 2026-03-30; https://spec.c2pa.org/specifications/specifications/2.4/security/Security_Considerations.html **39**
 38. Coalition for Content Provenance and Authenticity: C2PA technical specification. Technical Specification Version 2.4, Coalition for Content Provenance and Authenticity (2026). Release 2026-04-01; https://spec.c2pa.org/specifications/specifications/2.4/specs/C2PA_Specification.html **39**
 39. Cui, J., Li, H., Yao, Y., Zhu, H., Shang, H., Cheng, K., Zhou, H., Zhu, S., Wang, J.: Hallo2: Long-duration and high-resolution audio-driven portrait image animation. In: International Conference on Learning Representations (2025) **18, 20**
 40. Cui, J., Wu, J., Li, M., Yang, T., Li, X., Wang, R., Bai, A., Ban, Y., Hsieh, C.J.: Self-Forcing++: Towards minute-scale high-quality video generation. arXiv preprint arXiv:2510.02283 (2025) **18**
 41. Dao, T.: FlashAttention-2: Faster attention with better parallelism and work partitioning. In: International Conference on Learning Representations, vol. 2024, pp. 35549–35562 (2024) **7**
 42. Dao, T., Fu, D., Ermon, S., Rudra, A., Ré, C.: FlashAttention: Fast and memory-efficient exact attention with io-awareness. *Advances in neural information processing systems* **35**, 16344–16359 (2022) **7**
 43. Deng, J., Guo, J., Xue, N., Zafeiriou, S.: ArcFace: Additive angular margin loss for deep face recognition. In: Proceed-

- ings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 4690–4699 (2019) [35](#)
44. Deng, K., Guo, Y., Shen, L.: FlowTalk: Real-time audio-driven talking head synthesis via motion-space flow matching. In: Proceedings of the 7th ACM International Conference on Multimedia in Asia, pp. 1–6 (2025) [30](#)
 45. Deng, X., Zheng, Z., Zhang, Y., Sun, J., Xu, C., Yang, X., Wang, L., Liu, Y.: Ram-avatar: Real-time photo-realistic avatar from monocular videos with full-body control. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 1996–2007 (2024) [9](#)
 46. Dey, A., Alcoverro, M., Zabaleta, I., Fernández, C.: Interactive real-time visual avatar generation: A survey. Unpublished manuscript, Unith Research Lab (2026) [3](#)
 47. Dhano, H., Nie, Y., Moreau, A., Song, J., Shaw, R., Zhou, Y., Pérez-Pellitero, E.: HeadGaS: Real-time animatable head avatars via 3D gaussian splatting. In: European Conference on Computer Vision, pp. 459–476. Springer (2024) [28](#)
 48. Du, F., Li, T., Qiao, Q., Yu, T., Zhen, D., Zhang, Z., Jia, X., Yang, Y., Yin, S., Liu, S.: RAP: Real-time audio-driven portrait animation with video diffusion transformer. In: ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 1896–1900. IEEE (2026) [12](#), [13](#), [17](#), [18](#)
 49. Ekman, P., Friesen, W.V.: Facial action coding system. *Environmental Psychology & Nonverbal Behavior* (1978) [7](#)
 50. Esser, P., Kulal, S., Blattmann, A., Entezari, R., Müller, J., Saini, H., Levi, Y., Lorenz, D., Sauer, A., Boesel, F., et al.: Scaling rectified flow transformers for high-resolution image synthesis. In: Forty-first international conference on machine learning (2024) [7](#), [11](#)
 51. Fan, Y., Lin, Z., Saito, J., Wang, W., Komura, T.: Faceformer: Speech-driven 3d facial animation with transformers. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 18770–18780 (2022) [30](#)
 52. Fernandez, P., Elshahar, H., Yalniz, I.Z., Mourachko, A.: Video seal: Open and efficient video watermarking. arXiv preprint arXiv:2412.09492 (2024). DOI 10.48550/arXiv.2412.09492 [39](#)
 53. Gan, Y., Yang, Z., Yue, X., Sun, L., Yang, Y.: Efficient emotional adaptation for audio-driven talking-head generation. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 22634–22645 (2023) [9](#)
 54. Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y.: Generative adversarial networks. *Communications of the ACM* **63**(11), 139–144 (2020) [4](#), [6](#), [8](#)
 55. Gu, A., Dao, T.: Mamba: Linear-time sequence modeling with selective state spaces. arXiv preprint arXiv:2312.00752 (2023) [26](#)
 56. Guan, J., Xu, Z., Zhou, H., Wang, K., He, S., Zhang, Z., Liang, B., Feng, H., Ding, E., Liu, J., Wang, J., Zhao, Y., Liu, Z.: Resyncer: Rewiring style-based generator for unified audio-visually synced facial performer. In: European Conference on Computer Vision (ECCV), pp. 348–367. Springer (2024) [8](#)
 57. Guan, J., Zhang, Z., Zhou, H., Hu, T., Wang, K., He, D., Feng, H., Liu, J., Ding, E., Liu, Z., Wang, J.: Stylesync: High-fidelity generalized and personalized lip sync in style-based generator. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 1505–1515 (2023) [8](#)
 58. Guo, J., Zhang, D., Liu, X., Zhong, Z., Zhang, Y., Wan, P., Zhang, D.: LivePortrait: Efficient portrait animation with stitching and retargeting control. arXiv preprint arXiv:2407.03168 (2024) [1](#), [2](#), [3](#), [4](#), [26](#), [27](#), [29](#), [31](#), [32](#), [37](#), [39](#), [40](#)
 59. Guo, Y., Chen, K., Liang, S., Liu, Y.J., Bao, H., Zhang, J.: AD-NeRF: Audio driven neural radiance fields for talking head synthesis. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 5784–5794 (2021) [2](#), [3](#), [26](#), [27](#), [31](#), [32](#)
 60. Guo, Y., Liu, X., Zhen, C., Yan, P., Wei, X.: ARIG: Autoregressive interactive head generation for real-time conversations. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 12956–12965 (2025) [12](#), [19](#), [22](#), [23](#), [30](#), [31](#), [37](#)
 61. HaCohen, Y., Chiprut, N., Brazowski, B., Shalem, D., Moshe, D., Richardson, E., Levin, E., Shiran, G., Zabari, N., Gordon, O., et al.: Ltx-video: Realtime video latent diffusion. arXiv preprint arXiv:2501.00103 (2024) [13](#)
 62. He, T., Guo, J., Yu, R., Wang, Y., An, K., Li, L., Tan, X., Wang, C., Hu, H., Wu, H., et al.: GAIA: Zero-shot talking avatar generation. In: International Conference on Learning Representations, vol. 2024, pp. 2107–2124 (2024) [30](#)
 63. He, Y., Gu, X., Ye, X., Xu, C., Zhao, Z., Dong, Y., Yuan, W., Dong, Z., Bo, L.: LAM: Large avatar model for one-shot animatable gaussian head. In: Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers, pp. 1–13 (2025) [29](#), [31](#), [32](#), [39](#), [40](#)
 64. Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., Hochreiter, S.: Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems* **30** (2017) [35](#)
 65. Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models. *Advances in neural information processing systems* **33**, 6840–6851 (2020) [4](#), [7](#), [10](#)
 66. Hong, Y., Peng, B., Xiao, H., Liu, L., Zhang, J.: Head-NeRF: A real-time NeRF-based parametric head model. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 20342–20352 (2022). DOI 10.1109/CVPR52688.2022.01973 [27](#), [31](#)
 67. Hore, A., Ziou, D.: Image quality metrics: Psnr vs. ssim. In: 2010 20th international conference on pattern recognition, pp. 2366–2369. IEEE (2010) [35](#)
 68. Hu, L.: Animate anyone: Consistent and controllable image-to-video synthesis for character animation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 8153–8163 (2024) [4](#), [10](#)
 69. Hu, W., Li, S., Peng, Z., Zhang, H., Shi, F., Liu, X., Wan, P., Zhang, D., Tian, H.: GGTalker: Talking head synthesis with generalizable gaussian priors and identity-specific adaptation. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 10108–10117 (2025) [29](#), [39](#)
 70. Hu, Y., Liu, R., Ren, Y., Yin, X., Li, H.: Unitalker: Conversational speech-visual synthesis. In: Proceedings of the 33rd ACM International Conference on Multimedia, pp. 10248–10257 (2025) [22](#), [23](#)
 71. Hu, Y., Zhang, J.: Speculative decoding for autoregressive video generation. arXiv preprint arXiv:2604.17397 (2026) [15](#), [21](#), [25](#)
 72. Huang, L., Wu, Z., Wang, W., Shi, Y., Feng, M., He, J., Xie, C., Liu, Y., Zhou, J., Wang, A., et al.: Wan-Streamer v0.1: End-to-end real-time interactive foundation models. arXiv preprint arXiv:2606.25041 (2026) [17](#)

73. Huang, L., Wu, Z.F., Shi, Y., Wang, W., Feng, M., He, J., Xie, C.W., Liu, Y., Zhou, J., Wang, A., Zhang, B., Ai, B., Liang, C., Yu, C., Zhong, C., Qi, J., Zhu, K., Li, P., Zhang, P., Zhang, W., Cheng, X., Huang, Y., Zheng, Y., Bao, Y., Wang, Y., Bi, Z.: Wan-Streamer v0.2: Higher resolution, same latency (2026). URL <https://arxiv.org/abs/2607.04443> **17**, **23**, **24**, **25**
74. Huang, L., Wu, Z.F., Shi, Y., Wang, W., Feng, M., Yu, C., Liang, C., He, J., Xie, C.W., Liu, Y., et al.: Video= world+event stream. arXiv preprint arXiv:2607.15038 (2026) **23**, **24**
75. Huang, X., Li, Z., He, G., Zhou, M., Shechtman, E.: Self Forcing: Bridging the train-test gap in autoregressive video diffusion. *Advances in Neural Information Processing Systems* **38**, 167283–167308 (2026) **2**, **16**
76. Huang, Y., Guo, H., Wu, F., Wang, W., Zhang, S., Huang, S., Gan, Q., Liu, L., Zhao, S., Chen, E., et al.: Live Avatar: Streaming real-time audio-driven avatar generation with infinite length. arXiv preprint arXiv:2512.04677 (2025) **1**, **2**, **3**, **12**, **15**, **16**, **17**, **18**, **20**, **34**, **37**, **38**, **39**, **40**
77. Iandola, F., Pidhorskyi, S., Santesteban, I., Gupta, D., Pahuja, A., Bartolovic, N., Yu, F., Garbin, E., Simon, T., Saito, S.: SqueezeMe: Mobile-ready distillation of gaussian full-body avatars. In: *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference* Conference Papers, pp. 1–11 (2025) **29**, **40**
78. Ji, C., Qi, J., Xu, S., Zhang, P., Zhang, B.: PortraitDirector: A hierarchical disentanglement framework for controllable and real-time facial reenactment. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 32377–32388 (2026) **30**
79. Ji, Y., Zhong, Z., Zhang, J., Yang, Q., Jin, X., Qin, Y., Luo, W., Mao, S., Liu, W., Li, H.: Forcing-KV: Hybrid KV cache compression for efficient autoregressive video diffusion models. arXiv preprint arXiv:2605.09681 (2026) **16**
80. Kang, T., Oh, J., Lee, J., Park, S., Choo, J.: Expression domain translation network for cross-domain head reenactment. In: *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 7356–7359. IEEE (2024) **9**
81. Karras, T., Laine, S., Aila, T.: A style-based generator architecture for generative adversarial networks. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 4401–4410 (2019) **4**, **6**, **8**
82. Karras, T., Laine, S., Aittala, M., Hellsten, J., Lehtinen, J., Aila, T.: Analyzing and improving the image quality of stylegan. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8110–8119 (2020) **6**, **8**
83. Kerbl, B., Kopanas, G., Leimkühler, T., Drettakis, G., et al.: 3d gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.* **42**(4), 139–1 (2023) **3**, **4**, **7**, **26**, **27**
84. Kettunen, M., Härkönen, E., Lehtinen, J.: El-pips: robust perceptual image similarity via random transformation ensembles. arXiv preprint arXiv:1906.03973 (2019) **35**
85. Ki, T., Jang, S., Jo, J., Yoon, J., Hwang, S.J.: Avatar Forcing: Real-time interactive head avatar generation for natural conversation. arXiv preprint arXiv:2601.00664 (2026) **12**, **13**
86. Ki, T., Min, D.: Stylelipsync: Style-based personalized lip-sync video generation. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 22841–22850 (2023) **8**
87. Ki, T., Min, D., Chae, G.: FLOAT: Generative motion latent flow matching for audio-driven talking portrait. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 14699–14710 (2025) **30**
88. Kingma, D.P., Welling, M.: Auto-encoding variational bayes. arXiv preprint arXiv:1312.6114 (2013) **7**
89. Kirschstein, T., Qian, S., Giebenhain, S., Walter, T., Nießner, M.: NeRsemble: Multi-view radiance field reconstruction of human heads. *ACM Transactions on Graphics* **42**(4), 1–14 (2023). DOI 10.1145/3592455 **34**
90. Kong, W., Tian, Q., Zhang, Z., Min, R., Dai, Z., Zhou, J., Xiong, J., Li, X., Wu, B., Zhang, J., et al.: Hunyuanvideo: A systematic framework for large video generative models. arXiv preprint arXiv:2412.03603 (2024) **13**
91. Lee, J., Li, C., Tran, L., Wei, S.E., Saragih, J., Richard, A., Joo, H., Bai, S.: Audio driven real-time facial animation for social telepresence. In: *Proceedings of the SIGGRAPH Asia 2025 Conference Papers*, pp. 1–12 (2025) **12**, **13**
92. Lee, S., Moon, S., Lew, H.M., Kang, J.S., Park, G.M.: AudioAvatar: Personalized audio-driven whole-body talking avatars. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3998–4010 (2026) **29**, **36**, **39**
93. Leviathan, Y., Kalman, M., Matias, Y.: Fast inference from transformers via speculative decoding. In: *International Conference on Machine Learning*, pp. 19274–19286. PMLR (2023) **3**, **7**, **15**, **21**, **25**
94. Li, C., Huang, D., Lu, Z., Xiao, Y., Pei, Q., Bai, L.: A survey on long video generation: Challenges, methods, and prospects. arXiv preprint arXiv:2403.16407 (2024) **3**
95. Li, C., Li, J., Mei, R., Xia, H., Zhu, H., Wang, J., Zhu, S.: Hallo-Live: Real-time streaming joint audio-video avatar generation with asynchronous dual-stream and human-centric preference distillation. arXiv preprint arXiv:2604.23632 (2026) **15**, **17**, **18**
96. Li, C., Wang, R., Zhou, L., Feng, J., Luo, H., Zhang, H., Wu, Y., He, X.: JoyAvatar-Flash: Real-time and infinite audio-driven avatar generation with autoregressive diffusion. arXiv preprint arXiv:2512.11423 (2025) **18**, **20**
97. Li, H., Xu, M., Zhan, Y., Mu, S., Li, J., Cheng, K., Chen, Y., Chen, T., Ye, M., Wang, J., et al.: Openhumanvid: A large-scale high-quality dataset for enhancing human-centric video generation. In: *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 7752–7762 (2025) **33**, **34**
98. Li, J., Zhang, J., Bai, X., Zhou, J., Gu, L.: Efficient region-aware neural radiance fields for high-fidelity talking portrait synthesis. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 7568–7578 (2023). DOI 10.1109/ICCV51070.2023.00696 **27**, **28**, **31**, **32**
99. Li, T., Bolkart, T., Black, M.J., Li, H., Romero, J.: Learning a model of facial shape and expression from 4D scans. *ACM Transactions on Graphics* **36**(6), 194:1–194:17 (2017) **4**, **7**, **27**
100. Li, T., Zheng, R., Yang, M., Chen, J., Yang, M.: Ditto: Motion-space diffusion for controllable realtime talking head synthesis. In: *Proceedings of the 33rd ACM International Conference on Multimedia*, pp. 9704–9713 (2025) **30**
101. Li, X., Li, G., Lin, Z., Qian, Y., Yao, G., Jia, W., Wang, A., Chen, W., Wang, F.: Moda: Multi-modal diffusion architecture for talking head generation. arXiv preprint arXiv:2507.03256 (2025) **30**, **31**, **32**
102. Li, Z., Pun, C.M., Fang, C., Wang, J., Cun, X.: PersonaLive! expressive portrait image animation for live streaming. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 18118–18128 (2026) **1**, **3**, **14**, **17**, **18**, **20**

103. Liang, B., Chen, C., Lin, D., Somov, I., Zhao, J., Yuan, J., Zhang, J., Huang, J., Nolte, N., Haqiqi, P., et al.: Avatar V: Scaling video-reference avatar video generation. arXiv preprint arXiv:2606.13872 (2026) [12](#), [14](#), [16](#), [17](#), [20](#)
104. Liang, X., Fan, Y., Yang, Q., Wang, X., Gao, W., Li, G.: DGTalker: Disentangled generative latent space learning for audio-driven gaussian talking heads. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 11079–11088 (2025) [29](#), [39](#)
105. Lin, G., Jiang, J., Yang, J., Zheng, Z., Liang, C., Zhang, Y., Liu, J.: Omnihuman-1: Rethinking the scaling-up of one-stage conditioned human animation models. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 13847–13858 (2025) [10](#)
106. Lipman, Y., Chen, R.T., Ben-Hamu, H., Nickel, M., Le, M.: Flow matching for generative modeling. In: The Eleventh International Conference on Learning Representations (2023) [4](#), [7](#), [10](#), [11](#), [12](#)
107. Liu, F., Zhang, S., Wang, X., Wei, Y., Qiu, H., Zhao, Y., Zhang, Y., Ye, Q., Wan, F.: Timestep embedding tells: It's time to cache for video diffusion model. In: Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 7353–7363 (2025) [1](#), [7](#)
108. Liu, K., Hu, W., Xu, J., Shan, Y., Lu, S.: Rolling forcing: Autoregressive long video diffusion in real time. arXiv preprint arXiv:2509.25161 (2025) [16](#)
109. Liu, X., Gao, F., Zhang, X., Zhang, Y., Wei, X., Lei, Z., Zhu, X.: TurboTalk: Progressive distillation for one-step audio-driven talking avatar generation. arXiv preprint arXiv:2604.14580 (2026) [15](#)
110. Lombardi, S., Saragih, J., Simon, T., Sheikh, Y.: Deep appearance models for face rendering. ACM Transactions on Graphics **37**(4), 1–13 (2018) [27](#)
111. Long, J., Sun, W., Tu, R.C., Tao, D.: Lightning fast caching-based parallel denoising prediction for accelerating talking head generation. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 40, pp. 7503–7511 (2026) [15](#), [17](#)
112. Loper, M., Mahmood, N., Romero, J., Pons-Moll, G., Black, M.J.: SMPL: a skinned multi-person linear model. ACM Transactions on Graphics (TOG) **34**(6), 1–16 (2015) [7](#), [27](#)
113. Lu, Y., Chai, J., Cao, X.: Live speech portraits: real-time photorealistic talking-head animation. ACM Transactions on Graphics (ToG) **40**(6), 1–17 (2021) [2](#), [9](#), [11](#)
114. Lu, Y., Sun, J., Zhu, G., Cao, M.: AsymTalker: Identity-consistent long-term talking head generation via asymmetric distillation. arXiv preprint arXiv:2605.02948 (2026) [15](#), [18](#), [20](#)
115. Luo, C., Wang, J., Li, B., Song, S., Ghanem, B.: Omniresponse: Online multimodal conversational response generation in dyadic interactions. Advances in Neural Information Processing Systems **38**, 125869–125891 (2026) [22](#), [23](#)
116. Luo, S., Tan, Y., Patil, S., Gu, D., Von Platen, P., Passos, A., Huang, L., Li, J., Zhao, H.: Lcm-lora: A universal stable-diffusion acceleration module. arXiv preprint arXiv:2311.05556 (2023) [1](#), [7](#), [14](#)
117. Ma, J., Du, Y., Sun, Y., Xing, Z., Li, C., Tang, Z., Li, B., Jiang, P.T., Yuan, L., Zhou, D., et al.: SyncCache: Exploiting asymmetric dynamics for fast audio-driven portrait animation. arXiv preprint arXiv:2606.30849 (2026) [12](#), [15](#), [16](#)
118. Ma, X., Fang, G., Wang, X.: DeepCache: Accelerating diffusion models for free. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 15762–15772 (2024) [1](#), [7](#)
119. Ma, Y., Wang, S., Hu, Z., Fan, C., Lv, T., Ding, Y., Deng, Z., Yu, X.: StyleTalk: One-shot talking head generation with controllable speaking styles. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 37, pp. 1896–1904 (2023) [30](#)
120. Ma, Y., Yan, Z., Liu, H., Wang, H., Pan, H., He, Y., Yuan, J., Zeng, A., Cai, C., Shum, H.Y., et al.: Follow-Your-Emoji-Faster: Towards efficient, fine-controllable, and expressive freestyle portrait animation. International Journal of Computer Vision **134**(3), 130 (2026) [15](#), [17](#)
121. Ma, Y., Zheng, X., Xu, J., Xu, X., Ling, F., Zheng, X., Kuang, H., Li, H., Wang, X., Xiao, X., et al.: Flow caching for autoregressive video generation. arXiv preprint arXiv:2602.10825 (2026) [1](#), [7](#), [15](#)
122. Meng, D., Xiao, S., Zhang, X., Wang, G., Zhang, P., Wang, Q., Zhang, B., Bo, L.: MirrorMe: Towards real-time and high fidelity audio-driven halfbody animation. arXiv preprint arXiv:2506.22065 (2025) [30](#)
123. Meng, M., Zhao, Y., Zhang, B., Zhu, Y., Shi, W., Wen, M., Fan, Z.: A survey of talking head synthesis techniques: Portrait generation, driving mechanisms, and editing. ACM Computing Surveys **58**(7), 1–43 (2026) [1](#), [3](#), [4](#)
124. Meng, R., Wu, W., Li, Y., Ma, C.: EchoTorrent: Towards swift, sustained, and streaming multi-modal video generation. arXiv preprint arXiv:2602.13669 (2026) [17](#), [18](#), [20](#)
125. Meng, R., Zhang, X., Li, Y., Ma, C.: Echomimicv2: Towards striking, simplified, and semi-body human animation. In: Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 5489–5498 (2025) [33](#)
126. Mildenhall, B., Srinivasan, P.P., Tancik, M., Barron, J.T., Ramamoorthi, R., Ng, R.: NeRF: Representing scenes as neural radiance fields for view synthesis. In: European Conference on Computer Vision, pp. 405–421. Springer (2020) [3](#), [4](#), [7](#), [27](#)
127. Müller, T., Evans, A., Schied, C., Keller, A.: Instant neural graphics primitives with a multiresolution hash encoding. ACM transactions on graphics (TOG) **41**(4), 1–15 (2022) [7](#), [27](#)
128. Pan, Y., Liu, C., Xu, S., Tan, S., Yang, J.: VASA-Rig: Audio-driven 3D facial animation with “live” mood dynamics in virtual reality. IEEE Transactions on Visualization and Computer Graphics (2025) [30](#)
129. Pang, Y., Zhang, Y., Shao, R., Deng, X., Gao, F., Xiaoming, X., Wei, X., Liu, Y.: Unimo: Unifying 2d video and 3d human motion with an autoregressive framework. arXiv preprint arXiv:2512.03918 (2025) [22](#), [23](#), [24](#)
130. Papamakarios, G., Nalisnick, E., Rezende, D.J., Mohamed, S., Lakshminarayanan, B.: Normalizing flows for probabilistic modeling and inference. Journal of Machine Learning Research **22**(57), 1–64 (2021) [12](#)
131. Papantonioni, F.P., Filntisis, P.P., Maragos, P., Roussos, A.: Neural emotion director: Speech-preserving semantic control of facial expressions in in-the-wild videos. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 18781–18790 (2022) [9](#)
132. Park, S.J., Kim, M., Hong, J., Choi, J., Ro, Y.M.: Synctalk-face: Talking face generation with precise lip-syncing via audio-lip memory. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 36, pp. 2062–2070 (2022) [8](#)
133. Pavlakos, G., Choutas, V., Ghorbani, N., Bolkart, T., Osman, A.A.A., Tzionas, D., Black, M.J.: Expressive body capture: 3D hands, face, and body from a single image. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 10975–10985 (2019) [7](#), [27](#)

134. Peebles, W., Xie, S.: Scalable diffusion models with transformers. In: Proceedings of the IEEE/CVF international conference on computer vision, pp. 4195–4205 (2023) **11**, **13**
135. Peng, Z., Hu, W., Ma, J., Zhu, X., Zhang, X., Zhao, H., Tian, H., He, J., Liu, H., Fan, Z.: Synctalk++: High-fidelity and efficient synchronized talking heads synthesis using gaussian splatting. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025) **28**, **31**, **32**
136. Peng, Z., Hu, W., Shi, Y., Zhu, X., Zhang, X., Zhao, H., He, J., Liu, H., Fan, Z.: Synctalk: The devil is in the synchronization for talking head synthesis. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 666–676 (2024) **28**, **31**, **32**
137. Podell, D., English, Z., Lacey, K., Blattmann, A., Dockhorn, T., Müller, J., Penna, J., Rombach, R.: Sdxl: Improving latent diffusion models for high-resolution image synthesis. In: International Conference on Learning Representations, vol. 2024, pp. 1862–1874 (2024) **7**
138. Prajwal, K., Mukhopadhyay, R., Namboodiri, V.P., Jawahar, C.: A lip sync expert is all you need for speech to lip generation in the wild. In: Proceedings of the 28th ACM international conference on multimedia, pp. 484–492 (2020) **2**, **4**, **8**, **11**, **35**, **39**
139. Qi, J., Ji, C., Xu, S., Zhang, P., Zhang, B., Bo, L.: ChatAnyone: Stylized real-time portrait video generation with hierarchical motion diffusion model. *arXiv preprint arXiv:2503.21144* (2025) **30**
140. Qian, S., Kirschstein, T., Schoneveld, L., Davoli, D., Giebenhain, S., Nießner, M.: GaussianAvatars: Photorealistic head avatars with rigged 3D gaussians. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 20299–20309 (2024) **28**
141. Qian, Z., Wang, S., Mihajlovic, M., Geiger, A., Tang, S.: 3DGS-Avatar: Animatable avatars via deformable 3D gaussian splatting. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 5020–5030 (2024) **28**
142. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al.: Learning transferable visual models from natural language supervision. In: International conference on machine learning, pp. 8748–8763. *Pmlr* (2021) **35**
143. Radford, A., Narasimhan, K., Salimans, T., Sutskever, I., et al.: Improving language understanding by generative pre-training (2018) **7**
144. Rakesh, V.K., Mazumdar, S., Maity, R.P., Pal, S., Das, A., Samanta, T.: Advancements in talking head generation: a comprehensive review of techniques, metrics, and challenges: Advancements in talking head generation... *The Visual Computer* **42**(1) (2025) **3**
145. Richard, A., Lea, C., Ma, S., Gall, J., De la Torre, F., Sheikh, Y.: Audio- and gaze-driven facial animation of codec avatars. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pp. 41–50 (2021) **27**
146. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B.: High-resolution image synthesis with latent diffusion models. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 10684–10695 (2022) **7**
147. Salimans, T., Ho, J.: Progressive distillation for fast sampling of diffusion models. In: International Conference on Learning Representations (2022) **7**
148. Sha, T., Zhang, W., Shen, T., Li, Z., Mei, T.: Deep person generation: A survey from the perspective of face, pose, and cloth synthesis. *ACM Computing Surveys* **55**(12), 1–37 (2023) **3**
149. Shah, J., Bikshandi, G., Zhang, Y., Thakkar, V., Ramani, P., Dao, T.: FlashAttention-3: Fast and accurate attention with asynchrony and low-precision. *Advances in Neural Information Processing Systems* **37**, 68658–68685 (2024) **7**
150. Shao, Z., Wang, Z., Li, Z., Wang, D., Lin, X., Zhang, Y., Fan, M., Wang, Z.: SplattingAvatar: Realistic real-time human avatars with mesh-embedded gaussian splatting. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 1606–1616 (2024) **28**, **31**, **32**
151. Shen, H., Zhang, J., Xiong, B., Hu, R., Chen, S., Wan, Z., Wang, X., Zhang, Y., Gong, Z., Bao, G., et al.: Efficient diffusion models: A survey. *Transactions on Machine Learning Research* (2025) **3**, **4**, **7**
152. Shen, L., Qiao, Q., Yu, T., Zhou, K., Yu, T., Zhan, Y., Wang, Z., Tao, M., Yin, S., Liu, S.: SoulX-FlashTalk: Real-time infinite streaming of audio-driven avatars via self-correcting bidirectional distillation. *arXiv preprint arXiv:2512.23379* (2026). URL <https://arxiv.org/abs/2512.23379> **1**, **2**, **12**, **15**, **16**, **17**, **18**, **20**, **39**, **40**
153. Shin, J., Li, Z., Zhang, R., Zhu, J.Y., Park, J., Shechtman, E., Huang, X.: MotionStream: Real-time video generation with interactive motion controls. *arXiv preprint arXiv:2511.01266* (2025) **14**, **15**, **17**, **18**, **20**
154. Siarohin, A., Lathuilière, S., Tulyakov, S., Ricci, E., Sebe, N.: Animating arbitrary objects via deep motion transfer. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 2377–2386 (2019) **29**
155. Siarohin, A., Lathuilière, S., Tulyakov, S., Ricci, E., Sebe, N.: First order motion model for image animation. *Advances in Neural Information Processing Systems* **32** (2019) **27**, **29**
156. Siarohin, A., Woodford, O.J., Ren, J., Chai, M., Tulyakov, S.: Motion representations for articulated animation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 13653–13662 (2021) **29**
157. Song, H., Yoon, B., Cho, W., Woo, W.: Rc-smpl: Real-time cumulative smpl-based avatar body generation. In: 2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), pp. 89–98. *IEEE* (2023) **27**
158. Song, K., Cui, J., Zhang, J.: ProgressiveAvatars: Progressive animatable 3D gaussian avatars. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 32518–32527 (2026) **28**
159. Song, L., Chen, L., Liu, C., Liu, P., Xu, C.: Texttoon: Real-time text toonify head avatar from single video. In: SIGGRAPH Asia 2024 Conference Papers, pp. 1–11 (2024) **28**
160. Song, Y., Dhariwal, P., Chen, M., Sutskever, I.: Consistency models. In: International Conference on Machine Learning, pp. 32211–32252. *PMLR* (2023) **1**, **7**, **14**
161. Sun, Z., Peng, Z., Ma, Y., Chen, Y., Zhou, Z., Zhou, Z., Zhang, G., Zhang, Y., Zhou, Y., Lu, Q., et al.: StreamAvatar: Streaming diffusion models for real-time interactive human avatars. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 10887–10897 (2026) **12**, **15**, **16**, **17**, **18**, **20**, **38**
162. Tan, S., Gong, B., Ji, B., Pan, Y.: FixTalk: Taming identity leakage for high-quality talking head generation in extreme cases. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 24–36 (2025) **2**, **9**, **11**, **40**
163. Tan, S., Ji, B., Bi, M., Pan, Y.: EDTalk: Efficient disentanglement for emotional talking head synthesis. In: European

- Conference on Computer Vision, pp. 398–416. Springer (2024) [4](#), [9](#), [11](#)
164. Tan, S., Ji, B., Pan, Y.: FlowVQTalker: High-quality emotional talking face generation through normalizing flow and quantization. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 26317–26327 (2024) [9](#), [11](#)
 165. Tang, A., He, T., Tan, X., Ling, J., Song, L.: Memories are one-to-many mapping alleviators in talking face generation. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)* (2024) [8](#)
 166. Tang, J., Wang, K., Zhou, H., Chen, X., He, D., Hu, T., Liu, J., Liu, Z., Zeng, G., Wang, J.: Real-time neural radiance talking portrait synthesis via audio-spatial decomposition. *International Journal of Computer Vision* **133**(9), 6362–6373 (2025). DOI 10.1007/s11263-025-02481-9 [2](#), [28](#), [31](#), [32](#)
 167. Tian, L., Wang, Q., Zhang, B., Bo, L.: Emo: Emote portrait alive generating expressive portrait videos with audio2video diffusion model under weak conditions. In: European Conference on Computer Vision, pp. 244–260. Springer (2024) [1](#), [2](#), [4](#), [10](#)
 168. Tikhonova, A., Kravtsov, A., Ziganshin, D., Burkov, E., Balitskiy, G., Sherman, S., Lebedev, V., Poletaev, V.: AVTR-1 (2026). URL <https://avaturm.live> [28](#)
 169. Tu, S., Pan, Y., Huang, Y., Han, X., Xing, Z., Dai, Q., Luo, C., Wu, Z., Jiang, Y.G.: StableAvatar: Infinite-length audio-driven avatar video generation. *arXiv preprint arXiv:2508.08248* (2025) [17](#), [20](#)
 170. Unterthiner, T., Van Steenkiste, S., Kurach, K., Marinier, R., Michalski, M., Gelly, S.: Towards accurate generative models of video: A new metric & challenges. *arXiv preprint arXiv:1812.01717* (2018) [35](#)
 171. Van Den Oord, A., Vinyals, O., et al.: Neural discrete representation learning. *Advances in neural information processing systems* **30** (2017) [7](#), [22](#)
 172. Wan, T., Wang, A., Ai, B., Wen, B., Mao, C., Xie, C.W., Chen, D., Yu, F., Zhao, H., Yang, J., et al.: Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314* (2025) [13](#), [14](#)
 173. Wang, H., Weng, Y., Du, J., Xu, H., Wu, X., He, S., Yin, B., Liu, C., Gao, J., Liu, Q.: READ: Real-time and efficient asynchronous diffusion for audio-driven talking head generation. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 40, pp. 9766–9774 (2026) [12](#), [13](#), [18](#)
 174. Wang, H., Weng, Y., Li, Y., Guo, Z., Du, J., Niu, S., Ma, J., He, S., Wu, X., Hu, Q., Yin, B., Liu, C., Liu, Q.: EmotiveTalk: Expressive talking head generation through audio information decoupling and emotional video diffusion. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 26212–26221 (2025) [14](#)
 175. Wang, J., Qian, X., Zhang, M., Tan, R.T., Li, H.: Seeing what you said: Talking face generation guided by a lip reading expert. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 14653–14662 (2023) [9](#), [11](#)
 176. Wang, K., Wu, Q., Song, L., Yang, Z., Wu, W., Qian, C., He, R., Qiao, Y., Loy, C.C.: MEAD: A large-scale audio-visual dataset for emotional talking-face generation. In: Computer Vision – ECCV 2020, pp. 700–717 (2020). DOI 10.1007/978-3-030-58589-1_42 [33](#), [34](#)
 177. Wang, K., Zhou, H., Wu, Q., Tang, J., Xu, Z., Liang, B., Hu, T., Ding, E., Liu, J., Liu, Z., et al.: Efficient video portrait reenactment via grid-based codebook. In: ACM SIGGRAPH 2023 Conference Proceedings, pp. 1–9 (2023) [9](#)
 178. Wang, L., Zhao, X., Sun, J., Zhang, Y., Zhang, H., Yu, T., Liu, Y.: StyleAvatar: Real-time photo-realistic portrait avatar from a single video. In: ACM SIGGRAPH 2023 Conference Proceedings, pp. 1–10 (2023) [2](#), [9](#), [11](#)
 179. Wang, L., Zhu, Y., Ge, Z., Zheng, Y., Zhang, L., Hu, T., Qin, S., Luo, M., Zhang, J., Chen, X., et al.: FlowAct-R1: Towards interactive humanoid video generation. *arXiv preprint arXiv:2601.10103* (2026) [16](#), [17](#), [18](#), [19](#), [20](#)
 180. Wang, S., Li, L., Ding, Y., Fan, C., Yu, X.: Audio2Head: Audio-driven one-shot talking-head generation with natural head motion. In: Proceedings of the International Joint Conference on Artificial Intelligence, pp. 1098–1105 (2021) [29](#)
 181. Wang, S., Ma, Y., Ding, Y., Hu, Z., Fan, C., Lv, T., Deng, Z., Yu, X.: StyleTalk++: A unified framework for controlling the speaking styles of talking heads. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **46**(6), 4331–4347 (2024) [30](#)
 182. Wang, T.C., Mallya, A., Liu, M.Y.: One-shot free-view neural talking-head synthesis for video conferencing. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 10039–10049 (2021) [29](#)
 183. Wang, Y., Yang, D., Bremond, F., Dantcheva, A.: Latent image animator: Learning to animate images via latent space navigation. *arXiv preprint arXiv:2203.09043* (2022) [29](#)
 184. Wang, Z., Bovik, A.C., Sheikh, H.R., Simoncelli, E.P.: Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing* **13**(4), 600–612 (2004). DOI 10.1109/TIP.2003.819861 [35](#)
 185. Wang, Z., Wang, J., Ma, K., Lin, D., Zhou, B.: TalkVerse: Democratizing minute-long audio-driven video generation. *arXiv preprint arXiv:2512.14938* (2025). DOI 10.48550/arXiv.2512.14938 [33](#), [34](#)
 186. Wang, Z., Zhang, P., Qi, J., Xu, S., Zhang, B., et al.: OmniTalker: One-shot real-time text-driven talking audio-video generation with multimodal style mimicking. *Advances in Neural Information Processing Systems* **38**, 21834–21862 (2026) [12](#)
 187. Weng, Y., Wang, H., Yu, Y., Du, J., He, S., Wu, X., Xu, H.: Eartalking: End-to-end gpt-style autoregressive talking head synthesis with frame-wise control. *arXiv preprint arXiv:2603.20307* (2026) [4](#), [22](#), [23](#), [24](#)
 188. Xiang, J., Gao, X., Guo, Y., Zhang, J.: FlashAvatar: High-fidelity head avatar with efficient gaussian embedding. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 1802–1812 (2024). DOI 10.1109/CVPR52733.2024.00177 [2](#), [3](#), [27](#), [28](#), [31](#), [32](#), [39](#), [40](#)
 189. Xie, L., Wang, X., Zhang, H., Dong, C., Shan, Y.: VFHQ: A high-quality dataset and benchmark for video face super-resolution. In: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, pp. 656–665 (2022). DOI 10.1109/CVPRW56347.2022.00081 [33](#), [34](#)
 190. Xie, Y., Gu, T., Li, Z., Zhang, C., Song, G., Zhao, X., Liang, C., Jiang, J., Xu, H., Luo, L.: X-Streamer: Unified human world modeling with audiovisual interaction. *arXiv preprint arXiv:2509.21574* (2025) [12](#), [23](#), [24](#), [25](#)
 191. Xiong, J., Li, C., Zhang, P.: Voxface: Streaming audio-visual synthesis via relay-style multi-token prediction for interactive conversation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 3543–3552 (2026) [2](#)
 192. Xiong, J., Liu, G., Huang, L., Wu, C., Wu, T., Mu, Y., Yao, Y., Shen, H., Wan, Z., Huang, J., et al.: Autoregressive

- models in vision: A survey. *Transactions on Machine Learning Research* (2025) **3**, **4**, **7**, **21**
193. Xu, S., Chen, G., Guo, Y.X., Yang, J., Li, C., Zang, Z., Zhang, Y., Tong, X., Guo, B.: VASA-1: Lifelike audio-driven talking faces generated in real time. *Advances in Neural Information Processing Systems* **37**, 660–684 (2024) **1**, **2**, **3**, **4**, **30**, **31**, **32**, **39**, **40**
 194. Xu, S., Chen, G., Yang, J., Zhang, Y., Deng, Y., Lin, S., Guo, B.: VASA-3D: Lifelike audio-driven gaussian head avatars from a single image. *Advances in Neural Information Processing Systems* **38**, 150764–150786 (2026) **29**
 195. Xue, H., Luo, X., Hu, Z., Zhang, X., Xiang, X., Dai, Y., Liu, J., Zhang, Z., Li, M., Yang, J., et al.: Human motion video generation: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025) **1**, **3**, **4**
 196. Yan, Z., Zhang, Y., Yuan, X., Lyu, S., Wu, B.: DeepfakeBench: A comprehensive benchmark of deepfake detection. In: *Advances in Neural Information Processing Systems*, vol. 36 (2023) **39**
 197. Yang, F., Li, H., Li, P., Yuan, W., Qiu, L., Song, C., Chen, C., He, Y., Zhang, S., Han, X., et al.: ViSA: 3D-aware video shading for real-time upper-body avatar creation. *arXiv preprint arXiv:2512.07720* (2025) **18**
 198. Yang, H., Zhang, Z., Dong, Y., Qian, J., Yang, J.: GaussianEmoTalker: Real-time emotional talking head synthesis with audio-driven and blendshape-based 3D gaussian splatting. *arXiv preprint arXiv:2607.00959* (2026) **28**
 199. Yang, S., Huang, W., Chu, R., Xiao, Y., Zhao, Y., Wang, X., Li, M., Xie, E., Chen, Y., Lu, Y., Han, S., Chen, Y.: LongLive: Real-time interactive long video generation. *arXiv preprint arXiv:2509.22622* (2025) **3**, **18**
 200. Yang, S., Kong, Z., Gao, F., Cheng, M., Liu, X., Zhang, Y., Kang, Z., Luo, W., Cai, X., He, R., et al.: InfiniteTalk: Audio-driven video generation for sparse-frame video dubbing. *arXiv preprint arXiv:2508.14033* (2025) **17**, **18**
 201. Yang, S., Oh, S.W., Zhou, Y., Kim, S.J.: IF-MDM: Implicit face motion diffusion model for high-fidelity realtime talking head generation. *arXiv preprint arXiv:2412.04000* (2024) **12**, **13**
 202. Yang, S., Wang, B., Zhang, Y., Shen, Y., Kim, Y.: Parallelizing linear transformers with the delta rule over sequence length. *Advances in neural information processing systems* **37**, 115491–115522 (2024) **26**
 203. Yang, Z., Teng, J., Zheng, W., Ding, M., Huang, S., Xu, J., Yang, Y., Hong, W., Zhang, X., Feng, G., et al.: Cogvideox: Text-to-video diffusion models with an expert transformer. In: *International Conference on Learning Representations*, vol. 2025, pp. 83048–83077 (2025) **11**, **13**
 204. Yang, Z., Zeng, A., Yuan, C., Li, Y.: Effective whole-body pose estimation with two-stages distillation. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 4210–4220 (2023) **7**
 205. Ye, Z., Tan, X., Yin, A., Lin, H., Zhang, G., Sun, P., Li, Y., Chan, C.M., Ye, W., Zhang, S., et al.: Talker-t2av: Joint talking audio-video generation with autoregressive diffusion modeling. *arXiv preprint arXiv:2604.23586* (2026) **22**, **23**
 206. Yin, T., Gharbi, M., Park, T., Zhang, R., Shechtman, E., Durand, F., Freeman, W.T.: Improved distribution matching distillation for fast image synthesis. *Advances in neural information processing systems* **37**, 47455–47487 (2024) **14**
 207. Yin, T., Gharbi, M., Zhang, R., Shechtman, E., Durand, F., Freeman, W.T., Park, T.: One-step diffusion with distribution matching distillation. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 6613–6623 (2024) **1**, **7**, **14**
 208. Yin, T., Zhang, Q., Zhang, R., Freeman, W.T., Durand, F., Shechtman, E., Huang, X.: From slow bidirectional to fast autoregressive video diffusion models. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 22963–22974 (2025) **1**, **16**
 209. Yu, H., Qu, Z., Yu, Q., Chen, J., Jiang, Z., Chen, Z., Zhang, S., Xu, J., Wu, F., Lv, C., Yu, G.: GaussianTalker: Speaker-specific talking head synthesis via 3D gaussian splatting. In: *Proceedings of the 32nd ACM International Conference on Multimedia*, pp. 3548–3557 (2024). DOI 10.1145/3664647.3681675 **27**, **28**, **31**, **32**
 210. Yu, H., Wang, Z., Pan, Y., Cheng, M., Yang, H., Wang, C., Xie, T., Xu, X., Wei, X., Cai, X.: LLIA—enabling low-latency interactive avatars: Real-time audio-driven portrait video generation with diffusion models. *arXiv preprint arXiv:2506.05806* (2025) **15**, **17**
 211. Yu, J., Qin, Y., Che, H., Liu, Q., Wang, X., Wan, P., Zhang, D., Gai, K., Chen, H., Liu, X.: A survey of interactive generative video. *arXiv preprint arXiv:2504.21853* (2025) **3**, **4**
 212. Yu, T., Qiao, Q., Shen, L., Zhou, K., Hu, J., Sheng, D., Hu, B., Qin, H., Gao, J., Zhou, C., et al.: SoulX-FlashHead: Oracle-guided generation of infinite real-time streaming talking heads. *arXiv preprint arXiv:2602.07449* (2026) **17**, **18**, **20**
 213. Zeng, A., Yang, C., Ge, C., Zhang, E., Xu, G., Lin, G., Gu, G., Pi, J., Li, L., Shi, M., et al.: LPM 1.0: Video-based character performance model. *arXiv preprint arXiv:2604.07823* (2026) **3**, **18**, **19**, **20**
 214. Zhang, D., Liu, Y., Lin, L., Zhu, Y., Li, Y., Qin, M., Li, Y., Wang, H.: GUAVA: Generalizable upper body 3D gaussian avatar. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 14205–14217 (2025) **29**
 215. Zhang, J., Jiang, K., Chen, J., Wang, X., Luo, Y., Wang, Y., Chen, D., Li, J., Ye, C., Chen, M., Zhu, H., Zhao, M., Jiang, Y., Huang, Z., Xiang, C., Zheng, K., Wang, H., Wang, X., Jia, Q., Chen, X., Chen, Y., Jiang, Y., Fu, F., Deng, Z., Bao, F., Chen, J., Zhu, J.: Vidu S1: A real-time interactive video generation model (2026). URL <https://arxiv.org/abs/2607.03118> **3**, **17**
 216. Zhang, R., Isola, P., Efros, A.A., Shechtman, E., Wang, O.: The unreasonable effectiveness of deep features as a perceptual metric. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (2018) **35**
 217. Zhang, W., Cun, X., Wang, X., Zhang, Y., Shen, X., Guo, Y., Shan, Y., Wang, F.: Sadtalker: Learning realistic 3d motion coefficients for stylized audio-driven single image talking face animation. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8652–8661 (2023) **1**
 218. Zhang, Y., Li, Z., Wang, D., Zhang, J., Zhou, D., Yin, Z., Dai, X., Yu, G., Li, X.: SpeakerVid-5M: A large-scale high-quality dataset for audio-visual dyadic interactive human generation. *arXiv preprint arXiv:2507.09862* (2025). DOI 10.48550/arXiv.2507.09862 **33**, **34**
 219. Zhang, Y., Shen, K., Li, Y.: EmbodiedHead: Real-time listening and speaking avatar for conversational agents. *arXiv preprint arXiv:2604.17211* (2026) **12**, **13**
 220. Zhang, Y., Zhong, Z., Liu, M., Chen, Z., Wu, B., Zeng, Y., Zhan, C., He, Y., Huang, J., Zhou, W.: MuseTalk: Real-time high-fidelity video dubbing via spatio-temporal sampling. *arXiv preprint arXiv:2410.10122* (2024) **2**, **4**, **8**, **9**, **11**, **39**, **40**
 221. Zhang, Z., Hu, Z., Deng, W., Fan, C., Lv, T., Ding, Y.: Dinet: Deformation inpainting network for realistic face visually dubbing on high resolution video. In: *Proceedings*

- of the AAAI conference on artificial intelligence, vol. 37, pp. 3543–3551 (2023) [8](#)
222. Zhang, Z., Li, L., Ding, Y., Fan, C.: Flow-guided one-shot talking face generation with a high-resolution audio-visual dataset. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 3661–3670 (2021). DOI 10.1109/CVPR46437.2021.00366 [33](#), [34](#)
 223. Zhao, J., Zhang, H.: Thin-plate spline motion model for image animation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 3657–3666 (2022) [27](#), [29](#)
 224. Zhao, L., Lu, X., Fan, R., Im, S.K., Wang, L.: Gaussian-hand: Real-time 3d gaussian rendering for hand avatar animation. IEEE Transactions on Visualization and Computer Graphics **31**(9), 6484–6496 (2024) [28](#)
 225. Zhao, Q., Long, P., Zhang, Q., Qin, D., Liang, H., Zhang, L., Zhang, Y., Yu, J., Xu, L.: Media2Face: Co-speech facial animation generation with multi-modality guidance. In: ACM SIGGRAPH 2024 Conference Papers, pp. 1–13 (2024) [30](#)
 226. Zhao, S., Wang, C., Zhao, W., Li, Z., Li, M., Zhang, S.: REA-Listener: Real-time listening head generation with dynamic emotion modeling and flexible modality adaptation. In: Proceedings of the 33rd ACM International Conference on Multimedia, pp. 9733–9742 (2025) [12](#), [30](#), [31](#), [32](#)
 227. Zhao, Z., Bao, Z., Li, Q., Qiu, G., Liu, K.: PSAvatar: A point-based shape model for real-time head avatar animation with 3D gaussian splatting. IEEE Transactions on Visualization and Computer Graphics (2026) [28](#)
 228. Zhen, D., Yin, S., Qin, S., Yi, H., Zhang, Z., Liu, S., Qi, G., Tao, M.: Teller: Real-time streaming audio-driven portrait animation with autoregressive motion generation. In: Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 21075–21085 (2025) [12](#), [17](#), [22](#), [23](#), [30](#), [31](#), [37](#)
 229. Zhen, D., Zheng, X., Zhang, R., Jiang, Z., Yan, Y., Tao, M., Yin, S.: SoulX-LiveAct: Towards hour-scale real-time human animation with neighbor forcing and ConvKV memory. arXiv preprint arXiv:2603.11746 (2026) [16](#), [17](#), [18](#), [20](#)
 230. Zheng, Z., Zhao, X., Zhang, H., Liu, B., Liu, Y.: Avatarrex: Real-time expressive full-body avatars. ACM Transactions on Graphics (TOG) **42**(4), 1–19 (2023) [28](#), [31](#), [32](#)
 231. Zhong, T., Liang, C., Jiang, J., Lin, G., Yang, J., Zhao, Z.: FADA: Fast diffusion avatar synthesis with mixed-supervised multi-CFG distillation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 3101–3110 (2025) [2](#), [12](#), [14](#), [17](#)
 232. Zhong, W., Fang, C., Cai, Y., Wei, P., Zhao, G., Lin, L., Li, G.: Identity-preserving talking face generation with landmark and appearance priors. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 9729–9738 (2023) [8](#)
 233. Zhou, H., Sun, Y., Wu, W., Loy, C.C., Wang, X., Liu, Z.: Pose-controllable talking face generation by implicitly modularized audio-visual representation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 4176–4186 (2021) [9](#)
 234. Zhou, M., Bai, Y., Zhang, W., Yao, T., Zhao, T., Mei, T.: Responsive listening head generation: A benchmark dataset and baseline. In: Computer Vision – ECCV 2022, pp. 124–142 (2022). DOI 10.1007/978-3-031-19839-7_8 [33](#), [34](#)
 235. Zhou, Y., Han, X., Shechtman, E., Echevarria, J., Kalogerakis, E., Li, D.: Makeltalk: speaker-aware talking-head animation. ACM Transactions On Graphics (TOG) **39**(6), 1–15 (2020) [9](#)
 236. Zhu, H., Wu, W., Zhu, W., Jiang, L., Tang, S., Zhang, L., Liu, Z., Loy, C.C.: CelebV-HQ: A large-scale video facial attributes dataset. In: Computer Vision – ECCV 2022, pp. 650–667 (2022). DOI 10.1007/978-3-031-20071-7_38 [33](#), [34](#)
 237. Zhu, H., Zhao, M., He, G., Su, H., Li, C., Zhu, J.: Causal forcing: Autoregressive diffusion distillation done right for high-quality real-time interactive video generation. arXiv preprint arXiv:2602.02214 (2026) [16](#)
 238. Zhu, Y., Zhang, L., Rong, Z., Hu, T., Liang, S., Ge, Z.: INFP: Audio-driven interactive head generation in dyadic conversations. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 10667–10677 (2025) [12](#), [13](#)
 239. Zielonka, W., Kirschstein, T., Bolkart, T., Giebenhain, S., Sklyarova, V., Deng, X., Xiang, D., Saito, S., Liu, Y., Niessner, M., Thies, J.: How to build digital humans? from priors to photorealistic avatars (2026). URL <https://arxiv.org/abs/2607.04341> [3](#)